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PHYSICS

Optical Trapping of Cold Atoms with a Hollow-Core Fiber

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Abstract

This work presents the design, implementation and characterization of an Optical Dipole Trap (ODT) for capturing cold ^{87}Rb atoms from a Magneto-Optical Trap (MOT) and optical molasses setup.
The ODT has been realized by coupling 1064 nm laser light into a Hollow-Core Photonic Crystal Fiber (HCPCF). Due to the inherently multimodal nature of these fibers, light injection is highly sensitive to misalignment. Monitoring the near-field at the fiber output was found to be the most effective alignment procedure. For this purpose, a near-field imaging system was built to optimize the light injection, achieving a coupling efficiency of 96% with an LP-like fundamental mode profile in test fibers.
Subsequent characterization of the in-vacuum fiber revealed mode degradation and reduced transmission efficiency, likely due to mechanical stress from the vacuum seals.
The temperature of the optical molasses was minimized by systematically tuning the cooling beam detuning and intensity, and the compensation coil currents used to cancel external magnetic fields. A minimum temperature of $T = (9.6 \pm 0.1)\ \mu$K was achieved.
Finally, the loading efficiency from the molasses into the ODT was optimized. After the MOT loading phase, the compensation coils were used to shift the atomic cloud's position, increasing the spatial overlap with the ODT beam. A loading efficiency of $(0.160 \pm 0.005)\%$ was achieved. This result is of the same order of magnitude as the expected value of $(0.41 \pm 0.04)\%$, calculated using the non-shallow trap model proposed in this thesis. The obtained efficiency is sufficient for the subsequent realization of an optical conveyor belt to effectively load the atoms into the fiber's core.
This work establishes a reliable experimental starting point for introducing the second, counter-propagating 1064 nm beam, necessary to realize the optical conveyor belt.

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Introduction

The work presented in this thesis was conducted in the cold atoms laboratory of Professors Francesco Minardi and Marco Prevedelli at the Department of Physics and Astronomy "A. Righi" of the University of Bologna. The primary objective of the experiment is to study a quantum system consisting of a cold rubidium-87 (^{87}Rb) gas loaded into the core of a Hollow-Core Photonic Crystal Fiber (HCPCF). By confining atoms directly within their hollow core, HCPCFs enable tight spatial confinement and strong atom-light interactions over macroscopic distances, establishing a highly versatile platform for advanced quantum technologies. Potential applications for the fiber-atoms system include the realization of highly sensitive, miniaturized quantum sensors (such as atomic magnetometers and interferometers), as well as the realization of quantum memories, which constitute the fundamental building blocks for future quantum networks.

Loading cold atoms into an HCPCF requires an efficient intermediate trapping and transfer mechanism. As a starting point, the source of these cold atoms is provided by a pre-existing setup utilizing a Magneto-Optical Trap (MOT) followed by an optical molasses phase, preparing a cold, sub-Doppler atomic sample. To efficiently load the atoms into the HCPCF, an optical conveyor belt is required. This system is realized by using two counter-propagating Optical Dipole Trap (ODT) beams that interfere to create a standing wave. Tuning the relative frequency of the two branches allows the standing wave to become dynamic, enabling the controlled transport of the cold atoms. Furthermore, the standing wave provides the necessary axial confinement inside the optical fiber's core. This thesis details the implementation and optimization of the first 1064 nm ODT beam, which represents the starting point for the implementation of the complete transfer mechanism.

The thesis is structured as follows:

Chapter 1 establishes the theoretical framework underpinning the experiment. It begins with the physics of optical waveguides, deriving the Linearly Polarized (LP) modes for standard step-index fibers in the weakly guiding approximation. It then introduces HCPCFs, contrasting the Photonic Band Gap (PBG) and Inhibited Coupling (IC) guiding mechanisms. Finally, it reviews the semiclassical theory of the optical dipole force, deriving the conservative trapping potential and scattering rates for multi-level alkali atoms.

Chapter 2 details the experimental apparatus and the analytical methodologies. Following an overview of the vacuum chamber, the MOT, and the laser frequency stabilization systems, it describes the implementation of the 1064 nm ODT setup. This includes the spatial characterization of the ODT laser beam, the design of the optical injection line, and the development of a custom horizontal microscope to monitor the near-field mode to optimize the coupling into the HCPCF. It also describes two theoretical models to calculate a reference ODT capture efficiency in shallow and non-shallow trap regimes. Finally, this chapter outlines the experimental and analytical procedure employed for obtaining the results summarized in the last chapter.

Chapter 3 presents the experimental results and their discussion. It first reports the systematic minimization of the optical molasses temperature. Subsequently, it details the tuning of the compensation magnetic fields required to maximize the spatial overlap between the atomic cloud and the ODT beam, ultimately optimizing the capture efficiency and comparing the experimental results with the theoretical predictions.

Chapter 1

Theoretical Background

This chapter provides an overview of the theoretical background underlying the main topics addressed in this thesis.

The first part of the chapter focuses on conventional step-index optical fibers, with the aim of explaining the physics of mode confinement. Starting from the wave equation, the electromagnetic field behavior in the fibers is derived. This leads, under the weakly guiding approximation, to the description of linearly polarized LP modes. Subsequently, an overview of the hollow-core photonic crystal fibers is presented, highlighting the different guiding mechanisms with respect to the step-index fibers, briefly discussing their core modes.

The second part of the chapter is devoted to the theory of optical dipole trapping. The physical mechanism that allows optical fields to confine neutral atoms is discussed, and expressions for the trapping potential are derived for both ideal two-level atoms and realistic multi-level atoms.

1.1 Optical Fibers

Waveguides are devices that employ dielectric materials with a high refractive index to confine and guide electromagnetic radiation, preventing the diffraction-induced spreading that occurs during free-space propagation. The electromagnetic wave that propagates inside a waveguide is called guided mode.

Optical fibers are cylindrically symmetric waveguides, and represent the most widespread class of optical waveguides. Following chapter III of Yariv *et al.* [1], this section aims to explain the physical principles underlying light confinement in optical fibers. Particular attention is devoted to the characterization of the guided modes in the weakly guiding approximation, which leads to the so-called linearly polarized (LP) modes.

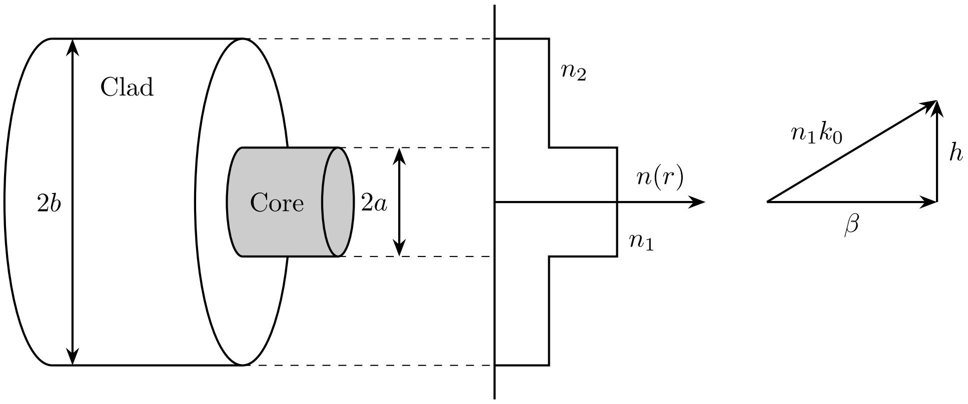


Figure 1.1: *Schematic representation of a step-index optical fiber (left), and wave vector diagram showing the geometric relationship between the wave number in the core* n_1k_0*, the longitudinal propagation constant* β*, and the transverse wave number* h *(right).*

1.1.1 Circularly symmetric waveguides

An optical fiber can be modeled as a cylindrical core of refractive index n_1 surrounded by a cylindrical cladding of refractive index n_2. The refractive index distribution is therefore described by the step-index profile

$$n(r) = \begin{cases} n_1, & 0 < r < a \\ n_2, & a < r < b \\ 1, & r > b, \end{cases} \tag{1.1}$$

where a is the radius of the core and b is the outer radius of the cladding. Usually the radius of the cladding is much larger than the radius of the core ($b \gg a$), thus the electromagnetic field can be considered zero at $r = b$. As an example, in a single-mode fiber guiding light with $\lambda = 1.5$ μm, the typical core radius is $a = 5$ μm while the cladding external radius is in the order of $b = 100$ μm.

1.1.1.1 The Cylindrical Wave Equation

Owing to the cylindrical symmetry of the system, it is convenient to work with the cylindrical coordinates:

$$\begin{cases} \mathbf{u_r} = & \hat{\mathbf{x}} \cos\phi + \hat{\mathbf{y}} \sin\phi \\ \mathbf{u}_\phi = & -\hat{\mathbf{x}} \sin\phi + \hat{\mathbf{y}} \cos\phi \\ \mathbf{u_z} = & \hat{\mathbf{z}}, \end{cases} \tag{1.2}$$

thus the electric and magnetic fields are expressed in terms of the components E_r, E_ϕ, E_z, H_r, H_ϕ and H_z. In cylindrical coordinates the unit vectors $\mathbf{u_r}$ and $\mathbf{u}_\phi$ are not constant vectors ($\partial\mathbf{u_r}/\partial\phi = \mathbf{u}_\phi$ and $\partial\mathbf{u}_\phi/\partial\phi = -\mathbf{u_r}$). This complicates the wave equation for the associated components of the field. Nevertheless, as $\mathbf{u_z} = \hat{\mathbf{z}}$ the equation for the z components remains simple. Assuming the dependence on the coordinate z and time t to be $e^{i(\omega t - \beta z)}$, namely:

$$\begin{bmatrix} \mathbf{E}(\mathbf{r}, t) \\ \mathbf{H}(\mathbf{r}, t) \end{bmatrix} = \begin{bmatrix} \mathbf{E}(r, \phi) \\ \mathbf{H}(r, \phi) \end{bmatrix} e^{i(\omega t - \beta z)}, \tag{1.3}$$

where β is referred to as the propagation constant of the mode, the wave equation for E_z and H_z is

$$(\nabla^2 + k^2) \begin{bmatrix} E_z \\ H_z \end{bmatrix} = 0, \tag{1.4}$$

where $k^2 = \omega^2 n^2 / c^2$ and ∇^2 is the Laplacian operator in cylindrical coordinates. To solve the cylindrical wave equation, the standard approach is to solve equation (1.4) and then express E_r, E_ϕ, H_r and H_ϕ in terms of E_z and H_z through Maxwell's curl equations in cylindrical coordinates (Appendix A of [1]).

Using (1.3) and explicitly writing the transverse cylindrical Laplacian in (1.4), one gets:

$$\left(\frac{\partial^2}{\partial r^2} + \frac{1}{r} \frac{\partial}{\partial r} + \frac{1}{r^2} \frac{\partial^2}{\partial \phi^2} + (k^2 - \beta^2) \right) \begin{bmatrix} E_z \\ H_z \end{bmatrix} = 0. \tag{1.5}$$

Equation (1.5) is solved by assuming the separation of variables

$$F_z(r, \phi) = R(r)\Phi(\phi), \tag{1.6}$$

where $F_z = E_z, H_z$, obtaining

$$\Phi \frac{d^2 R}{dr^2} + \frac{1}{r} \Phi \frac{dR}{dr} + \frac{1}{r^2} R \frac{d^2 \Phi}{d\phi^2} + (k^2 - \beta^2) R(r) \Phi(\phi) = 0. \tag{1.7}$$

Dividing by $R\Phi$ and rearranging terms leads to

$$r^2 \left(\frac{1}{R} \frac{d^2 R}{dr^2} + \frac{1}{r} \frac{1}{R} \frac{dR}{dr} + (k^2 - \beta^2) \right) = -\frac{1}{\Phi} \frac{d^2 \Phi}{d\phi^2}. \tag{1.8}$$

The last equation is in the form $f(r) = g(\phi)$. Therefore, since the left-hand side depends only on r and the right-hand side only on ϕ, both sides must be equal to the same constant:

$$\begin{cases} r^2 \left(\frac{1}{R} \frac{d^2 R}{dr^2} + \frac{1}{r} \frac{1}{R} \frac{dR}{dr} + (k^2 - \beta^2) \right) = l^2 \\ -\frac{1}{\Phi} \frac{d^2 \Phi}{d\phi^2} = l^2 \end{cases} . \tag{1.9}$$

The angular part has a simple exponential solution:

$$\Phi(\phi) = C_1 e^{+il\phi} + C_2 e^{-il\phi}, \tag{1.10}$$

where C_1 and C_2 are constants that are determined by the boundary conditions of the problem. To be physically meaningful, $\Phi(\phi)$ must be single valued ($\Phi(\phi) = \Phi(\phi + 2\pi)$), thus the value of l is constrained to

$$l = 0, 1, 2, ... \tag{1.11}$$

A set of elementary solutions can therefore be expressed in the form

$$\begin{bmatrix} E_z \\ H_z \end{bmatrix} = R(r)e^{\pm il\phi},$$

where the differential equation that describes the radial part is

$$\frac{d^2R}{dr^2} + \frac{1}{r}\frac{dR}{dr} + \left(k^2 - \beta^2 - \frac{l^2}{r^2}\right) R = 0, \tag{1.12}$$

which is recognized as Bessel's differential equation, and its solutions are a linear combination of Bessel functions of order l.

1.1.1.2 The Bessel functions

The radial equation (1.12) admits different classes of solutions depending on the sign of the quantity $k^2 - \beta^2$:

- for $k^2 - \beta^2 > 0$, the solutions are given by the ordinary Bessel functions of the first and second kind of order l:

$$R(r) = \bar{C}_1 J_l(hr) + \bar{C}_2 Y_l(hr); \tag{1.13}$$

- for $k^2 - \beta^2 < 0$, the solutions are given by the modified Bessel functions of the first and second kind, of order l:

$$R(r) = \bar{C}_1 I_l(qr) + \bar{C}_2 K_l(qr), \tag{1.14}$$

where $h^2 = k^2 - \beta^2$ and $q^2 = \beta^2 - k^2$, and $\bar{C}_1$ and $\bar{C}_2$ are arbitrary constants determined by the boundary conditions.

The behavior of the ordinary and modified Bessel functions is illustrated in Figure 1.2. To proceed with the physical solution, it is useful to consider their asymptotic forms for both large and small arguments.

For $x \gg 1, l$, the asymptotic expressions are

$$\begin{aligned}
J_l(x) &\to \left(\frac{2}{\pi x}\right)^{1/2} \cos\left(x - \frac{l\pi}{2} - \frac{\pi}{4}\right), \\
Y_l(x) &\to \left(\frac{2}{\pi x}\right)^{1/2} \sin\left(x - \frac{l\pi}{2} - \frac{\pi}{4}\right), \\
I_l(x) &\to \left(\frac{1}{2\pi x}\right)^{1/2} e^{x}, \\
K_l(x) &\to \left(\frac{\pi}{2x}\right)^{1/2} e^{-x}.
\end{aligned} \tag{1.15}$$

For $x \ll 1$, the corresponding asymptotic expressions are

$$\begin{aligned}
J_l(x) &\to \frac{1}{l!}\left(\frac{x}{2}\right)^{l}, \\
Y_0(x) &\to \frac{2}{\pi}\left(\ln\frac{x}{2} + \gamma_E\right), \\
Y_l(x) &\to -\frac{(l-1)!}{\pi}\left(\frac{2}{x}\right)^{l} \quad l = 1, 2, 3, \ldots, \\
I_l(x) &\to \frac{1}{l!}\left(\frac{x}{2}\right)^{l}, \\
K_0(x) &\to -\left(\ln\frac{x}{2} + \gamma_E\right), \\
K_l(x) &\to \frac{(l-1)!}{2}\left(\frac{2}{x}\right)^{l} \quad l = 1, 2, 3, \ldots,
\end{aligned} \tag{1.16}$$

where γ_E is the Euler–Mascheroni constant.

These asymptotic behaviors play a crucial role in the construction of guided-mode solutions. In particular, the functions $Y_l(x)$ and $K_l(x)$ diverge as $x \to 0$, whereas $K_l(x)$ decays exponentially for large arguments. Consequently, only specific combinations of Bessel functions satisfy the physical requirements for confined modes.

Another important property is that the derivative of a Bessel function of order l can be expressed as a combination of Bessel functions of orders $l-1$ and $l+1$ [2]. Since the transverse field components E_r, E_ϕ, H_r, and H_ϕ are obtained from spatial derivatives of E_z and H_z, they are generally described by combinations of Bessel functions of different orders.

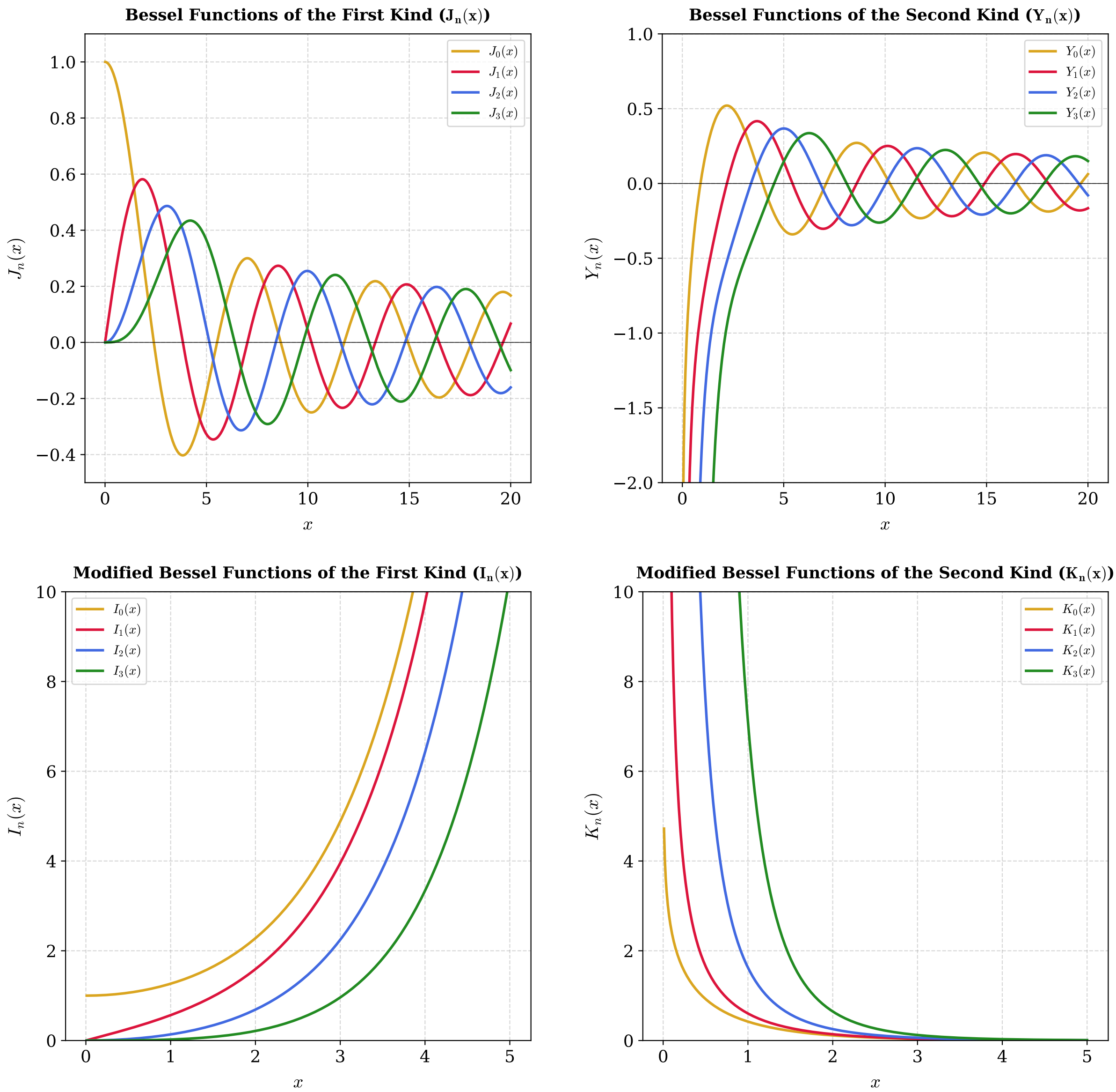


Figure 1.2: *Behavior of ordinary and modified Bessel functions for the first four integer orders, $n \in \{0, 1, 2, 3\}$. Top panels: Ordinary Bessel functions of the first kind, $J_n(x)$ (left), and second kind, $Y_n(x)$ (right). Bottom panels: Modified Bessel functions of the first kind, $I_n(x)$ (left), and second kind, $K_n(x)$ (right).*

1.1.2 The Weakly Guiding Approximation

The solution of the complete problem of electromagnetic guided modes in circular waveguides is cumbersome, and it leads to the so-called hybrid modes indicated by EH and HE, which are a combination of modes with the electric field transverse to the propagation of the wave (TE) and modes with the magnetic field transverse to the propagation direction (TM) [1].

However, for a large variety of fibers, the core refractive index is only slightly higher than the refractive index of the cladding:

$$\frac{n_1 - n_2}{n_1} \ll 1, \tag{1.17}$$

a condition commonly referred to as weakly guiding approximation. Under this assumption, the exact modes can be accurately approximated by a simpler set of modes, known as linearly polarized (LP) modes, first introduced in the work of Gloge [3].

Using the relation $k^2 = (n(r)\,k_0)^2$ where $k_0 = \omega/c$, equation (1.5) can be rewritten as

$$[\nabla_t^2 + ((n(r)\,k_0)^2 - \beta^2)] \begin{bmatrix} E_z \\ H_z \end{bmatrix} = 0. \tag{1.18}$$

Looking for a guided mode in the fiber is equivalent to seeking a solution that exhibits an oscillatory behavior in the core while exhibiting evanescent decay in the cladding. From the properties of second-order differential equations, these requirements imply:

$$\begin{aligned} r > a \quad &\text{vanishing} \quad &\Longrightarrow (n_2 k_0)^2 - \beta^2 < 0, \\ r < a \quad &\text{oscillating} \quad &\Longrightarrow (n_1 k_0)^2 - \beta^2 > 0. \end{aligned} \tag{1.19}$$

Consequently, the propagation constant of a guided mode must satisfy

$$n_2 k_0 < \beta < n_1 k_0, \tag{1.20}$$

which in weakly guiding regime becomes

$$n_2 k_0 \simeq \beta \simeq n_1 k_0. \tag{1.21}$$

At this point, it is convenient to introduce the transverse wave numbers in the core (h) and in the cladding (q):

$$h = \sqrt{n_1^2\,k_0^2 - \beta^2}, \tag{1.22}$$

$$q = \sqrt{\beta^2 - n_2^2\,k_0^2}. \tag{1.23}$$

The parameter h describes the transverse oscillation of the field within the core (see Figure 1.1), whereas q characterizes the decay of the mode in the cladding.
Equation (1.21) implies that

$$h, q \ll \beta. \tag{1.24}$$

Physically this means that the rays are approximately parallel to the fiber axis. Therefore, the longitudinal components of the electric and magnetic field are far weaker than the transverse ones, and the guided modes are approximately transverse electromagnetic (TEM) [4].

The objective is now to identify a complete set of approximate modal solutions in the weakly guiding limit. The approach is to focus on solutions where either the x or the y component of the field vanishes, thus looking for x-polarized or y-polarized solutions. The approximation made by Gloge [3] is based on postulating for the transverse component a simple dependence on a single Bessel function. Matching the requirements (1.19) with the Bessel equation solution condition (1.13) and (1.14), one deduces that the mode must be described by ordinary Bessel functions inside the core and by modified Bessel functions in the cladding. Furthermore, physically acceptable solutions must remain finite at the fiber axis and vanish at large radial distances. By comparison with the asymptotic behavior in (1.15) and (1.16), only J_l is admissible in the core region, whereas only K_l satisfies the required decay condition in the cladding.

For a mode polarized along the (y)-direction, the transverse electric field is therefore postulated as

$$\begin{aligned} E_x &= 0 \\ E_y &= \begin{cases} AJ_l(hr)e^{\pm il\phi}e^{i(\omega t-\beta z)}, & r < a \\ BK_l(qr)e^{\pm il\phi}e^{i(\omega t-\beta z)}, & r > a \end{cases}, \end{aligned} \tag{1.25}$$

where A and B are constants that are determined from the continuity condition at the core–cladding interface.
It is worth noting that, for each value of l, equation (1.25) describes two independent azimuthal solutions associated to $e^{+il\phi}$ and $e^{-il\phi}$. Taking into account the two possible independent linear polarizations, a total of four independent basis functions are associated with each value of l.

The remaining electromagnetic field components can be obtained from Maxwell's equations. In particular, the magnetic field is given by [1]:

$$\begin{cases} H_x = \frac{-i}{\omega\mu}\frac{\partial}{\partial z}E_y = -\frac{\beta}{\omega\mu}E_y \\ H_y \simeq 0 \\ H_z = \frac{i}{\omega\mu}\frac{\partial}{\partial x}E_y \end{cases}, \tag{1.26}$$

and similarly, the longitudinal electric field component can be expressed as

$$E_z = \frac{i}{\omega\epsilon}\frac{\partial}{\partial y}H_x = \frac{-i\beta}{\omega^2\mu\epsilon}\frac{\partial}{\partial y}E_y. \tag{1.27}$$

Solving the differential equations for E_z and H_z yields the complete set of basis fields in the weakly guiding approximation [1].

In the core ($r < a$):

$$\begin{cases} E_x = 0 \\ E_y = AJ_l(hr)e^{\pm il\phi}e^{i(\omega t-\beta z)} \\ E_z = \pm\frac{h}{\beta}\frac{A}{2}[J_{l+1}(hr)e^{\pm i(l+1)\phi} + J_{l-1}(hr)e^{\pm i(l-1)\phi}]e^{i(\omega t-\beta z)} \end{cases}, \tag{1.28}$$

$$\begin{cases} H_x = -\frac{\beta}{\omega\mu}AJ_l(hr)e^{\pm il\phi}e^{i(\omega t-\beta z)} \\ H_y = 0 \\ H_z = -\frac{ih}{\omega\mu}\frac{A}{2}[J_{l+1}(hr)e^{\pm i(l+1)\phi} - J_{l-1}(hr)e^{\pm i(l-1)\phi}]e^{i(\omega t-\beta z)} \end{cases}. \tag{1.29}$$

In the cladding ($r > a$):

$$\begin{cases} E_x = 0 \\ E_y = BK_l(qr)e^{\pm il\phi}e^{i(\omega t-\beta z)} \\ E_z = \frac{q}{\beta}\frac{B}{2}[K_{l+1}(qr)e^{\pm i(l+1)\phi} - K_{l-1}(qr)e^{\pm i(l-1)\phi}]e^{i(\omega t-\beta z)} \end{cases}, \tag{1.30}$$

$$\begin{cases} H_x = -\frac{\beta}{\omega\mu}BK_l(qr)e^{\pm il\phi}e^{i(\omega t-\beta z)} \\ H_y = 0 \\ H_z = -\frac{iq}{\omega\mu}\frac{B}{2}[K_{l+1}(qr)e^{\pm i(l+1)\phi} + K_{l-1}(qr)e^{\pm i(l-1)\phi}]e^{i(\omega t-\beta z)} \end{cases}. \tag{1.31}$$

Considering that $E_z/E_y \propto h/\beta$, $H_z/H_x \propto h/\beta$ in the core and $E_z/E_y \propto q/\beta$, $H_z/H_x \propto q/\beta$ in the cladding, since $h, q \ll \beta$, E_y and H_x are the dominant components, and the mode is approximately TEM, consistently with the considerations given above.

Considering that $E_\phi = -E_x \sin(\phi) + E_y \cos(\phi)$, for a y-polarized mode $E_\phi \propto E_y$, thus the radial continuity condition at $r = a$ reduces to the continuity condition of E_y, which leads to

$$B = A\frac{J_l(ha)}{K_l(qa)}. \tag{1.32}$$

An identical condition is obtained from the continuity of H_ϕ.

On the other hand, the continuity condition of E_z at $r = a$ must hold for every value of the azimuthal coordinate ϕ. Consequently, the coefficients of $\exp[\phi(l+1)\phi]$ and $\exp[\pm i(l-1)\phi]$

must be matched separately. This leads to

$$ha\frac{J_{l+1}(ha)}{J_l(ha)} = qa\frac{K_{l+1}(qa)}{K_l(qa)}, \tag{1.33}$$

$$ha\frac{J_{l-1}(ha)}{J_l(ha)} = -qa\frac{K_{l-1}(qa)}{K_l(qa)}, \tag{1.34}$$

which are equivalent as a consequence of the recurrence relations satisfied by the Bessel functions [1]. The same characteristic equations are obtained by imposing continuity of H_z.

Finally, repeating the same procedure for an x-polarized mode leads to identical conditions, demonstrating that both polarizations share the same modal eigenvalues.

1.1.3 The LP modes

Equation (1.33) depends only on β through the transverse wave numbers h and q. It yields an infinite set of discrete solutions for each value of l. Therefore, the propagation constants are labeled as:

$$\beta_{lm} \quad : \quad l = 0, 1, 2, ... \quad m = 1, 2, 3, ... \tag{1.35}$$

Their corresponding modes are called Linearly Polarized modes, denoted by:

$$LP_{lm} \quad : \quad l = 0, 1, 2, ... \quad m = 1, 2, 3, ... \tag{1.36}$$

Since the condition in (1.33) holds for both x-polarized and y-polarized waves, the modes LP_{lm-x} and LP_{lm-y} are degenerate sharing the same propagation constant β_{lm}. This degeneracy is a direct consequence of the circular symmetry of the fiber. Furthermore, accounting for the degeneracy associated with the azimuthal phase term $e^{\pm il\phi}$, the propagation constant β_{lm} and the mode LP_{lm} are four-fold degenerate. A notable exception occurs for the fundamental case $l = 0$, for which the exponential is equal to unity independently of the sign of $l\phi$, reducing β_{0m} and LP_{0m} to a two-fold degeneracy.

1.1.3.1 Characteristic equation of the LP modes

In order to determine the allowed values of β_{lm}, the characteristic equation (1.33) must be solved. Since the transverse wave numbers h and q are interdependent, it is convenient to express them in terms of a dimensionless parameter. For this purpose, the normalized frequency, also called fiber V parameter, is introduced and defined as:

$$V = k_0 a\sqrt{n_1^2 - n_2^2} = \frac{2\pi a}{\lambda}\sqrt{n_1^2 - n_2^2} = \sqrt{(ha)^2 + (qa)^2}. \tag{1.37}$$

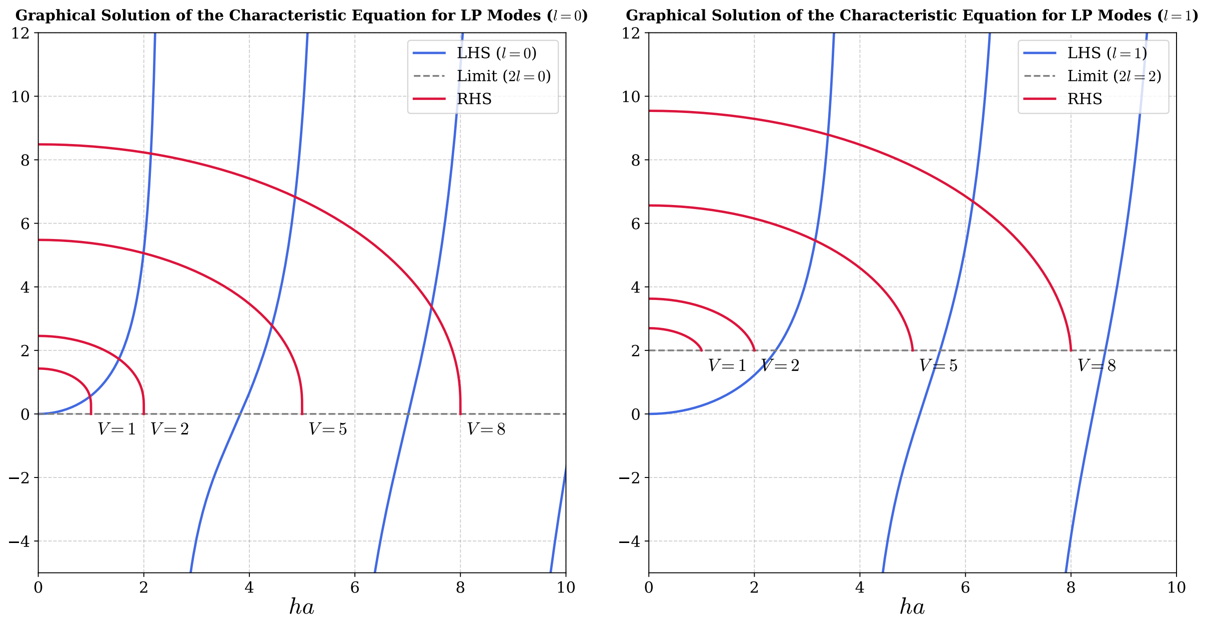


Figure 1.3: *Graphical solution of the characteristic equation* (1.33) *of the LP modes, for* $l = 0$ *(left) and* $l = 1$ *(right). The blue curves represent the left-hand side (LHS) of equation* (1.33) *for the given* l. *The red curves represent the right-hand side (RHS) for* $V = 2,\ 5,\ 8$. *The gray dashed line indicates the limit* $qa \to 0$ *of the LHS.*

The V parameter is an intrinsic property of the fiber determined by its structural and material characteristics. As shown below, it governs the number of guided modes and determines their respective confinement factor.

Using the V parameter, the cladding variable qa can be expressed as:

$$qa = \sqrt{V^2 - (ha)^2}, \tag{1.38}$$

and substituting it back in equation (1.33), the characteristic equation is reduced to a function of the single variable ha, allowing for a graphical or numerical solution. Moreover, equation (1.38) explicitly restricts the domain of the left-hand side of (1.33) to $ha < V$.

Figure 1.3 shows the graphical solution for $l = 0$ (left) and $l = 1$ (right). For $l = 0$, the left-hand side of equation (1.33) vanishes at the roots of $J_1(ha)$, located at $ha = 0,\ 3.832,\ 7.016\ \ldots$, whereas the denominator vanishes at the roots of $J_0(ha)$ which occur at $ha = 2.405,\ 5.520,\ 8.654\ \ldots$ corresponding to vertical asymptotes. Therefore, in the interval $ha \in [0, 2.405]$ the left-hand side is a strictly increasing positive function going from 0 to $+\infty$. Conversely, the right-hand side, at $ha = 0$ assumes the positive value $VK_1(V)/K_0(V)$ and monotonically decreases to 0 at $ha = V$. Consequently, an intersection point always exists, even for very small values of V, implying that the fundamental LP_{01} mode is always guided by an optical fiber regardless of its V parameter.

Using (1.16), one derives that for $l \geq 1$, the right-hand side asymptotically approaches $2l$ as $ha \to V$. Under these conditions, an intersection is no longer guaranteed for every

value of V. In particular for $V < 2.405$ the only possible guided mode is the LP_{01}. As V increases beyond 2.405, also the LP_{11} appears, therefore $V = 2.405$ is called the V cutoff value of the mode LP_{11}.
The cutoff value V_{cutoff} for a given mode can be determined by considering the condition at which the field is no longer evanescent in the cladding, corresponding to $q = 0$. Since $V = ha$ when $q = 0$, the cutoff condition can be obtained by imposing $q = 0$ in equation (1.34), yielding

$$V_{\text{cutoff}} \frac{J_{l-1}(V_{\text{cutoff}})}{J_l(V_{\text{cutoff}})} = 0. \tag{1.39}$$

For $l \geq 1$, in the limit $V_{\text{cutoff}} \to 0$ (see (1.16)) the left-hand side evaluates to $2l$ therefore for $l \geq 1$ the cutoff is strictly larger than zero. For $l = 0$ using the property $J_{-1}(V) = -J_1(V)$, for $V_{\text{cutoff}} \to 0$ one gets $0 = 0$, confirming that the mode LP_{01} is guided independently of the value of V.
Excluding the special case of the LP_{01}, equation (1.39) leads to

$$J_{l-1}(V_{\text{cutoff}}) = 0. \tag{1.40}$$

Therefore, the cutoff value of the LP_{lm} mode is given by the m-th zero of the Bessel function J_{l-1} if $l \neq 0$, and by the $(m-1)$-th zero of J_{l-1} if $l = 0$, while the LP_{01} mode is always guided. Here, the zero at the origin is not included in the enumeration of the zeros of the Bessel functions. The cutoff values of several low-order LP modes are reported in Table 1.1.
It is worth noting that equation (1.34) is employed to obtain the cutoff condition rather than (1.33) because its right-hand side conveniently vanished as $ha \to V$, making the cutoff directly associated to the roots of ordinary Bessel functions of the first kind.

Table 1.1: *Cutoff values of V for low-order LP modes.*

V_{cutoff}	$m = 1$	$m = 2$	$m = 3$	$m = 4$
$l = 0$	0	3.832	7.016	10.173
$l = 1$	2.405	5.520	8.654	11.792
$l = 2$	3.832	7.016	10.173	13.323
$l = 3$	5.136	8.417	11.620	14.796
$l = 4$	6.379	9.760	13.017	16.224

1.1.3.2 Power Flow of the LP modes

Prior to evaluating the power flow of the LP modes, a clarification regarding the basis chosen for the spatial representation of the field is necessary. So far, following the formalism in [1], a basis in which the azimuthal dependence is expressed by the exponential $e^{\pm il\phi}$, for the two degenerate solution $\pm l$ has been utilized. This representation is convenient for calculation, however, it leads to a basis in which the modes have an azimuthal phase dependence, or equivalently modes that carry a defined orbital angular momentum (OAM) $l\hbar$ per photon [5]. In typical experimental conditions, however, optical fibers are coupled with standard Gaussian beams, which carry zero macroscopic OAM due to their lack of azimuthal phase gradients. Therefore, to conserve the net angular momentum at the fiber input, the coupling excites the two azimuthal counter-propagating spatial modes $+l$ and $-l$ with identical amplitudes. Consequently, a basis employing trigonometric functions ($\cos(l\phi)$ and $\sin(l\phi)$) is physically more representative of typical intensity profiles and is widely adopted in the literature. These functions represent a linear superposition of counter-propagating azimuthal modes carrying opposite OAM, yielding a net-zero OAM state:

$$\begin{aligned} \cos(l\phi) &= \frac{e^{il\phi} + e^{-il\phi}}{2}, \\ \sin(l\phi) &= \frac{e^{il\phi} - e^{-il\phi}}{2i}. \end{aligned} \tag{1.41}$$

Therefore, the trigonometric basis is adopted for the remainder of this analysis. While a pure $e^{\pm il\phi}$ basis implies a ring-shaped intensity profile devoid of azimuthal modulation, the trigonometric basis captures the interference between opposite OAM states, producing the characteristic $2l$-lobe intensity patterns routinely observed experimentally.

The power flow of the LP modes is evaluated via the longitudinal component of the time-averaged Poynting vector:

$$S_z = \frac{1}{2}\,\mathrm{Re}[E_x H_y^* - E_y H_x^*]. \tag{1.42}$$

Using the field expressions (1.28), (1.29), (1.30) and (1.31), the trigonometric azimuthal basis, the power flux density becomes:

$$S_z^{(lm)} = \frac{\beta}{2\omega\mu} \begin{cases} |A|^2 J_l^2(hr)\cos^2(l\phi) & r < a \\ \\ |B|^2 K_l^2(qr)\cos^2(l\phi) & r > a \end{cases}, \tag{1.43}$$

where the dependence on m is implicitly contained in h and q. The power guided in the core and cladding is obtained by integrating the Poynting vector over the respective cross-sections (using infinite cladding approximation):

$$P_{\text{core}} = \int_0^{2\pi}\int_0^a S_z r\, dr\, d\phi, \tag{1.44}$$

$$P_{\text{clad}} = \int_0^{2\pi}\int_a^\infty S_z r\, dr\, d\phi. \tag{1.45}$$

Evaluating these integrals [1], and substituting B from (1.32), the result is:

$$P_{\text{core}} = \frac{\beta}{2\omega\mu}\pi a^2 |A|^2 \left[J_l^2(ha) - J_{l-1}(ha)J_{l+1}(ha)\right], \tag{1.46}$$

$$P_{\text{clad}} = \frac{\beta}{2\omega\mu}\pi a^2|A|^2 \left[-J_l^2(ha) - \left(\frac{h}{q}\right)^2 J_{l-1}(ha)J_{l+1}(ha)\right]. \tag{1.47}$$

Summing these contributions gives the total power flow:

$$P_{\text{tot}} = \frac{\beta}{2\omega\mu}\pi a^2|A|^2 \left(1 + \frac{h^2}{q^2}\right)\left[-J_{l-1}(ha)J_{l+1}(ha)\right]. \tag{1.48}$$

By multiplying equations (1.33) and (1.34) side by side, one gets:

$$(ha)^2 \frac{J_{l+1}(ha)J_{l-1}(ha)}{J_l^2(ha)} = -(qa)^2 \frac{K_{l+1}(qa)K_{l-1}(qa)}{K_l^2(qa)}. \tag{1.49}$$

Given that the modified Bessel functions K_l are always positive (see Figure 1.2), the quantity $J_{l-1}(ha)J_{l+1}(ha)$ is always negative for the allowed values of ha, therefore the total power flow (1.48) is always positive as physically expected.
The core confinement factor is defined as $\Gamma_1 = P_{\text{core}}/P_{\text{tot}}$

$$\Gamma_1 = \frac{P_{\text{core}}}{P_{\text{tot}}} = \frac{1}{V^2}\left((qa)^2 - \frac{(qa)^2 J_l^2(ha)}{J_{l-1}(ha)J_{l+1}(ha)}\right). \tag{1.50}$$

Γ_1 gives a measure of how much of the mode is travelling in the core rather than in the cladding. This parameter is important for calculating the dispersion and attenuation properties of the fiber, which are critical parameters in the design of fibers for telecommunication systems. The dependence of Γ_1 on V is not trivial as both h and q depend on V through the solution of the characteristic equation (1.33). Generally, the confinement factor Γ_1 increases monotonically with V. The confinement factor at cutoff is obtained by taking the limit $q \to 0$ in (1.50), obtaining:

$$\Gamma_1 = \begin{cases} 0 & \text{for} \quad l = 0, 1 \\ 1 - \frac{1}{l} & \text{for} \quad l > 1 \end{cases}. \tag{1.51}$$

For example the mode LP_{21} at cutoff has a confinement efficiency of 0.5. Figure 1.4 illustrates the behavior of the confinement factor as a function of the V parameter for some low-order LP modes.

Finally, the calculated intensity profiles for selected LP modes are mapped in Figure 1.5. The figure highlights how the azimuthal index l governs the number of diametric nodal lines crossing the beam axis, whereas the radial index m dictates the number of concentric local intensity maxima along the radial coordinate.

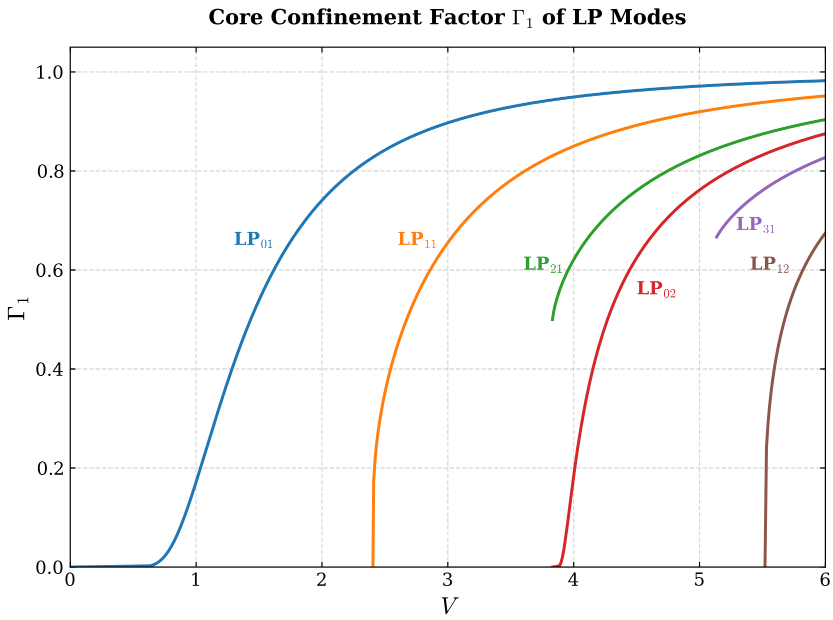


Figure 1.4: *Confinement factor Γ_1 of LP modes as a function of the fiber V parameter. Note that for the modes with $l > 1$ (LP_{21}, LP_{31}) the confinement at cutoff is different from 0.*

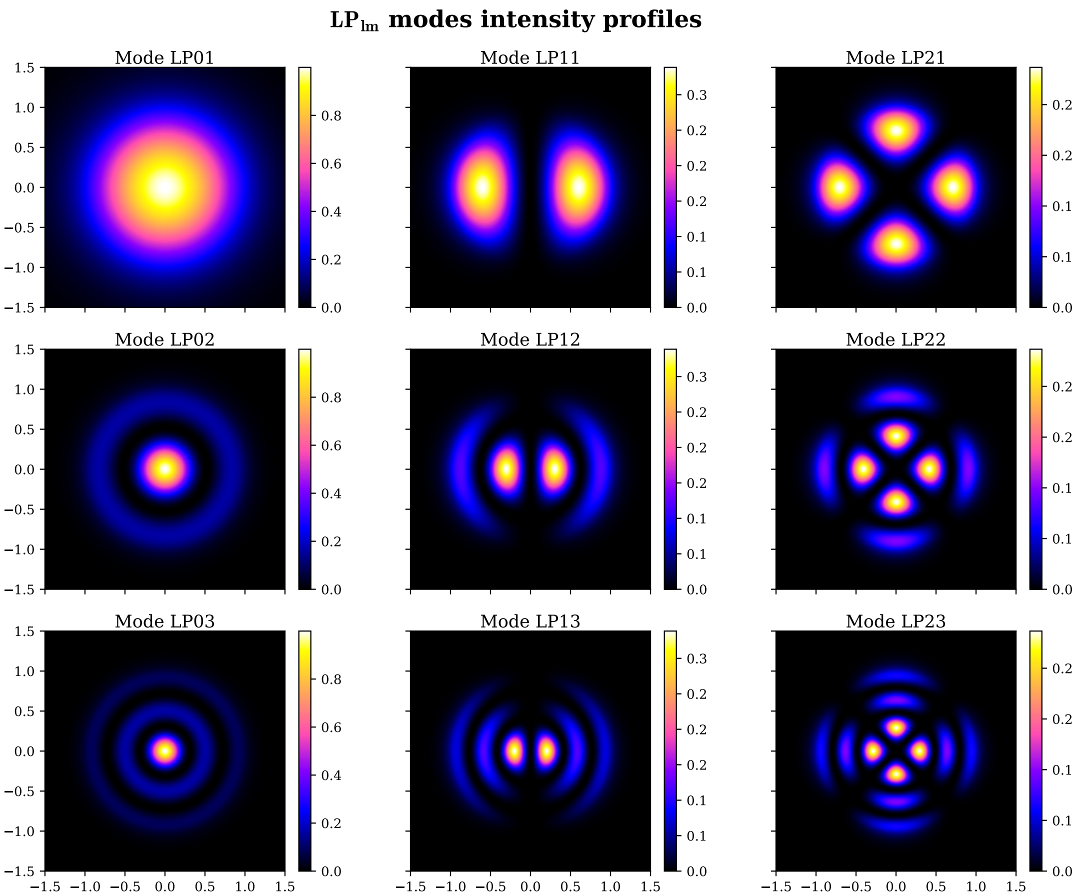


Figure 1.5: *Intensity profiles of the LP modes with $l = 0, 1, 2$ and $m = 1, 2, 3$. Each mode LP_{lm} is calculated with $V^{(lm)} = V^{(lm)}_{cutoff} + 2$. The x, y coordinates are expressed in units of fiber core radius.*

1.2 Hollow-Core Photonic Crystal Fibers

The previous section focussed only on light propagation in conventional step-index fibers. However, technological developments in the field of optical fibers have led to novel low-loss optical waveguides that are very promising for both telecommunication and fundamental research. Those are the so-called Photonic Crystal Fibers (PCFs) first introduced by Knight *et al.* [6]. For the purposes of this work, particular interest lies in the subclass of Hollow-Core Photonic Crystal Fibers (HCPCFs) that were introduced a few years later in 1999 by Cregan *et al.* [7].

In conventional fibers, light is confined within a solid core surrounded by a bulk cladding, requiring $n_{\text{core}} > n_{\text{clad}}$. By contrast, the core of an HCPCF is physically hollow (typically filled with air) and surrounded by a periodical microstructured cladding, resulting in $n_{\text{core}} < n_{\text{clad}}$. Figure 1.6 presents Scanning Electron Microscope (SEM) images illustrating various cladding microstructures utilized in HCPCFs.

This section proposes a review of the physical principles governing HCPCFs, based on the work of Debord *et al.* [8].

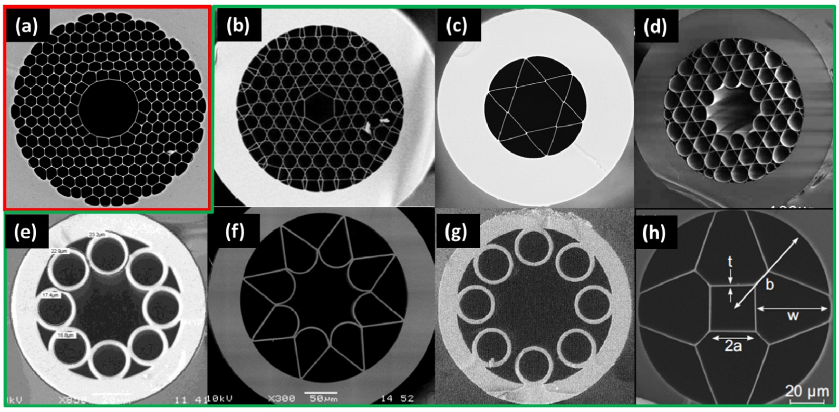


Figure 1.6: *SEM images of various hollow core fibers: (a) photonic band gap fiber; (b-h) inhibited coupling fibers. Credits [9].*

1.2.1 Introduction to HCPCF Guiding Principles

The guiding mechanism of conventional fibers can be explained employing the principle of total internal reflection (TIR), which strictly requires the condition $n_{\text{core}} > n_{\text{clad}}$. This condition is expressed in the wave optic treatment by Equation (1.20) as a requirement for a mode to be guided (oscillatory in the core and evanescent in the cladding). As previously mentioned, in an HCPCF $n_{\text{core}} < n_{\text{clad}}$, therefore the guiding mechanism must have a different physical nature.

The theoretical method used to explain the physics of these fibers is heavily drawn from condensed matter physics. In this approach, the periodic dielectric microstructure of the cladding (referred to as *photonic crystals*) is treated in analogy to a periodic potential in a crystal, while the core is treated as a defect of the periodicity. The propagation of guided light is mathematically solved by casting the electromagnetic wave equation as an eigenvalue problem in the frequency-wavevector space, in analogy to the solution in the k-space for electronic states in a crystal. Consequently, concepts and notations typical of quantum mechanics and condensed matter physics arise from the solutions, such as Bloch states, density of states and band gaps.

The crucial physical quantity for understanding the different guiding mechanisms is the modal spectrum of both the core and the cladding, represented by the Density Of Photonic States (DOPS) in the frequency-wavevector space (ω, β). Here, β is the propagation constant of the mode, namely the z component of the wave vector $\mathbf{k}$ of the mode, equivalent to the β defined in Section 1.1. In the literature, the DOPS is conventionally expressed in terms of the effective refractive index, $n_{\text{eff}} = \beta/k$.

Analogous to electronic band structures in solids, the DOPS (ω, n_{eff}) delineates regions where there are photonic states available corresponding to effective indices for which light can propagate, and void regions where there are no states available for light propagation. The latter regions are called Photonic Band Gaps (PBGs).

By comparing the core and cladding DOPS, two distinct guiding mechanisms emerge: The first is choosing a core index and a geometry so that at least a part of its supported mode (ω, n_{eff}) lies in the cladding PBG. In this way the core modes cannot leak out because there are no available modes in the cladding to excite at that specific effective index. The second relies on cladding modes and core modes that can coexist at the same n_{eff} but have a small spatial overlap so that the core modes cannot leak in the cladding because their coupling with the cladding mode continuum is suppressed.

These mechanisms divide the HCPCFs into two different classes: Photonic Band Gap (PBG) fibers and Inhibited Coupling (IC) fibers.

1.2.2 PBG and IC Guiding Mechanisms Comparison

Figure 1.7 summarizes the core and cladding modal content for the three different considered fiber architectures: TIR, PBG and IC. First, in TIR fibers the cladding is made of a uniform material with the refractive index of the glass n_{g}, therefore it can guide modes for $n_{\text{eff}} < n_{\text{g}}$, represented in Figure 1.7a as a continuum of cladding modes. Above this continuum no cladding modes exist effectively creating a gap. Since the core is made of a uniform material with the refractive index of doped glass $n_{\text{dg}} > n_{\text{g}}$, modes in the core exist for $n_{\text{eff}} < n_{\text{dg}}$, therefore, the guidance occurs for $n_{\text{g}} < n_{\text{eff}} < n_{\text{dg}}$. From this perspective, TIR fibers are just a form of PBG fibers, achieved with a core defect that has a uniform refractive index higher than that of the cladding.

Moving to a microstructured cladding, the requirement of a higher refractive index in the

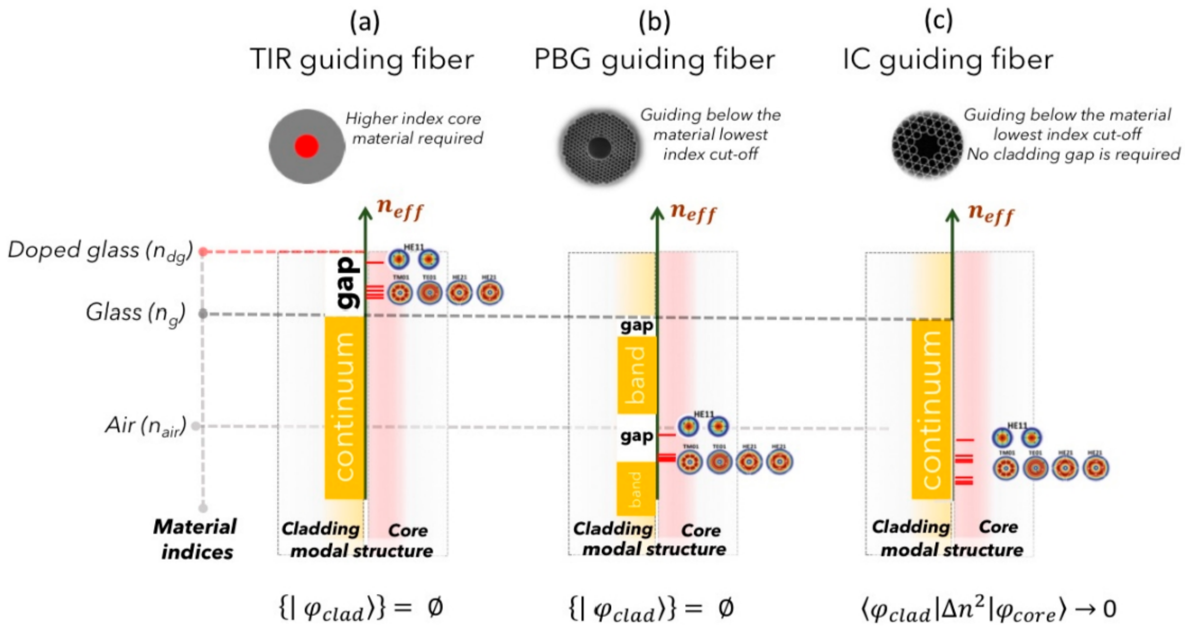


Figure 1.7: *Modal content representation of optical guiding fibers: (a) Total internal reflection (TIR); (b) Photonic Ban Gap (PBG); and (c) Inhibited Coupling (IC). The vertical axis represent the effective index n_{eff}, with the indices involved represented as horizontal dashed lines. Each graph represents the photonic bands of the cladding mode as a function of n_{eff}, with continuum (orange-filled rectangles) or gaps region (white-filled rectangles), highlighting where the core guidance happens. For both TIR and PBG fibers, guided core modes appear in correspondence of gaps. For IC fibers, no band gaps are present, and guidance occurs alongside the continuum of cladding modes. Credits [8].*

core is lifted. In this way the cladding photonic band structure acquires a more complex pattern made of an alternation of continuum regions and gap regions (Figure 1.7b). By a correct engineering of the microstructures, the gaps can be in the region $n_{\text{eff}} < n_{\text{air}}$, allowing optical guidance in the air filled core.

Finally, IC fibers exhibit a fundamentally different guiding mechanism, as they are the only fiber type not requiring a cladding PBG in correspondence of the core's guiding region (Figure 1.7c). In these fibers the cladding continuum and the core continuum of modes can coexist without leaking of the core modes to the cladding. The guiding mechanism is explained through the concept brought from quantum mechanics of Bound States in the Continuum [10, 11]. According to this model, the coupling between the field of the core mode, $|\phi_{\text{core}}\rangle$, and the cladding mode, $|\phi_{\text{clad}}\rangle$, is strongly reduced. Mathematically, this translates to the condition $\langle\phi_{\text{clad}}|\Delta n|\phi_{\text{core}}\rangle \longrightarrow 0$, with Δn being the transverse index profile function [8]. This decoupling can be achieved by either having little spatial intersection between the fields of the $|\phi_{\text{clad}}\rangle$ and $|\phi_{\text{core}}\rangle$ photonic states, or by having a strong mismatch in their respective transverse spatial phases. This low overlap also explains why IC-HCPCF, among the fibers, are those that achieve the lowest loss. Notably, since the guidance occurs at a continuum of cladding modes, the IC fibers do not have

any cutoff frequency, therefore, their nature is intrinsically multimodal.
To further reduce the overlap between core and cladding modes, Wang *et al.* [12] introduced an optimized core-contour shape known as a hypocycloid. Also referred to as a negative-curvature core contour, it consists of a series of alternating negative-curvature cusps defined by an inner radius R_{in} and an outer radius R_{out}. Figure 1.8 shows the difference between a standard core kagome IC-HCPCF and a hypocycloid variant. In particular Figure 1.8 right shows a hypocycloid kagome 7-cell (K7) fiber, very similar to the one employed in this work (see Section 2.2.3).

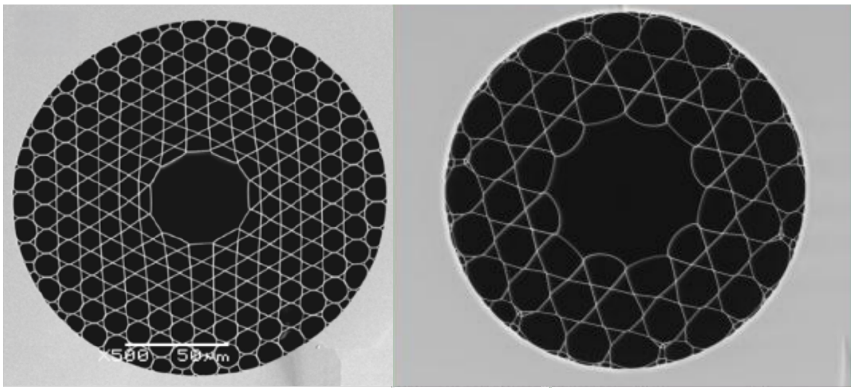


Figure 1.8: *Comparison between the SEM images of a kagome IC-HCPCF with standard core-contour (left) and a kagome IC-HCPCF with hypocycloid core-contour (right). Credits [13, 8]*

1.2.3 HCPCF core guided modes

Despite the substantial physical difference in the guiding mechanisms of PBG and IC with respect to the TIR, the core modes do not deviate significantly from the modes in conventional step-index optical fibers [8]. More precisely, the propagation constants and the mode fields do not differ much from the results obtained by Marcatili and Schmeltzer [15] for electromagnetic modes in hollow dielectric capillaries, which, in turn, share the same qualitative shapes as the complete solution TE, TM, EH and HE modes of step-index optical fibers. For instance, Digonnet *et al.* [16] show how the $LP_{l,m}$ modes remain a good approximation for the core mode of PBG fibers and LP mode based approximation can effectively yield estimates for the number of supported modes, their cutoffs, and their effective indices. However, for more accurate, quantitative predictions and a rigorous description, the complete solutions must be computed [17].
Figure 1.9 left presents the calculated LP-like modes for a kagome IC-HCPCF from [8]. The modal composition of a fiber can be experimentally characterized using Spatially and Spectrally resolved imaging (S^2) technique, which was first introduced by Nicholson *et al.* [18]. This technique relies on the acquisition and analysis of the optical interference spectrum generated by the superposition of multiple guided modes. By applying a Fourier

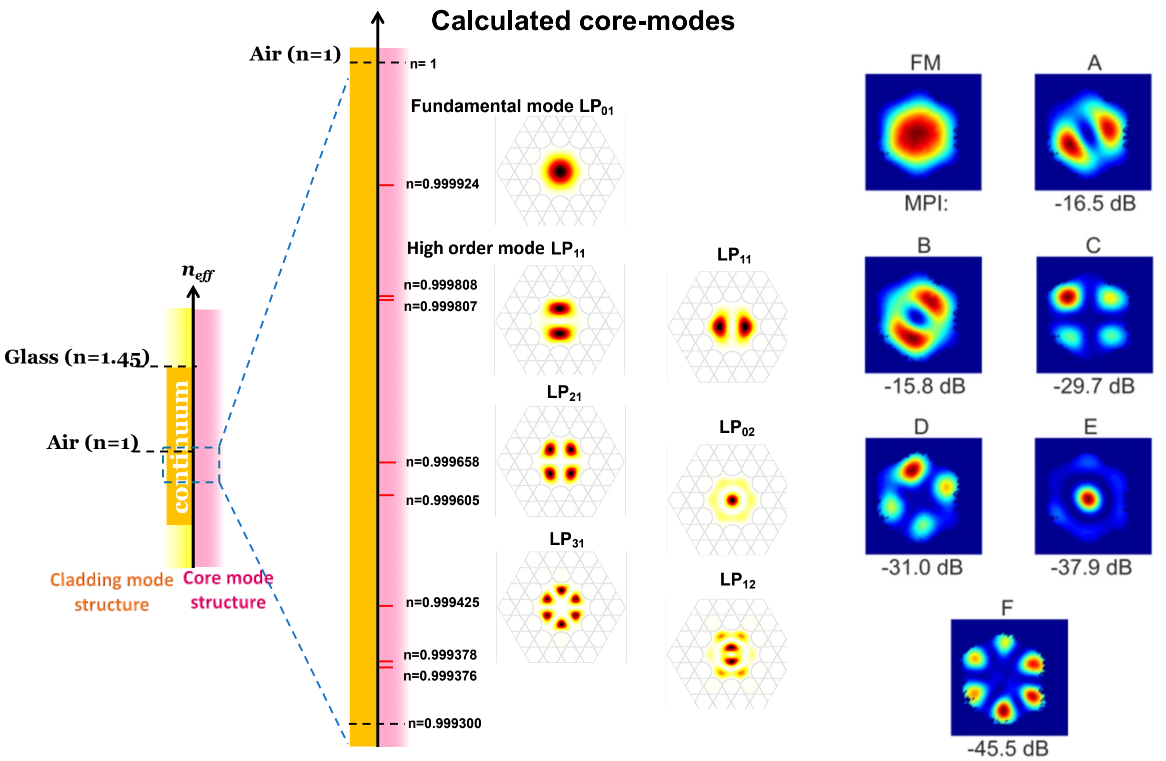


Figure 1.9: *LP-like modes in HCPCFs. Left: effective refractive index spectrum and intensity profiles of the calculated LP modes for the core of a Kagome fiber. Credits [8]. Right: modal profiles experimentally measured via the S^2 technique on a 7-cell (K7) Kagome fiber, with the corresponding Multipath Interference (MPI). The image reports the fundamental mode (FM); two different orientation of LP_{11}-like modes (A, B); two different orientation of LP_{12}-like modes (C,D); an LP_{02}-like mode (E); and an LP_{31}-like mode (F). Credits [14].*

transform to the measured spectrum, the differential group delays between the propagating modes can be extracted, enabling the simultaneous characterization of both their spatial profiles and their spectral properties. Figure 1.9 right displays the experimentally measured mode profiles for a K7 IC-HCPCF, obtained using the S^2 technique by Bradley *et al.* [14]. The measured mode profiles are clearly similar to both the LP modes of the conventional step-index fibers, and to the theoretically calculated ones in Figure 1.9 left. Beneath each mode profile, Figure 1.9 right also reports the value of the Multipath Interference (MPI), defined as the ratio, expressed in dB, of the power guided by the high-order mode under consideration (P_{HOM}) to the power guided by the fundamental mode (P_{FM}): $\mathrm{MPI} = 10\log(P_{\mathrm{HOM}}/P_{\mathrm{FM}})$. It is worth noting that the guided light is mainly in the fundamental core mode as the lowest MPI, represented by the LP_{11}-like modes, which, when combined, yield ~ -13 dB, indicating that the high-order modes carry roughly $1/20$ of the total transmitted power. According to [8], an IC-HCPCF can effectively operate as a single-mode fiber if engineered to exhibit high losses for higher-order modes while maintaining low loss for the fundamental mode.

1.3 Optical Dipole Trap

Optical traps are experimental tools that allow the confinement of atom or ions within a localized region of space, under the condition that these particles have a kinetic energy lower than the trap potential depth (typically below 1mK). For the work presented in this thesis, among all possible trapping techniques, the focus will be strictly placed on Optical Dipole Traps (ODTs) for neutral atoms.

Optical dipole traps rely on the electric dipole interaction of the atoms with far-detuned light. Following the work of Grimm *et al.* [19], this section reviews the theory behind the working mechanism of this technique, particularly in the case of red-detuned ODTs.

1.3.1 Oscillator model

The optical dipole force arises from the interaction of the induced electric dipole moment with the intensity gradient of the light field. It is a conservative force, thus in the following its potential energy will be derived modelling the atom as an oscillator subjected to a classical oscillating electric field.

1.3.1.1 Interaction between induced dipole moment and driving field

The laser field is modeled as the oscillating electric field:

$$\mathbf{E}(\mathbf{r},t) = \hat{\mathbf{e}}\mathcal{E}(\mathbf{r})e^{-i\omega t} + c.c. \ , \tag{1.52}$$

where $\hat{\mathbf{e}}$ is the unitary polarization vector and $\mathcal{E}(r)$ is the complex amplitude of the field. Denoting the atomic polarizability as α, the induced dipole momentum of the atom is given by:

$$\mathbf{p}(\mathbf{r},t) = \alpha\mathbf{E}(\mathbf{r},t) = \hat{\mathbf{e}}\alpha\mathcal{E}(\mathbf{r})e^{-i\omega t} + c.c. \tag{1.53}$$

In the semiclassical framework, the interaction of a neutral atom with a laser light field can be described using the dipole approximation. The interaction Hamiltonian takes the form $H_{\mathrm{dip}} = -\mathbf{p}\cdot\mathbf{E}$. For an induced dipole, where the dipole moment depends linearly on the external field, the potential energy is obtained by integrating the work required to induce the dipole as the electric field is increased from zero to its final value:

$$U_{\mathrm{dip}}(t) = \int_0^{\mathbf{E}} -\alpha\mathbf{E}'\cdot d\mathbf{E}' = -\frac{1}{2}\alpha E^2 = -\frac{1}{2}\mathbf{p}\cdot\mathbf{E}\,. \tag{1.54}$$

Averaging over a field period [19]:

$$U_{\mathrm{dip}} = -\frac{1}{2}\langle \mathbf{p}\cdot\mathbf{E}\rangle = -\frac{1}{2\epsilon_0 c}\mathrm{Re}[\alpha]I, \tag{1.55}$$

where $I = 2\epsilon_0 c|\mathcal{E}|^2$ is the time-averaged field intensity.
This result shows that an atom in an oscillating laser field experiences a force proportional to the field intensity I, and to the real part of the polarizability, which represent the in-phase component of the dipole oscillation, responsible for the dispersive properties of the interaction.

It follows that the dipole force is a conservative force proportional to the electric field intensity gradient:

$$\mathbf{F}_{\mathrm{dip}}(\mathbf{r}) = -\nabla U_{\mathrm{dip}} = \frac{1}{2\epsilon_0 c}\mathrm{Re}[\alpha]\nabla I(\mathbf{r}). \tag{1.56}$$

The dipole oscillation absorbs a power

$$P_{\mathrm{abs}} = \langle \dot{\mathbf{p}}\mathbf{E}\rangle = 2\omega\mathrm{Im}[\alpha\mathcal{E}^2] = \frac{\omega}{\epsilon_0 c}\mathrm{Im}[\alpha]I, \tag{1.57}$$

which is proportional to the out-of-phase component of the oscillating dipole. The absorbed power can be seen as series of absorption and spontaneous emission of photons with energy $\hbar\omega$. Therefore, the total scattering rate is given by

$$\Gamma_{\mathrm{sc}}(\mathbf{r}) = \frac{P_{\mathrm{abs}}}{\hbar\omega} = \frac{1}{\hbar\epsilon_0 c}\mathrm{Im}[\alpha]I(\mathbf{r}). \tag{1.58}$$

1.3.1.2 Atomic polarizability

The polarizability α can be calculated by considering the classical Lorentz oscillator model for the atom, according to which the polarizability can be calculated by solving the differential equation

$$\ddot{x} + \Gamma(\omega)\dot{x} + \omega_0^2 x = -\frac{eE(t)}{m_e}, \tag{1.59}$$

Using $E(t) = \mathcal{E}_0\exp(-i\omega t)$, thus looking for a solution in the form $x(t) = x_0\exp(-i\omega t)$, equation (1.59) gives

$$x_0 = -\frac{e\mathcal{E}_0}{m_e}\frac{1}{\omega_0^2 - \omega^2 - i\omega\Gamma(\omega)}. \tag{1.60}$$

Using $p = -ex(t) = -ex_0\exp(-i\omega t)$ and (1.60), one can easily obtain the polarizability

$$\alpha(\omega) = \frac{e^2}{m_e}\frac{1}{\omega_0^2 - \omega^2 - i\omega\Gamma(\omega)}, \tag{1.61}$$

where $\Gamma(\omega)$ is the classical damping rate due to the radiative energy loss:

$$\Gamma^{\text{class.}}(\omega) = \frac{e^2\omega^2}{6\pi\epsilon_0 m_e c^3}. \tag{1.62}$$

Defining the on-resonance spontaneous emission rate $\Gamma \equiv \Gamma^{\text{class.}}(\omega_0) = (\omega_0/\omega)^2\,\Gamma^{\text{class.}}(\omega)$, the polarizability becomes

$$\alpha(\omega) = 6\pi\epsilon_0 c^3 \frac{\Gamma/\omega_0^2}{\omega_0^2 - \omega^2 - i(\omega^3/\omega_0^2)\Gamma}. \tag{1.63}$$

When considering a full quantum picture of a two level system subjected to classical radiation, Γ should be substituted with the spontaneous emission rate

$$\Gamma = \frac{\omega_0^3}{3\pi\epsilon_0\hbar c^3}|\langle e|\,\hat{d}\,|g\rangle|^2, \tag{1.64}$$

where $\hat{d}$ is the dipole moment operator.
Nevertheless, optical dipole traps work in far-detuned low saturation regime. Consequently, the transitions to the excited state are strongly suppressed, and the atom can be considered as occupying the ground state only. In this limit, the atomic equation of motion reduces to a classical damped harmonic oscillator model described by equation 1.59 (for a rigorous derivation see Chapter V of Steck [20]). Therefore, the classical polarizability given by equation 1.63 represent a well suited approximation for ODTs.

1.3.1.3 Dipole potential and scattering rate

Substituting α from (1.63) in (1.55) and (1.58) one gets the expressions of the dipole potential and the scattering rate of a dipole trap as explicit functions of ω, ω_0 and Γ:

$$U_{\text{dip}}(\mathbf{r}) = -\frac{3\pi c^2}{2\omega_0^3}\left(\frac{\Gamma}{\omega_0-\omega} + \frac{\Gamma}{\omega_0+\omega}\right) I(\mathbf{r}), \tag{1.65}$$

$$\Gamma_{\text{sc}}(\mathbf{r}) = \frac{3\pi c^2}{2\hbar\omega_0^3}\left(\frac{\omega}{\omega_0}\right)^3\left(\frac{\Gamma}{\omega_0-\omega} + \frac{\Gamma}{\omega_0+\omega}\right)^2 I(\mathbf{r}). \tag{1.66}$$

These equations show two resonance frequencies for $\omega = \pm\omega_0$. Introducing the detuning $\Delta = \omega - \omega_0$, in typical experimental conditions, it holds that $|\Delta| \ll \omega_0$. This allows the introduction of the well-known rotating-wave approximation, according to which the anti-resonant term can be neglected, further simplifying the equations to

$$U_{\text{dip}}(\mathbf{r}) = \frac{3\pi c^2}{2\omega_0^3}\frac{\Gamma}{\Delta} I(\mathbf{r}) \tag{1.67}$$

$$\Gamma_{\mathrm{sc}}(\mathbf{r}) = \frac{3\pi c^2}{2\hbar\omega_0^3}\left(\frac{\Gamma}{\Delta}\right)^2 I(\mathbf{r}) = \frac{\Gamma}{\hbar\Delta}U_{\mathrm{dip}} \tag{1.68}$$

Equations (1.67) and (1.68) are the fundamental equations governing the behavior of optical dipole traps and yield important physical considerations for their implementation. The sign of U_{dip} is determined by the sign of the detuning Δ. For red detuning ($\Delta < 0$) the potential is negative, confining atoms at intensity maxima, whereas for blue detuning ($\Delta > 0$), the potential is positive pushing atoms to intensity minima. Because the potential depth scales as I/Δ while the scattering rate scales as I/Δ^2, optical traps are typically realized with high intensity field and large detunings, in this way, the trap depth is maximized while maintaining a low scattering rate.

1.3.2 Multi-level atoms

Moving from the two-level atom to a real atom, the electronic transitions become more complex, and the calculation of the dipole potential must take into consideration all the possible allowed transitions. This lead to a dipole potential that depends on the particular electronic substate.

1.3.2.1 Ground-state light shifts

Labelling as ε_i the energy levels of the multi-level atom, the effect of the laser light on the levels can be described by means of the non-degenerate time-independent second order perturbation theory. Considering the electric dipole interaction Hamiltonian $\mathcal{H}_I = -\hat{\mathbf{d}}\cdot\mathbf{E}$, the perturbative correction on the i-th energy level is

$$\Delta E_i = \sum_{j\neq i}\frac{|\langle j|\mathcal{H}_I|i\rangle|^2}{\varepsilon_i - \varepsilon_j}. \tag{1.69}$$

The calculation can be developed by employing the dressed atom approach, in which the quantum states describe the combined atom-field system. Defining the unperturbed atomic ground state energy as zero, in this framework an unperturbed dressed state has an energy $\varepsilon_i = n\hbar\omega$ that depends solely on the number of photons n in the field. The absorption of a photon transitions the system to another dressed state characterized by the energy $\varepsilon_j = \hbar\omega_j + (n-1)\hbar\omega$, where $\hbar\omega_j$ represent the bare atomic transition energy. In such a way, the denominator of equation (1.69) reduces to

$$\varepsilon_i - \varepsilon_j = -\hbar\omega_j + \hbar\omega = -\hbar\Delta_{ij}, \tag{1.70}$$

where Δ_{ij} is the detuning of the $i \to j$ transition with respect to the laser field.
Considering a two-level atom first, (1.69) has only one term, using equations (1.64) and

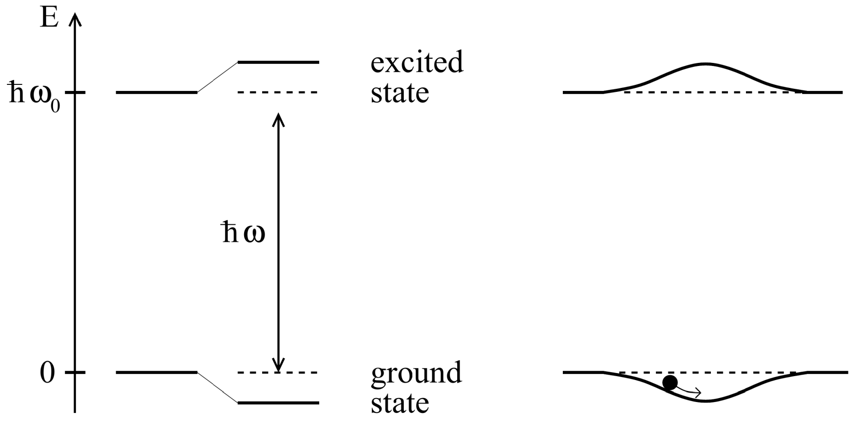


Figure 1.10: *Left: light-shifted energy levels for a two-level atom for a red detuning ($\Delta < 0$). Right: a spatially inhomogeneous field produces a potential well that traps the atoms. Credits [19].*

(1.70) and $I = 2\epsilon_0 c|\mathcal{E}|^2$, one gets:

$$\Delta E = \pm\frac{|\langle e|\mu|g\rangle|^2}{\Delta}|E|^2 = \pm\frac{3\pi c^2}{2\omega_0^3}\frac{\Gamma}{\Delta}I, \tag{1.71}$$

for the ground and excited state (upper and lower sign, respectively). Equation (1.71) shows that the energy shift of the ground state (the solution with the sign +) exactly correspond to the dipole potential (1.67). This allows the reinterpretation of the semi-classical result obtain for the external degrees of freedom (position and momentum) in terms of the dynamic of the internal degrees of freedom. In presence of the laser field, the energy levels are shifted by a quantity proportional to the field intensity with a sign that depends on the sign of the detuning. Assuming a low saturation regime, corresponding to $s \ll 1$, where the saturation parameter s is given by

$$s = \frac{\Omega^2/2}{\Delta^2 + (\Gamma^2/4)}, \tag{1.72}$$

and Ω is the Rabi frequency, defined as

$$\Omega = \frac{|\mathcal{E}_0|}{\hbar}\langle e|\,\hat{d}\,|g\rangle\,, \tag{1.73}$$

the atomic population can be considered almost entirely confined to the ground state. Consequently, the light-shifted ground state becomes the effective potential that governs the motion of the atoms, driving them toward regions of minima of ΔE (Figure 1.10).

For a multi-level atom Equation (1.69) contains many terms similar to equation (1.71), accounting for all the allowed transitions from any ground state sublevel $|g_i\rangle$ to any excited state $|e_j\rangle$. Each of these terms depends on the dipole matrix element $d_{ij} = \langle g_i|\,\hat{d}\,|e_j\rangle$.

Applying the Wigner–Eckart theorem [21], this matrix element can be factored as:

$$d_{ij} = c_{ij}||d|| \tag{1.74}$$

where $||d||$ is the reduced matrix element given by (1.64) and c_{ij} are real Clebsch-Gordan coefficients describing the selection rules and coupling strengths of the transitions. They depend on the light polarization and the angular momentum quantum numbers of the involved hyperfine states.
Therefore, the energy shift (1.69) for the ground state $|g_i\rangle$ is

$$\Delta E_i = \frac{3\pi c^2 \Gamma}{2\omega_0^3} I \times \sum_j \frac{c_{ij}^2}{\Delta_{ij}} \tag{1.75}$$

and according to the interpretation given from the two-level atoms, the dipole potential at low saturation is

$$U_{\text{dip}} = \Delta E_i. \tag{1.76}$$

Thus, the dipole potential in a multi-level atom arises from the contribution of all the allowed transitions, each weighted by the ratio between its line strengths c_{ij}^2 and its detuning Δ_{ij}.

1.3.2.2 Fine and Hyperfine structures contribution for Alkali atoms

Many laser cooling experiments utilize alkali metal atoms. This preference arises because their principal transition frequencies fall within the visible and infrared (VIS-IR) range making them easily accessible with standard laser technologies. Moreover, their relatively simple atomic structures is particularly convenient, allowing easy identification of closed transitions necessary for continuous cooling cycles, while minimizing the number of repumper lasers needed to address the population leaking to dark states.
When considering the fine structure arising from the spin-orbit interaction, the electronic ground state of alkali atoms is ${}^2S_{1/2}$. From this state, the electric dipole selection rules ($\Delta L = \pm 1$, $\Delta S = 0$) allow for two primary transitions, known as $D-$lines:

- D_1: ${}^2S_{1/2} \to {}^2P_{1/2}$;
- D_2: ${}^2S_{1/2} \to {}^2P_{3/2}$.

Beyond the fine structure, the energy levels undergo further splitting due to the coupling between the nuclear spin and the total electronic angular momentum. The resulting configuration is called hyperfine structure. Denoting the fine structure splitting as $\hbar\Delta_{FS}$, the ground state hyperfine splitting as $\hbar\Delta_{HFS}$, and the excited state hyperfine splitting

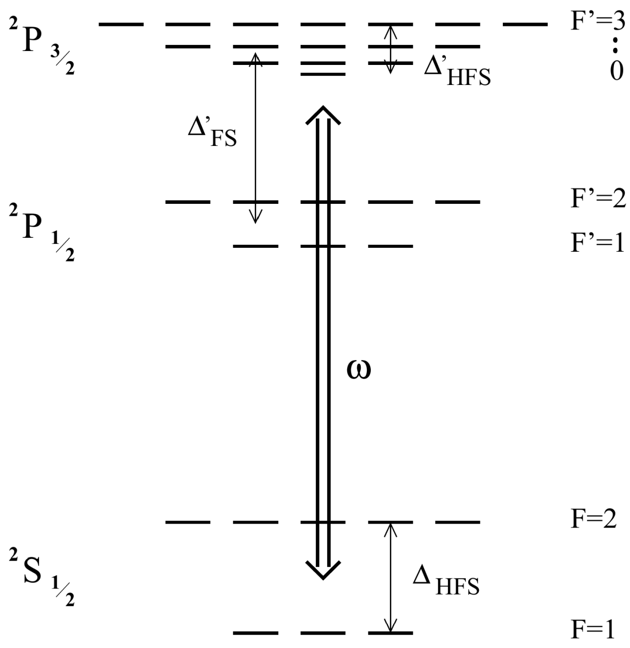


Figure 1.11: *Energy level scheme of an alkali atom with $I = 3/2$ such as ^{87}Rb. Credits [19].*

as $\hbar\Delta'_{HFS}$, these characteristic energy scales obey $\Delta_{FS} \gg \Delta_{HFS} \gg \Delta'_{HFS}$. The complete electronic energy level scheme for an alkali atom with nuclear spin $I = 3/2$ is shown in figure 1.11.

Assuming that all optical detunings stay large compared with the excited-state hyperfine splitting Δ'_{HFS}, equation (1.75) can be calculated for an alkali atom in the ground state of total angular momentum F and magnetic quantum number m_F, obtaining [19]:

$$U_{\rm dip}(\mathbf{r}) = \frac{\pi c^2 \Gamma}{2\omega_0^3}\left(\frac{2+\mathcal{P}g_F m_F}{\Delta_{2,F}} + \frac{1-\mathcal{P}g_F m_F}{\Delta_{1,F}}\right) I(\mathbf{r}), \tag{1.77}$$

where the two term in the parentheses represent respectively the contribution of the D_2 and D_1 line. Here, g_F is the Landé factor, $\mathcal{P}$ characterizes the laser polarization ($\mathcal{P} = 0, \pm 1$ for linear polarization and circular $\sigma^{\pm}$ respectively), and $\Delta_{2,F}$ and $\Delta_{1,F}$ denote the detunings of the laser field relative to the transitions from the $^2S_{1/2}, F$ ground state to the center of respectively the $^2P_{3/2}$ and $^2P_{1/2}$ hyperfine excited state.

If the detunings are large with respect to the fine-structure splitting ($|\Delta_{1,F}|, |\Delta_{2_F}| \gg \Delta'_{FS}$), introducing the detuning Δ with respect to the center of the $D-$line doublet, expanding equation (1.77) at first order in the small parameter Δ'_{FS}/Δ, one gets [19]:

$$U_{\rm dip}(\mathbf{r}) = \frac{3\pi c^2}{2\omega_0^3}\frac{\Gamma}{\Delta}\left(1 + \frac{1}{3}\mathcal{P}g_F m_F \frac{\Delta'_{\rm FS}}{\Delta}\right) I(\mathbf{r}). \tag{1.78}$$

The zeroth order term corresponds exactly to the dipole potential of a simple two-level atom (see Eq. (1.67)). The first-order term introduces a small, spin-dependent correction depending on the laser polarization $\mathcal{P}$ and the magnetic quantum number m_F.

Physically, equation (1.78) describes that, when the fine structure is completely unresolved due to the large laser detuning, the atom effectively acts as a generic two-level system undergoing an $s \to p$ transition. Consequently, the resulting dipole potential is dominated by the scalar term, matching the two-level model previously discussed.

Since in the unresolved fine structure case, the polarization-dependent term is heavily suppressed by the small parameter Δ'_{FS}/Δ (and vanishes entirely for linearly polarized light), the atom effectively acquires only the scalar light shift of the unperturbed s state. Consequently, all hyperfine and magnetic sublevels experience a uniform trapping potential that confines the atom while preserving its internal spin degeneracy.

In a more general case, when the fine structure is resolved, but the hyperfine structure is unresolved ($|\Delta_{1,F}|, |\Delta_{2,F}| \gg \Delta_{HFS} \gg \Delta'_{HFS}$), only linearly polarized light guarantees that the dipole potential becomes completely independent of m_F and F [19]. Therefore, linearly polarized light is usually preferred for dipole trap implementation.

1.3.3 Heating mechanisms

Typical trap depths for optical dipole traps are usually below 1 mK. Therefore, capturing a substantial number of atoms requires a preliminary cooling stage. For instance, the experiment discussed in this thesis employs a magneto-optical trap (MOT) followed by an optical molasses phase.

Given these shallow confining potentials, even minor heating rates can largely limit the trap lifetime. It is therefore important to outline the mechanisms responsible for the parasitic heating induced by the trapping light.

1.3.3.1 Heating source

In general, when atoms interact with light and absorb photons, when the subsequent spontaneous emission event occur, the randomness of the emitted photons' direction causes a diffusion in the phase space that corresponds to a heating processes. This discrete momentum transfer is the fundamental phenomenon that establishes lower limits for laser cooling mechanisms [22, 23].

At large detuning the scattering process follows a Poisson statistic. In this regime, the absorption of a photon from the dipole beam introduces a heating energy equivalent to the recoil energy $E_{\text{rec}} = k_B T_{\text{rec}}/2$ per scattering event, where T_{rec} is the recoil temperature. Since the absorptions occur strictly along the propagation axis the momentum diffusion occurs only along this axis, making the associated heating effect anisotropic. On the other hand, the subsequent spontaneous emission also increase the thermal energy by one recoil energy E_{rec} per event, however the emission directions are random, thus the effect is spread over all the three spatial direction.

Therefore, in a single scattering cycle, the atom gains E_{rec} of thermal energy from the absorption and then E_{rec} from the spontaneous emission, for a total of $2E_{\text{rec}}$, in a time $\bar{\Gamma}_{\text{sc}}^{-1}$. It leads to the total heating power

$$P_{\text{heat}} = 2E_{\text{rec}}\bar{\Gamma}_{\text{sc}} = k_B T_{\text{rec}}\bar{\Gamma}_{\text{sc}}, \tag{1.79}$$

the heating depends on the mean scattering rate $\bar{\Gamma}_{\text{sc}}$ and on the recoil E_{rec} which depends on the laser frequency.

Other fundamental source of heating, such as the redistribution of photons between different travelling wave, should be quenched by the far-detuned working frequency [19].

In addition to the fundamental heating mechanism, it must be mentioned also the presence of technical heating caused by fluctuations in the laser power and in its pointing direction. In particular power fluctuations at twice the characteristic trap frequencies are relevant, as they can parametrically drive the oscillatory atomic motion [24].

1.3.3.2 Heating rate

Assuming that the trapped atoms are at thermal equilibrium, the mean energy can be expressed through the energy equipartition theorem. Assuming the trap as a three-dimensional harmonic potential well, there is a contribution of three quadratic degrees of freedom from the kinetic energy, and other three from the harmonic potential. According to the energy partition theorem, the mean energy is

$$\bar{E} = 3k_B T. \tag{1.80}$$

Using (1.79) in (1.80), one can obtain an equation that describes the temperature change with time:

$$\dot{T} = \frac{1}{3}T_{\text{rec}}\bar{\Gamma}_{\text{sc}}. \tag{1.81}$$

Expressing the dipole potential as

$$\bar{U}_{\text{dip}} = U_0 + \bar{E}_{\text{pot}} = U_0 + \frac{3}{2}k_B T, \tag{1.82}$$

where U_0 is the potential energy minimum.

Using equation (1.68), $\bar{U}_{\text{dip}}$ can be expressed as:

$$\bar{U}_{\text{dip}} = \frac{\hbar\Delta\Gamma_{\text{sc}}}{\Gamma}. \tag{1.83}$$

Substituting this into Equation (1.82) yields the mean scattering rate:

$$\bar{\Gamma}_{\text{sc}} = \frac{\Gamma}{\hbar\Delta}(U_0 + \frac{3}{2}k_B T). \tag{1.84}$$

It is important to notice that, for a blue-detuned trap, $U_0 = 0$ and the trap depth $\hat{U}$ is given by the maximum value of the potential surrounding the minimum, by contrast for a red-detuned trap $\hat{U} = |U_0|$. Finally, using equation (1.84) in (1.81), with a depth $\hat{U} \gg k_B T$, one gets the following temperature changing rate for a red and a blue trap:

$$\dot{T}_{\text{red}} = \frac{1}{3}T_{\text{rec}}\frac{\Gamma}{\hbar|\Delta|}\hat{U}, \tag{1.85a}$$

$$\dot{T}_{\text{blue}} = \frac{1}{2}T_{\text{rec}}\frac{\Gamma}{\hbar\Delta}k_B T. \tag{1.85b}$$

Equations (1.85) describe the increasing in temperature of the atomic cloud if subjected to the dipole trap only. It highlights another blue trap limitation: while a red trap has a linear heating rate (as long as $\hat{U} \gg k_B T$), the blue trap shows an exponential heating law.

Chapter 2

Materials And Method

This chapter describes the experimental apparatus used in this work. First, a brief overview of the implementation of the Magneto-Optical Trap (MOT) for rubidium-87 atoms (^{87}Rb) is presented (this setup was already available prior to the present work). The chapter then focuses on the work carried out during this thesis, namely the implementation of the Optical Dipole Trap setup employed to transfer and trap atoms from the MOT. A particular focus is put on the coupling between the ODT laser source and the Hollow-Core Photonic Crystal Fiber (HCPCF), that delivers the ODT light in the vacuum chamber.

2.1 The Experimental Apparatus

In this section the experimental apparatus used to produce, cool, and trap ^{87}Rb atoms is presented. The preparation of the atomic sample relies on a well-established sequence consisting of a Magneto-Optical Trap followed by an optical molasses phase. These initial stages are essential to produce a cold atomic cloud, which provides the atoms for loading the Optical Dipole Trap which is the core addition and focus of this thesis work.

The MOT provides the initial spatial confinement and cooling of the background rubidium vapor. It operates by exploiting the scattering force exerted by three pairs of red-detuned, counter-propagating laser beams, spatially tuned by a quadrupole magnetic field gradient [23]. This stage effectively compresses the cloud and cools it down through Doppler cooling mechanism. To further decrease the temperature and increase the phase-space density prior to the ODT loading, the magnetic field gradient is switched off, leaving the atoms exposed only to the cooling light in an optical molasses configuration. In this regime, sub-Doppler cooling is triggered by the σ^+-σ^- configuration of the MOT beams [23, 25].

2.1.1 Setup Description

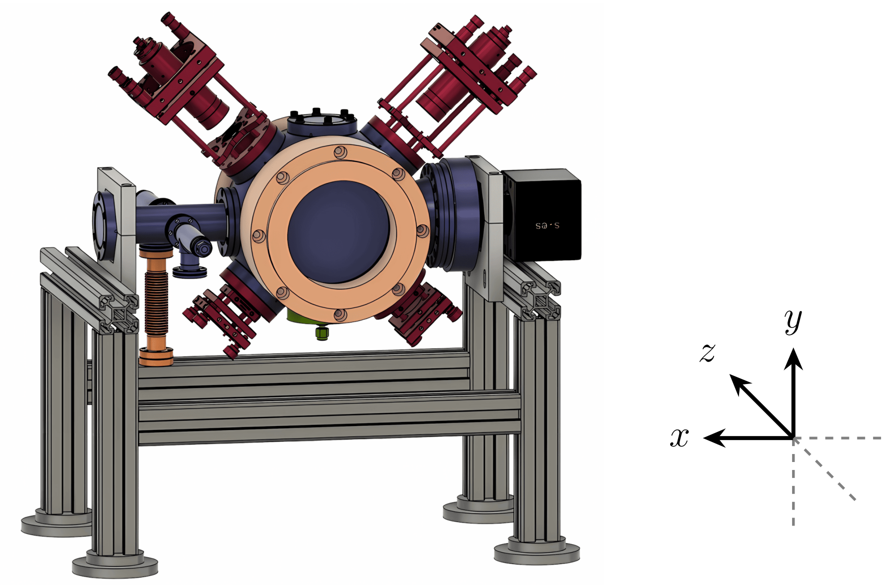


Figure 2.1: *CAD rendering of the vacuum chamber and surrounding experimental setup. The color coding highlights the main vacuum chamber (purple), MOT optics (red), MOT coils (orange), ion pump (black), rubidium source (dark orange, left), and the HCPCF port (green) located below the chamber. The compensation coils are not drawn. Credits [26, 27].*

The core component of the experimental apparatus is the vacuum chamber operating under Ultra-High Vacuum (UHV) conditions, inside which the atoms are trapped and cooled. The chamber is secured by a clamping structure to an optical table. A CAD rendering of the structure is displayed in Figure 2.1 together with the surrounding experimental elements.

The UHV is achieved through the employment of an internal pump (SAES Getter NEX-Torr® Z200), composed of an active ion-pump and a passive non-evaporable getter (NEG). The internal pump is activated once the system reaches $\sim 10^{-7}$ mbar, which is initially attained using an external dual diaphragm-turbo pump (Pfeiffer Vacuum HiCube 80 Eco DN63). With the internal pump the system reaches a pressure of $\sim 10^{-10}$ mbar, as read from the ion-pump monitor.

The chamber is equipped with eight windows. Two large windows (mounted on CF100 flanges, diameter $\sim$ 100 mm) along the z direction, and the two pairs of smaller windows (mounted on CF40 flanges, diameter $\sim$ 40 mm) and oriented at 45° with respect to the optical table surface, are coated with 780 nm anti-reflective coating, to optimize the transmission of the six MOT beams. The remaining two CF40 windows are equipped with regular glass. One is mounted on the top side of the chamber (positive y-axis side), and the other on the positive x-axis side connected through a 4-way cross connector. The top window is utilized for the injection of the second beam of the ODT, whereas the x-axis window was intended for optical imaging in the chamber. However, this viewpoint directly faces the ion-pump facet (see Figure 2.1), introducing a significant parasitic background in the images due to reflection. For this reason, in this thesis work, the cameras were mounted on supports to observe the system from the two large windows on the z-axis ensuring a highly non-reflective background during image acquisition.

The rubidium vapor inside the chamber is provided by a 1 g sample of pure ^{87}Rb, connected through a valve on one of the four flanges of the 4-way cross connector, while the fourth flange is connected to the external vacuum pump. The rubidium sample is heated by a heating band powered by a variac. It is worth noting that opening the rubidium valve increases the pressure in the chamber from $\sim 10^{-10}$ mbar to $\sim 10^{-8}$ mbar.

The MOT also needs a magnetic field gradient to induce a spatially dependent Zeeman shift of the atomic energy levels. This shift induces a spatially dependent radiation pressure that pushes the atoms towards the magnetic field minimum. The magnetic field is provided by two circular coils (110 mm internal diameter) mounted on the $+z$ and $-z$ sides of the chamber. The coils are driven in an anti-Helmholtz configuration so that they create a quadrupole magnetic field. The quadrupole field is zero exactly midway between the coils (at the center of the chamber), and exhibits a constant gradient in its proximity. The measured axial gradient is $1.33\ \mathrm{G \cdot cm^{-1}/A}$. Due to Maxwell's divergence law $\nabla \cdot \mathbf{B} = 0$, the radial magnetic field gradient is half the axial gradient with opposite sign, conferring the typical ellipsoidal shape to the MOT cloud.

Furthermore, the experimental apparatus is provided by three pairs of compensation coils, commonly referred to as *shim coils*. These are square-shaped coils with an internal side of $\sim$ 650 mm, arranged in a mutually orthogonal configuration along the x, y, and z axes. Driven in a Helmholtz configuration, they produce a uniform magnetic field at the center of the chamber in any desired orientation. The compensation coils are primarily intended to cancel external magnetic fields, such as the Earth's one. However, they were also employed to shift the MOT position to increase its spatial overlap with the ODT beam.

The magnetic field generated by the compensation coils depends linearly on the applied current, with the following calibration factors [26]:

$$\begin{cases} B_z \approx 1.4\,\mathrm{mG/mA}, \\ B_x \approx 1.3\,\mathrm{mG/mA}, \\ B_y \approx 1.5\,\mathrm{mG/mA}. \end{cases} \tag{2.1}$$

On the bottom of the vacuum chamber (negative y-axis) a CF40 blind flange fitted with a 1/8" *Swagelok* feedthrough is mounted. A Teflon ferrule with a 0.5 mm diameter hole is inserted inside the Swagelok to host the HCPCF while ensuring proper sealing. Inside the chamber, the fiber is supported by a custom-designed mount to maintain strict linear

alignment, with its tip positioned approximately 10 mm below the chamber center.
Finally, the setup features three collimators (Schäfter + Kirchhoff, 60FC-Q780-4-M100S-37) that inject the MOT beams inside the chamber. These MOT beams pass through the windows and are retro-reflected by mirrors. In front of each mirror a $\lambda/4$ wave plate is placed so that the beams pass through it twice, ensuring that the two counter-propagating beams of each branch possess opposite circular polarization (σ^+ and σ^-). This polarization configuration is necessary for the realization of the MOT, and for the subsequent sub-Doppler cooling mechanism in the molasses phase. In this arrangement, the retro-reflected beams have an intensity of $\sim 88\%$ of the incoming ones. Although not perfectly balanced, the forces applied on the atomic cloud from opposite directions are similar enough to guarantee adequate trapping [26].
A key characteristic of this setup is that, except for the master laser locking system, the ODT injection line developed in this work, and the MOT beams, the entire apparatus is fully fiber-coupled. The system employs polarization-maintaining and absorption-reducing (PANDA) fibers [28] to ensure polarization stability. This approach offers several advantages: it minimizes misalignment caused by thermal drifts, makes the optical setups significantly more compact, and lowers safety risks associated with laser light propagating in free space. These features are particularly relevant in view of future integration of similar systems in other experimental platforms and for potential technology-transfer applications.

2.1.2 Laser System and Frequency Stabilization

The cooling system operates on the D_2 line transition $5^2S_{1/2} \rightarrow 5^2P_{3/2}$ of ^{87}Rb, which occurs at a wavelength of approximately 780 nm. To fully describe the cooling mechanism, it is necessary to consider the hyperfine structure characterized by the total angular momentum quantum number $F = I + J$, where J is the total electronic angular momentum and I is the nuclear spin. The hyperfine structure with the employed laser coupled transitions is shown in Figure 2.2 for both ^{85}Rb and ^{87}Rb.
The cooling transition is $F = 2 \rightarrow F' = 3$. At thermal equilibrium, prior to the illumination of the MOT beams, the ^{87}Rb atoms in the vapor are distributed between the $F = 1$ and $F = 2$ hyperfine ground states in a ratio of $3 : 5$, governed by their statistical weights. When the MOT beams are turned on the cooling laser starts to excite atoms to the $F' = 3$ state. According to the dipole transition selection rules $\Delta F = 0, \pm 1$, the only allowed decay is $F' = 3 \rightarrow F = 2$. Since in an absorption–emission cycle the atom starts and ends in the $F = 2$ state, the transition is referred to as a *closed transition*.
However, starting from $F = 2$, there is also a small probability that the atom undergoes the off-resonance excitation $F = 2 \rightarrow F' = 2$, and from $F' = 2$ the atom can spontaneously decay either to $F = 2$ or to $F = 1$. The latter is a *dark state* that does not interact with

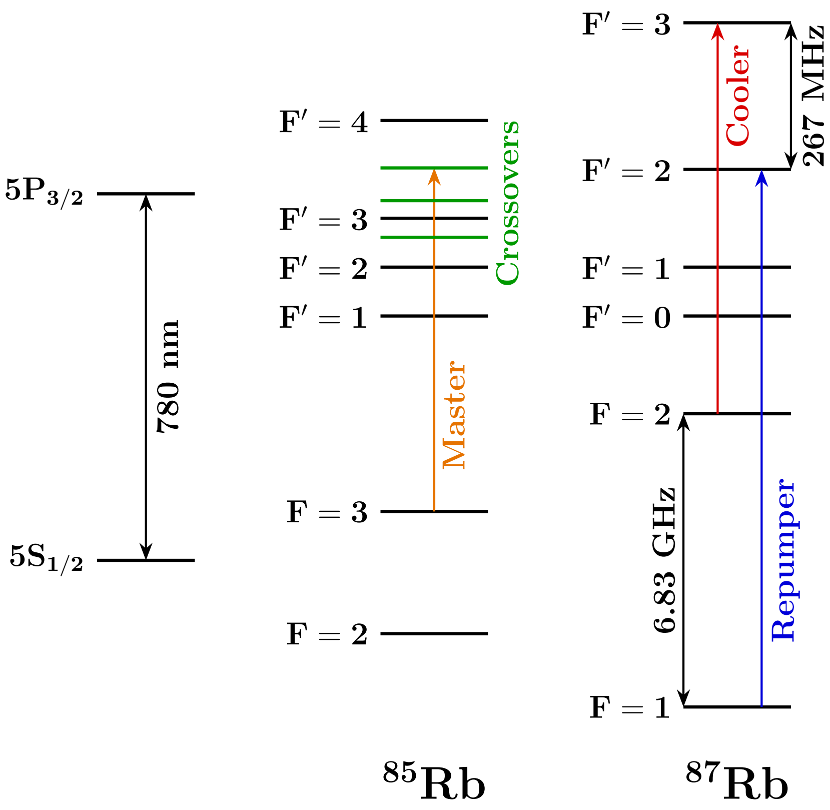


Figure 2.2: *Level scheme of the* ^{85}*Rb and* ^{87}*Rb* D_2 *line. Reference transitions are indicated for the master laser (orange), cooler laser (red), and repumper laser (blue). For* ^{85}*Rb, the crossover transition frequency used for saturation spectroscopy locking is also shown. Detailed data are available in [29, 30].*

the cooling laser, therefore, an atom in $F = 1$ cannot be cooled or trapped anymore. To address this leakage, a repumping laser is introduced on resonance with the transition $F = 1 \rightarrow F' = 2$, allowing the atoms to eventually decay back to $F = 2$ and re-enter the cooling cycle. The use of the repumper also brings the initial thermal population from the $F = 1$ state into the cooling cycle, maximizing the trap loading efficiency from the vapor.

In order to work properly, both the cooling and repumping lasers must be precisely frequency-locked to their target transition (with the appropriate detuning). The locking mechanism is briefly discussed in the following sections. A more exhaustive description of the laser system can be found in [31, 26].

2.1.2.1 Master laser

The cooling and the repumping lasers are locked to their target frequency by taking as a reference a third laser, known as the *master laser*. The master laser itself requires a precise frequency stabilization, which is achieved via *saturation spectroscopy* [20]. Saturation spectroscopy allows resolving spectral features much narrower than the Doppler width of a thermal vapor. For experimental convenience, the master frequency is chosen to lie

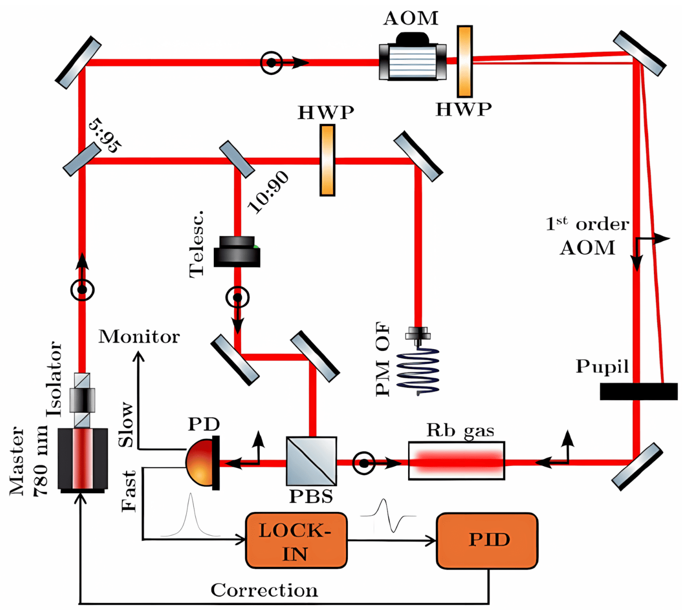


Figure 2.3: *Master laser locking setup scheme*

between the cooling and repumping frequencies, Therefore, it is locked to the $(F = 3 \rightarrow F' = 3) \times (F = 3 \rightarrow F' = 4)$ crossover transition of the ^{85}Rb isotope (see Figure 2.2).

The locking scheme is illustrated in Figure 2.3. The light source is a 780 nm Distributed Feedback (DFB) laser diode (Toptica EYP-DFB-0780-00040-1500-BFW11-0005), current-driven and temperature-stabilized by custom electronics. The beam is split by a 5 : 95 beam splitter into two branches. The 5% branch takes the name of *probe beam*. The 95% branch encounters a subsequent 10 : 90 beam splitter: the 90% is routed to the cooler and repumper locking setup, while the 10% is used for the saturation spectroscopy and takes the name of *pump beam*.

The pump beam passes through a Polarization Beam splitter (PBS) where it is fully reflected before propagating through a ^{85}Rb heated vapor cell. On the other branch, the probe beam passes through an Acousto-Optic Modulator (AOM), driven by an amplified RF signal generated by a Direct Digital Synthesizer (DDS), controlled by the experiment electronics. Only the 1st diffraction order is retained, while the 0th order is removed by an iris. This AOM is used to frequency-modulate the probe beam as required by the locking protocol.

After the AOM, the probe beam passes through a half-wave plate (HWP) to adjust its polarization. It then traverses the rubidium cell, counter-propagating with respect to the pump beam. Upon exiting the cell, the polarized probe beam is fully transmitted by the aforementioned PBS and incident on a photodiode. The photodiode features two outputs: a slow port, connected to an oscilloscope for visual monitoring of the spectrum, and a fast port, whose signal is demodulated to extract the dispersive error signal. Finally, a PID controller processes this error signal to generate a feedback correction for the laser diode, actively stabilizing the laser frequency.

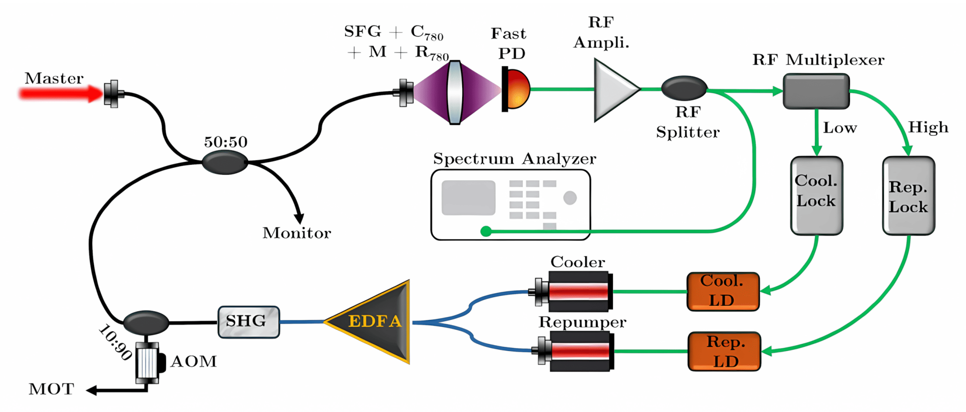


Figure 2.4: *Cooling laser and repumping laser setup scheme.*

2.1.2.2 Cooler and repumper lasers

The cooling and repumping light is generated by two 1560 nm DFB laser diodes (Eblana EP1562-0-NLW-B26-100FM) that are fiber-coupled and combined into a single fiber, which is then directed into an Erbium-Doped Fiber Amplifier (EDFA). Under standard operating conditions, the EDFA receives an input power of $\sim$ 5.2 mW, delivering an output of $\sim$ 500 mW, as read from its internal sensors.
Following the EDFA, the amplified light encounters a nonlinear crystal for Second Harmonic Generation (SHG), which converts the input wavelength from 1560 nm to 780 nm. The SHG crystal, together with the doubling of both cooling and repumping frequencies, also generates light at the sum frequency via *Sum-Frequency Generation* (SFG). This additional frequency is a byproduct and is not utilized in the experiment.
After the SHG stage, the light is split by a 10 : 90 beam splitter. The 90% fraction passes through an AOM, that acts as a fast switch, and goes to the three MOT beam collimators. The remaining 10% is combined with the Master laser light via a 2×2 fiber splitter to feed the cooler and repumper locking system. It should be noted that during this thesis work, the 10 : 90 splitter was damaged and was temporarily replaced by a 25 : 75 splitter. The superimposed optical signal (composed of Master, Cooler, Repumper, and SFG frequencies) is detected by a fast photodiode with a high bandwidth of $\sim$ 10 GHz. The interference between these fields yields a signal that presents amplitude modulation at the beat note frequencies. In particular, the beat notes relevant for the frequency stabilization are the Master-Cooler beat at $\sim$ 1 GHz and the Master-Repumper beat at $\sim$ 5.5 GHz. The photodiode output signal is amplified and split into low- and high-frequency bands using RF filters to isolate the Cooler and Repumper beat signals. These signals are fed into two independent PID controllers, which drive the Laser Drivers (LDs) to actively stabilize the frequencies. This technique is known as *frequency offset locking*, or *beat note locking*.

2.2 Optical dipole trap

This section describes the implementation of the Optical Dipole Trap (ODT) setup used to transfer atoms from the MOT into a conservative optical potential. The ODT beam is guided through a Hollow-Core Photonic Crystal Fiber (HCPCF). As mentioned in Section 2.1, the HCPCF integrated into the experimental apparatus was already present prior to this thesis work. However, a test HCPCF was employed during the development of the setup in order to optimize the alignment procedure.
During the implementation of the system, dedicated diagnostic tools were also developed to investigate the near-field beam profile and improve the fiber-coupling alignment. The following sections describe the ODT optical configuration and the experimental procedures adopted for the alignment and loading of the trap.

2.2.1 1064 nm laser characterization

The ODT is realized by using an IPG Photonics YLR-20-1064-LP-SF laser, a linearly polarized Ytterbium fiber laser that emits 1064 nm light, at a maximum nominal power of 20 W. For this laser model, the beam shape changes significantly upon changes of the nominal power set. Therefore, the mode profile has been characterized as a function of the power set in order to allow a correct realization of the HCPCF injection setup. This calibration also provides a useful reference for future optimization of the HCPCF injection.

2.2.1.1 Methodology

To characterize the beam properties, its spatial profile is modeled as a Gaussian beam. Therefore, the beam intensity distribution can be described by [4]:

$$I(r,z) = I_0 \left(\frac{w_0}{w(z)} \right)^2 \exp\left(\frac{-2r^2}{w(z)^2} \right), \tag{2.2}$$

where

- z is the axial distance from the beam waist;
- r is the radial distance from the center of the beam;
- $w(z)$ is the beam radius, defined as the radius at which the intensity decreases of a factor $1/e^2$ with respect to the intensity at the beam center;
- $w_0 = w(0)$ is the beam waist, i.e. the beam radius at its focus;
- I_0 is the intensity at the center of the beam at its waist.

The transverse intensity profile follows a Gaussian shape, hence the name of Gaussian beam.

At any axial position z, the beam radius is given by [4]:

$$w(z) = w_0\sqrt{1 + \left(M^2\,\frac{z}{z_R}\right)^2} \tag{2.3}$$

where

$$z_R = \frac{\pi w_0^2}{\lambda} \tag{2.4}$$

is the Rayleigh range, which represent the characteristic distance around the waist over which the beam remains approximately collimated, and $M^2 \geq 1$ is the beam quality factor, accounting for deviations from an ideal Gaussian beam. By acquiring intensity images at multiple longitudinal positions and extracting the corresponding beam radii, Equation (2.3) can be fitted to determine both the waist radius and the waist position.

The experimental setup used for the intensity measurements corresponds to the initial section of the ODT injection setup (see Section 2.2.2.1). With reference to Figure 2.7, measurements were performed on the beam exiting the reflected branch of the second polarization beam splitter (PBS2). The power was adjusted through the regulation of the half-wave plates HWP2 and HWP3, together with the addition of optical density filters when needed. A Complementary Metal-Oxide-Semiconductor (CMOS) camera was employed to take 17 intensity images separated by 25 mm along the propagation axis for the nominal laser powers of 4 W, 7 W, 10 W, 13 W, 16 W and 19 W.

The straightforward approach of extracting the beam radius from each image consists of fitting a 2D asymmetric Gaussian profile and extract the radius along the x and y axis as

$$w_x(z) = 2\sigma_x, \qquad w_y(z) = 2\sigma_y. \tag{2.5}$$

However, fitting equation (2.3) using these extracted radii led to values of $M^2 < 1$ for all the power settings except 4 W, which is unphysical. However, enforcing the constraint $M^2 \geq 1$ resulted in visibly poor agreement between the fitted curves and the measured data. A likely explanation is that the presence of several optical components before the camera introduce aberrations or beam clipping that make equation (2.3) incorrect.

For this reason a second method called *D4σ* was tested [32]. In this method, the beam radius is determined from the second moment of the beam intensity distribution with respect to the intensity centroid. For the x-axis:

$$\hat{\sigma}_x^2 = \frac{\int_{-\infty}^{\infty}(x - x_0)^2 I(x,y)\,dx\,dy}{\int_{-\infty}^{\infty} I(x,y)\,dx\,dy} \tag{2.6}$$

where x_0 is the intensity centroid x-coordinate, and $I(x,y)$ is the beam intensity profile.

Equation (2.3) is then fitted using the radii

$$w_x = 2\hat{\sigma}_x, \qquad w_y = 2\hat{\sigma}_y \, . \tag{2.7}$$

The $D4\sigma$ method should be more robust than the Gaussian fitting when dealing with non-ideal beam profiles, since it accounts for intensity contributions in the beam tails that may not be properly captured by a Gaussian fit. However, the $D4\sigma$ is very sensitive to anisotropic background noise, which can bias both the centroid and second-moment calculations. In the present setup, the presence of the PBS1 and the beam dump that dissipates a large quantity of optical power, generates a non-uniform scattering background. This asymmetry introduces systematic distortions in the $D4\sigma$ analysis. Consequently, even this method did not consistently yield $M^2 \geq 1$ without constraints, although it performed better than the Gaussian fit, giving naturally $M^2 \geq 1$ for 4 W, 16 W and 19 W.

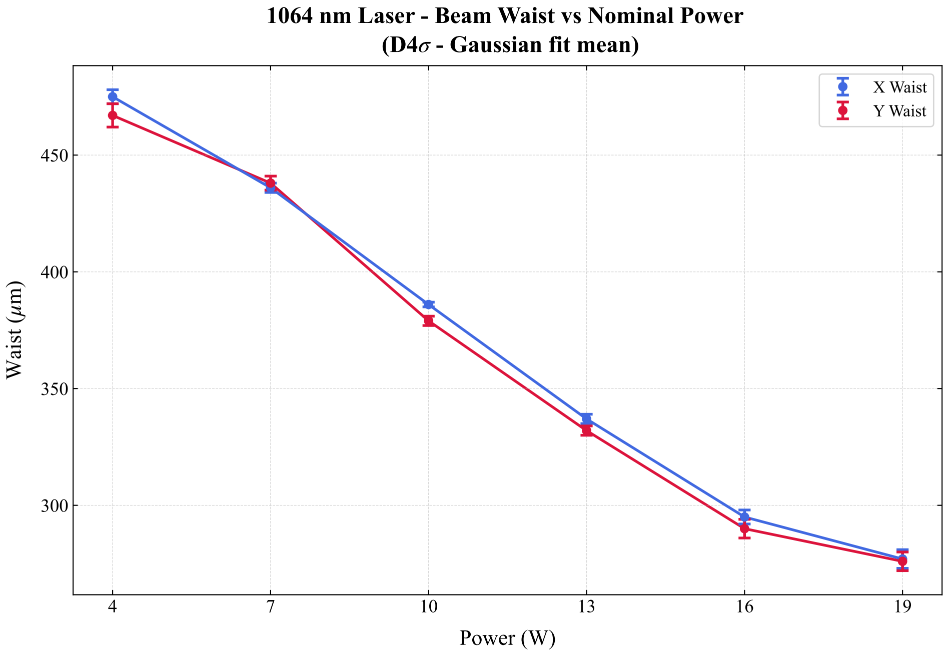
For both analysis methods, a preliminary Gaussian fit was first performed to find approximately the beam center position. A region of interest (ROI) extending to 5σ around the beam center was then selected. Subsequently, either a refined Gaussian fit was performed or the centroid and second moments were calculated to determine the beam radius before fitting Equation (2.3).

2.2.1.2 Results

The beam waist radii and waist positions obtained with the two analysis methods are summarized in Tables 2.1 and 2.2. The fit of Equation (2.3) was performed using the Trust Region Reflective (TRF) algorithm while imposing the constraint $M^2 \geq 1$. The final reported values were obtained by averaging the results of the Gaussian fit and $D4\sigma$ methods.

The waist radius as a function of the nominal laser power is plotted in Figure 2.5 top. Except for 4 W and 7 W, the results reveal a small astigmatism, with the waist measured along the x direction exceeding that along the y direction by approximately 1–2%. Furthermore, the graph shows that the waist radius decreases approximately linearly from 7 W to 19 W, indicating a significant power dependence of the laser mode size.

The waist longitudinal position as a function of the nominal laser power is plotted in Figure 2.5 bottom. These values should be interpreted with caution. Imposing $M^2 \geq 1$ allows the fits to better match the middle and far points which depends asymptotically on solely w_0 (limit $z \gg z_R$ in equation 2.3), but strongly miss the prediction on the point at short distance that give the information on the waist position (see Figure 2.6). Although the extracted values carry significant systematic uncertainty, the observed trend appears physically plausible, with the beam waist shifting downstream as the nominal laser power increases from 4 W to 13 W.



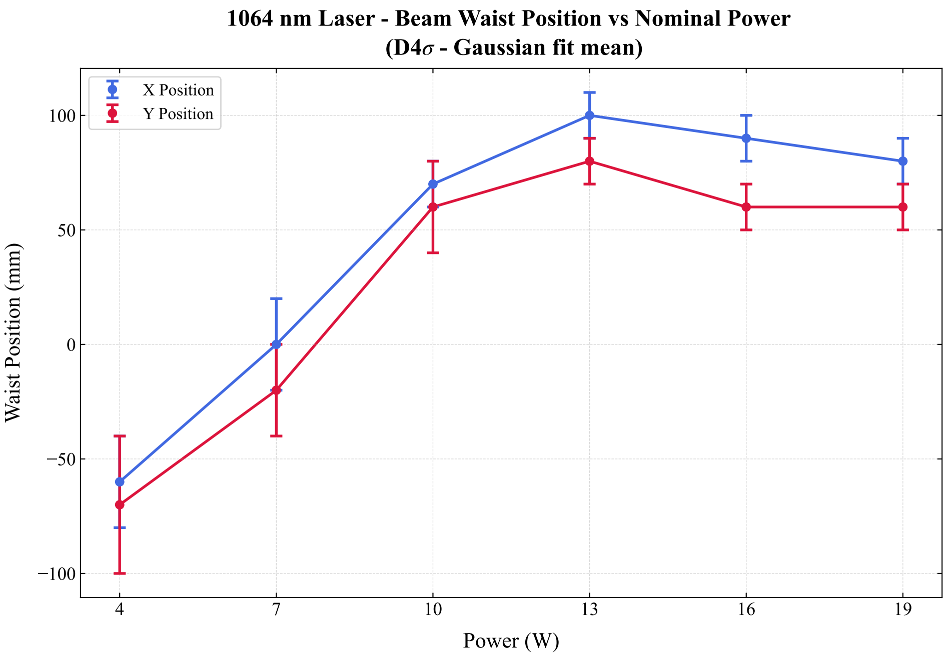


Figure 2.5: *Characterization of the 1064 nm laser beam parameters as a function of nominal output power. (Top) Transverse beam waist size versus power. (Bottom) Longitudinal position of the waist. The reported values correspond to the average of the results obtained using the D4σ and Gaussian-fit methods. Blue and red markers denote measurements along the orthogonal x and y transverse axes, respectively.*

Table 2.1: *Comparison of the beam waist w_0 obtained via the D4σ method and the Gaussian fit method. The final values correspond to the average of the two estimates.*

Nominal Power (W)	Estimated waist w_0 (μm)					
	$D4\sigma$-x	$D4\sigma$-y	Gaus-x	Gaus-y	Avg-x	Avg-y
4	481 ± 5	474 ± 7	469 ± 4	460 ± 7	475 ± 3	467 ± 5
7	454 ± 3	452 ± 4	419 ± 3	423 ± 5	436 ± 2	438 ± 3
10	403 ± 2	394 ± 3	370 ± 2	364 ± 3	386 ± 1	379 ± 2
13	351 ± 2	343 ± 3	323 ± 4	321 ± 4	337 ± 2	332 ± 2
16	307 ± 2	298 ± 4	283 ± 6	282 ± 6	295 ± 3	290 ± 4
19	288 ± 3	284 ± 4	266 ± 7	268 ± 6	277 ± 4	276 ± 4

Table 2.2: *Comparison of the beam waist position z_0 extracted via the D4σ method and the Gaussian fit method. The final values correspond to the average of the two estimates.*

Nominal Power (W)	Estimated waist position z_0 (mm)					
	$D4\sigma$-x	$D4\sigma$-y	Gaus-x	Gaus-y	Avg-x	Avg-y
4	-90 ± 40	-80 ± 50	-40 ± 20	-60 ± 40	-60 ± 20	-70 ± 30
7	20 ± 30	-3 ± 40	-20 ± 20	-40 ± 20	0 ± 20	-20 ± 20
10	90 ± 20	70 ± 30	50 ± 10	40 ± 10	70 ± 10	60 ± 20
13	110 ± 10	90 ± 20	90 ± 20	70 ± 20	100 ± 10	80 ± 10
16	72 ± 9	50 ± 20	100 ± 20	80 ± 20	90 ± 10	60 ± 10
19	70 ± 10	40 ± 10	90 ± 20	70 ± 20	80 ± 10	60 ± 10

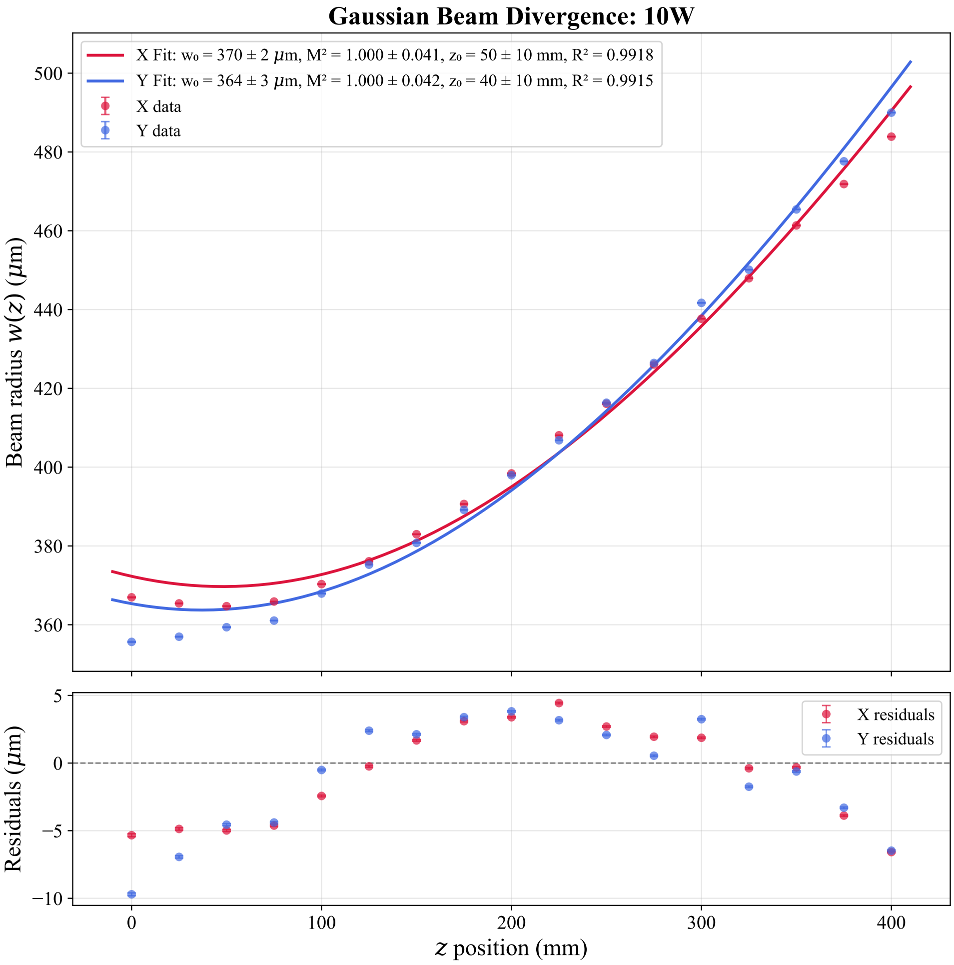


Figure 2.6: *Gaussian beam divergence profile measured at a nominal laser power of 10 W. The beam radius $w(z)$ is shown as a function of the longitudinal position z along the x (red) and y (blue) axes. Solid lines denote Gaussian propagation fits performed under the constraint $M^2 \geq 1$. The systematic discrepancy observed at short propagation distances ($z < 100mm$) results from the imposed M^2 constraint, which prioritizes agreement with the intermediate- and far-field data at the expense of the near-field region. Consequently, the accuracy of the fitted beam waist position z_0 is reduced.*

2.2.2 Optical setup

In this experiment the ODT light is guided through a HCPCF, which introduces a critical complication in the injection procedure. Indeed, in contrast with standard telecom fibers, HCPCFs are multi-mode fibers. As a consequence, even small deviations of the injected beam from the ideal injection angle excites higher order modes in addition to the fundamental one. This results in distorted output beam profiles, which are not well suited for the realization of an optical conveyor belt based on the superposition of the beam guided by the HCPCF and a beam guided through a conventional telecom fiber supporting only the fundamental LP_{01} mode.
For this reason, particular care was devoted to the alignment procedure. A significant improvement in the quality of the transmitted mode was achieved by optimizing the injection while inspecting the near-field of the transmitted beam, rather than simply maximizing the transmission efficiency. The following sections describe the injection setup and the horizontal microscope built for the near-field inspection.

2.2.2.1 Injection setup

Figure 2.7 shows the optical setup scheme used for the injection of the laser light in the HCPCF. The scheme is composed of the following elements:

- **Half Wave Plate 1** (HWP1) and **Optical Isolator**. The optical isolator prevents back reflections from propagating towards the laser cavity. High-power lasers are particularly sensitive to back reflections, which can introduce power instabilities and potentially damage the laser system. Since optical isolators operate correctly only for a specific input polarization, HWP1 is used to optimize the polarization of the incoming beam and maximize the transmission through the isolator.

- **HWP2** and **Polarization Beam Splitter 1** (PBS1). These components are used to regulate the optical power delivered to the subsequent stages of the setup. The unwanted fraction of the beam is directed onto a beam dump. This configuration is particularly useful during alignment procedures, where the optical power is reduced to a few milliwatts for safety reasons.

- **HWP3** and **PBS2**. PBS2 splits the beam into two different branches. The first is employed for the HCPCF injection, while the second will be injected into a PANDA polarization-maintaining fiber, which eventually will provide the second beam for the realization of the optical conveyor belt. HWP3 is adjusted to have approximately 50/50 power splitting ratio, although this value can be modified if required.

- **Acousto-Optic Modulator** (AOM). The AOM is employed as a fast switch for the ODT beam. The device operates using an acoustic wave at 110 MHz, generated by a

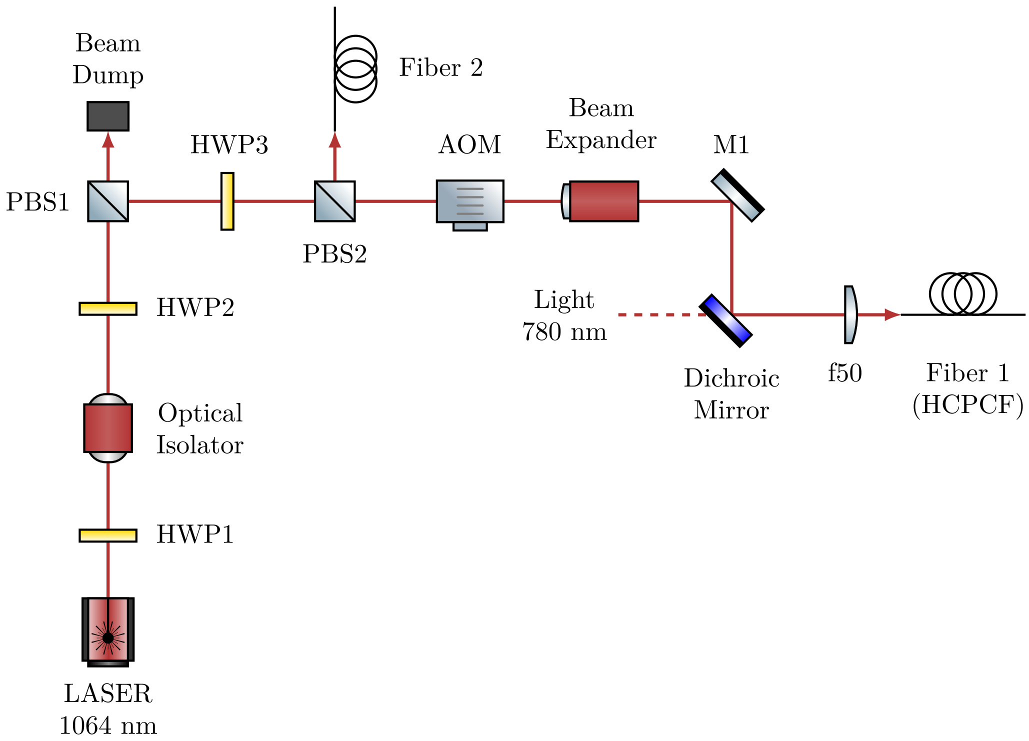


Figure 2.7: *Scheme of the optical setup for 1064 nm ODT beam injection into the HCPCF. The Acousto-Optic Modulator (AOM) operates at 110 MHz, and the subsequent setup utilizes the first diffraction order. The zero-th order is blocked by a beam dump (not drawn). The 780 nm light setup is not drawn, its presence will be important in the future stages of the experiment for the characterization of the atoms inside the HCPCF core. Fiber 2 injection setup is not drawn as well.*

Direct Digital Synthesizer (DDS), causing the first diffraction order to be frequency shifted by the same amount. Although this shift is negligible with respect to the trap depth of the ODT, it is essential for the realization of the optical conveyor belt.

- **Beam expander**. A ~3X beam expander is employed to increase the beam diameter and collimate it.

- **Mirror 1** (M1) and **Dichroic Mirror**. Both mirrors are mounted on tilt stages and their angular adjustment is used for optimizing the fiber injection alignment. The dichroic mirror also combines the 1064 nm ODT beam with a 780 nm beam that will be used for the characterization of the atoms trapped inside the HCPCF core.

- **Focusing Lens** (f50). A lens with focal length $f = 50$ mm focuses the beam into the HCPCF. It is mounted on a horizontal micrometric translation stage, providing an additional degree of freedom for optimizing the injection alignment.

- **Fiber 1** (HCPCF). The HCPCF is mounted on an x-y translation stage and terminated with an FC/APC connector. The translation stage allows transverse displacement of the fiber, providing additional alignment degrees of freedom during the injection procedure, although these degrees of freedom were found to have limited impact on the final optimization.

Using the beam waist measurements in Section 2.2.1.2, a nominal laser power of 10 W was selected as the optimal operating point. Since the fitted waist positions were found to be affected by large systematic uncertainty, the beam waist was assumed to be located at the output face of PBS2. The radius at the entrance of the beam expander can be estimated using (2.3), and the measured distance of approximately 175 mm between PBS2 and the beam expander (see figure 2.9). Assuming an ideal beam $M^2 = 1$, using the x waist value for 10 W of $w_0 = 384$ μm, the beam radius at the entrance of the beam expander is estimated to be w(beam expander entrance) $\simeq 414$ μm.
The beam expander provides a magnification factor of three, yielding w(expander exit) $\simeq$ 1240 μm. At this stage the corresponding Rayleigh range, calculated using 2.4 is $z_R \simeq$ 4.5 m. Since the remaining propagation distance before the injection optics is negligible compared to the Rayleigh range, the beam is considered approximately collimated. Under this assumption, the beam waist after the injection lens can be estimated using [4]:

$$w_{\text{out}} = \frac{\lambda f}{\pi w_{\text{in}}}, \tag{2.8}$$

where $f = 50$ mm is the focal length of the injection lens. Propagating the uncertainty, one gets:

$$w_{\text{inj}} \simeq (13.6 \pm 0.1)\ \mu\text{m}.$$

This result is obtained in the ideal case, the real waist at the injection is expected to be closer to the target value of approximately ~ 18 μm. For instance, assuming a more realistic beam quality factor of $M^2 = 1.2$ leads to

$$\tilde{w}_{\text{inj}} \simeq (15.1 \pm 0.1)\ \mu\text{m}$$

Additional deviations from the ideal prediction are expected due to the AOM, which distorts the beam profile by compressing it along the y direction and expanding it along the x direction, resulting in an expected beam radii ratio $w_y/w_x \approx 0.85$ at AOM exit, which is reversed at injection by the relation (2.8).
To verify these predictions, the beam profile after the injection lens was characterized following a procedure analogous to that described in Section 2.2.1, yielding

$$w_{\text{inj-}x} = (17.1 \pm 0.7\ \mu\text{m}) \qquad w_{\text{inj-}y} = (19.3 \pm 0.4\ \mu\text{m}),$$

sufficiently close to the target waist.

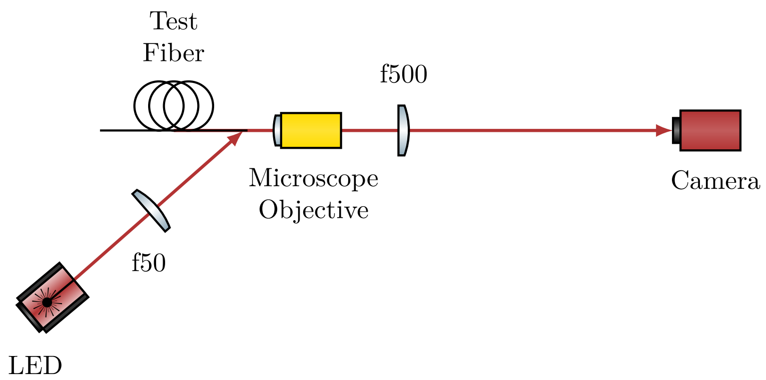


Figure 2.8: *Scheme of the optical setup for the horizontal microscope.*

2.2.2.2 Horizontal microscope

Initially, the injection into the HCPCF installed inside the vacuum chamber was attempted by using a CMOS camera to monitor the beam profile. However, despite the effort, strongly deformed mode profiles were obtained.

In order to better investigate the injection procedure and simplify the alignment process, a separate test HCPCF was employed outside the vacuum chamber. The test fiber consisted of an approximately 50 cm HCPCF segment with the same FC/APC connector as the fiber in the chamber. The test fiber allowed an easy measurement of the efficiency on the exit facet of the beam without any window in the path, and being outside the vacuum chamber allowed for easier mode imaging. Moreover, since the two fibers shared the same connector, the optimal alignment conditions identified with the test fiber were expected to provide a good starting point for the alignment of the HCPCF installed inside the chamber. Nevertheless, even with the test fiber, optimization based solely on transmission efficiency and far-field inspection still resulted in poor beam quality.

The fiber manufacturer suggested that the optimal injection condition should be determined by inspecting the shape of the near-field of the transmitted mode. For this purpose, a horizontal microscope system was developed to inspect the HCPCF facet and to investigate the near-field profile of the transmitted mode. The optical scheme of the microscope is shown in Figure 2.8.

The fiber was positioned inside a 90° V-groove of depth 350 μm. The fiber was fixed inside the groove by means of a thin foam layer attached to a tiltable mount using adhesive tape. The coating was removed from the fiber end, leaving approximately 15 mm of suspended fiber in order to allow lateral illumination of the facet.

The HCPCF facet was illuminated by a red LED (wavelength $\lambda = 635$ nm) focused by an $f = 50$ mm lens onto the final suspended portion of the fiber. Part of the red light propagated through the silica structure forming the kagome lattice, making it visible on the microscope camera. The observation of the kagome structure was used to optimize the microscope focus and to inspect the integrity and cleanliness of the fiber facet.

The light emerging from the fiber was collected by an optical microscope objective lens (Olympus PLN10X objective, nominal focal length $f = 18$ mm, working distance $d = 10.6$ mm) and subsequently re-imaged onto a CMOS camera by means of an $f = 500$ mm lens. The expected magnification is $M = f_{\text{tube}}/f_{\text{lens}} \approx 27.8$. The objective is mounted on a horizontal translation stage used for the focussing. The camera was connected to a computer for real-time imaging of the transmitted near-field mode. In addition, a powermeter could be inserted between the fiber output and the objective lens in order to evaluate the coupling efficiency by comparing the transmitted power with the optical power measured before the injection stage. Finally, a black paper tube was placed between the imaging $f = 500$ mm lens and the camera to reduce background light from the LED illumination and from ambient room light.

The alignment procedure was carried out by optimizing the transmitted near-field profile in order to obtain a mode as close as possible to a single-lobed structure, results and further details are found in Section 2.2.3.

Figure 2.9 shows the top-down photographs of the complete experimental apparatus, including both the injection setup and the horizontal microscope system, and Figure 2.10 shows a detailed view of the horizontal microscope.

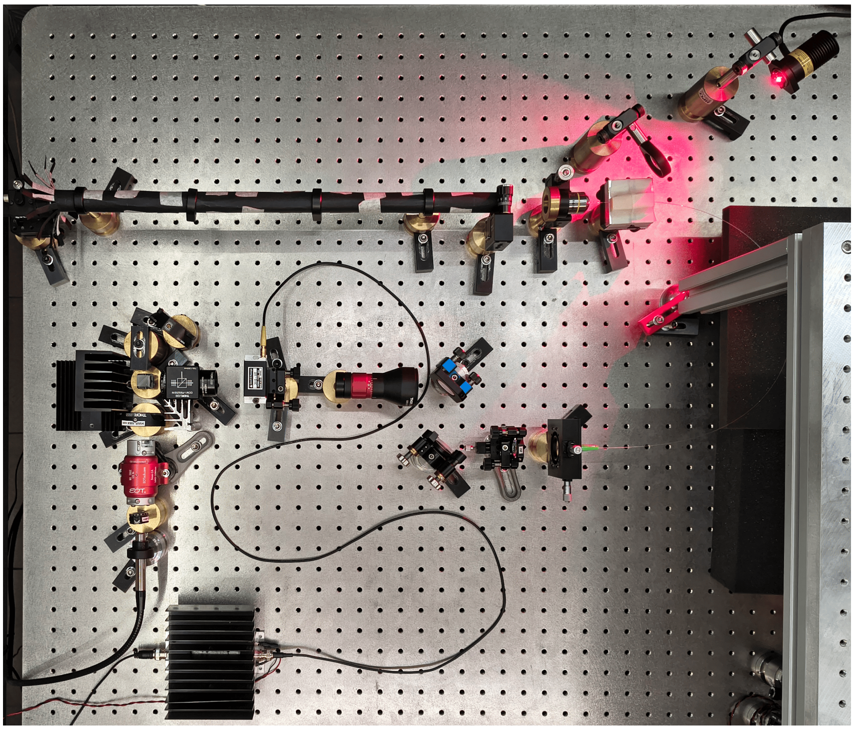


Figure 2.9: *Top-down photograph of the complete experimental apparatus. The horizontal microscope, shielded by the black paper tube, is visible at the top of the image. The optical injection setup is located in the center, while the RF amplifier driving the AOM is placed at the bottom. For reference, the distance between the screw holes on the breadboard is 25 mm.*

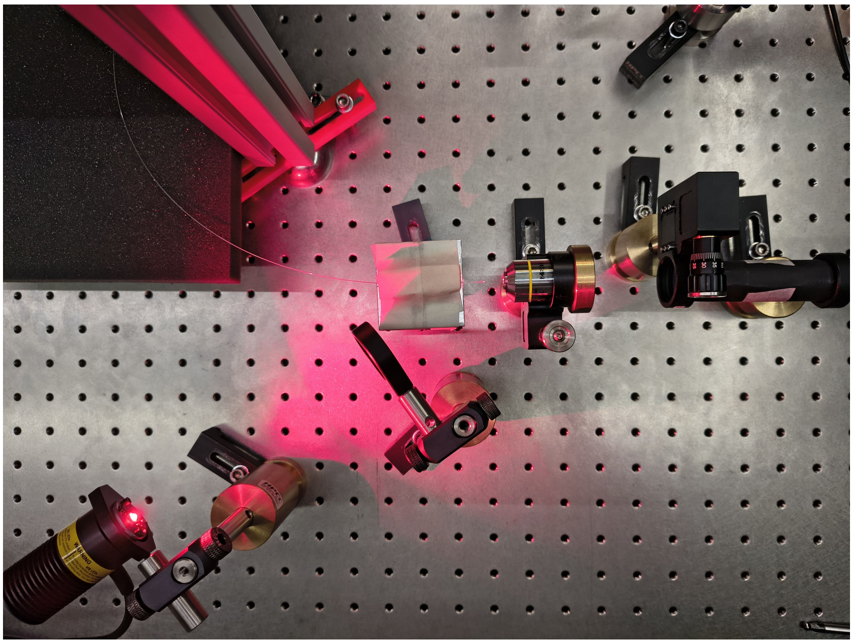


Figure 2.10: *Detailed top-down photograph of the horizontal microscope. The setup highlights the test fiber, the red LED providing lateral illumination to the fiber facet, the microscope objective lens and the refocusing lens.*

2.2.2.3 Chamber fiber optical imaging setup

After the optimization and the validation of the injection alignment procedure using the test fiber, the same approach was applied to the fiber inside the vacuum chamber.

To investigate the near-field profile, a dedicated optical imaging setup was built on a breadboard positioned above the vacuum chamber (Figure 2.11). Since the optical fiber is located inside the chamber, a first stage microscope was designed to create a virtual image of the fiber facet outside the chamber. For this purpose a f150 lens ($f = 150$ mm) was placed approximately 150 mm above the fiber facet, directly above the chamber window, such that the light emerging from the fiber is collimated. The collimated beam propagate through an opening and is redirected horizontally by a mirror (M1) tilted at 45°. A second lens f300 ($f = 300$ mm) refocuses the beam, forming an inverted image of the fiber facet with a magnification of $\times 2$.

A second imaging stage, consisting of lenses L3 and L4, acted as a microscope for this intermediate image and projected a magnified image onto the sensor of a camera. The focal lengths of L3 and L4 were modified several times throughout the experiment to optimize the field of view and magnification. The configuration used most frequently employed an L3 lens with $f = 100$ mm and an L4 lens with $f = 500$ mm, resulting in an additional magnification of $\times 5$. Combined with the first imaging stage, this yielded an overall magnification of approximately $\times 10$, which was used for the majority of the measurements.

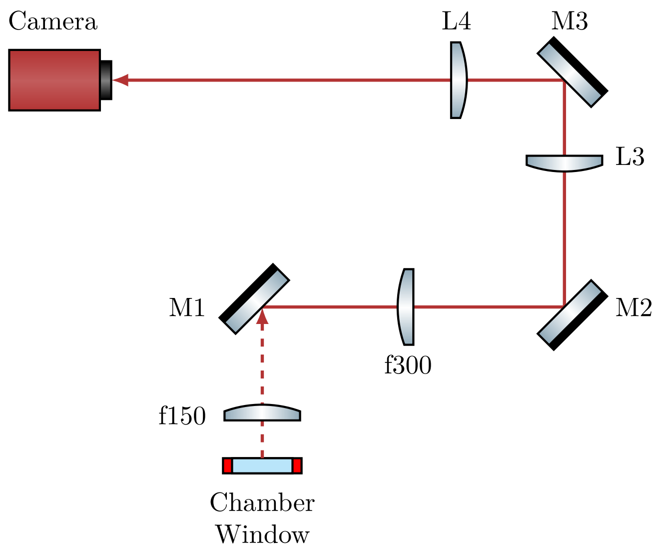


Figure 2.11: *Scheme of the optical setup used to image the fiber inside the vacuum chamber. The optical path between the chamber window and the mirror M1 is vertical (characterized by the dashed beam line), whereas the rest of the setup is arranged horizontally on a breadboard mounted above the chamber. The focal length of lenses L3 and L4 were varied several times during the experiment. Most images where acquired using an* $f = 100$ mm *and an* $f = 500$ mm *providing an overall magnification of approximately* $\times 10$ *combined with the first imaging stage.*

2.2.3 Injection alignment and near-field imaging

For a conventional single-mode telecom fiber, the alignment procedure is straightforward: it is sufficient to inject a low-power beam of few milliwatts, monitor the transmitted power at the fiber exit, and adjust the injected beam to maximize the guided optical power.

By contrast, aligning the injection of a multimodal HCPCF is significantly more complex. The silica jacket that surrounds the kagome core can guide the light with coupling efficiencies as high as 60%, although with badly shaped mode (see figure 2.12). Considering also that the jacket is large compared to the core, monitoring just the coupling efficiency is particularly ineffective, as the optimization can easily get trapped in a local maximum associated with jacket guiding.

As suggested by the manufacturer, the best procedure for the injection alignment resulted to be monitoring the near-field mode shape at the fiber exit. The complete alignment procedure begins with the use of a powermeter to ensure the impinging beam hits the fiber, even on the jacket. Then switch to live monitoring of the near-field with a camera and optimize its shape by tuning the degrees of freedom of the injection setup. Coupling efficiency is measured only after the mode profile has been optimized.

Applying this protocol, the test fiber injection was optimized employing the injection optical setup and the horizontal microscope described in Section 2.2.2. This approach yielded to a good result and thus was subsequently applied to align the injection into the

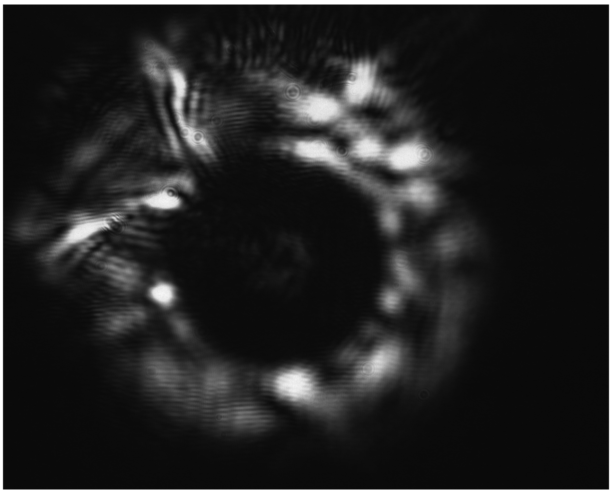


Figure 2.12: *Example of the near-field of a mode guided by the test fiber jacket. The light is mainly guided by the silica jacket surrounding the kagome structure, while only a weak signal is visible in the core. Note that this strongly deformed mode had a coupling efficiency of* $\sim 60\%$.

fiber in the vacuum chamber.

2.2.3.1 Test fiber

The test fiber was cleaved prior to inspection. Since the horizontal microscope has a sufficient resolution to resolve the kagome structure of the fiber, it was exploited for the focusing procedure. By monitoring the camera imaging in real-time, the kagome structure was focussed using the translation stage of the microscope objective. The objective refractive index can be considered roughly constant across the visible spectrum and into the near-infrared (NIR). This guarantees that when the structure (illuminated by red light) is in focus, the guided mode (1064 nm NIR light) is also in focus. Figure 2.13 shows an image of solely the kagome structure alongside an image of the kagome structure together with some guided light. The image shows that the kagome structure is intact confirming that the cleaving was correctly performed. Furthermore, the mode is clearly guided by the core and exhibits a single lobe, however, the shape is not symmetric with respect to the fiber axis, indicating the excitation of higher order LP_{11} mode alongside the fundamental LP_{01}.

The mode seen in Figure 2.13 right was associated with a transmission efficiency of 63%. By optimizing the alignment using the horizontal microscope the mode shape was optimized, and an efficiency of 80% has been achieved. For this optimization two screws of the mirror mounts were adjusted at the same time as a continuous beam walking, therefore performing an optimization remaining as long as possible in a good injection condition. The resulting mode is shown in Figure 2.14 left. Figure 2.14 right displays another mode obtained with nearly 80% of injection efficiency, where the surrounding kagome structure

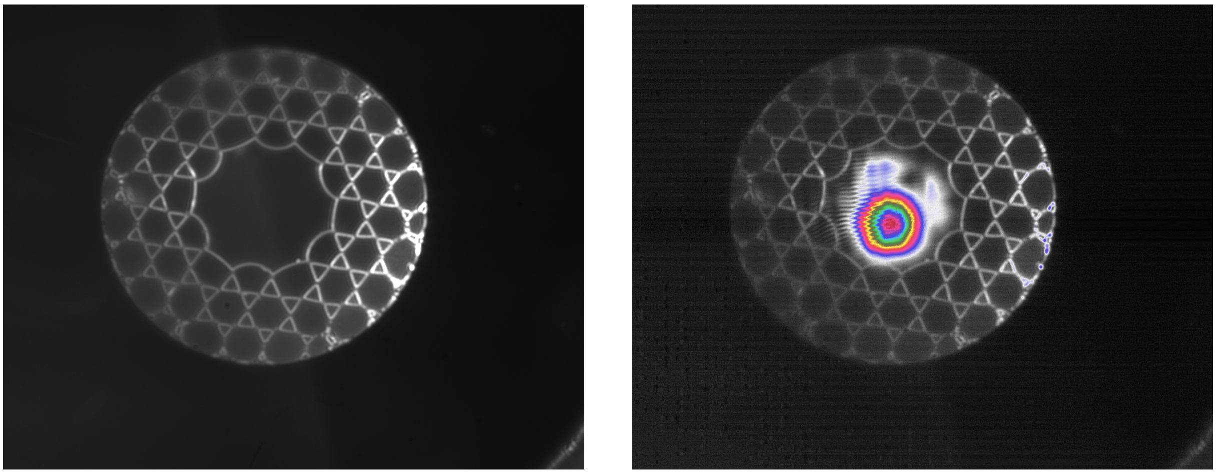


Figure 2.13: *Images taken with a preliminary horizontal microscope setup of the HCPCF kagome structure (left) and of the transmitted mode together with the kagome structure, with a pseudo color map (right). These images were taken with a preliminary setup of the horizontal microscope that provided a more homogeneous illumination, with the red light incident from two direction, making the kagome lattice distinctly visible.*

remains visible.

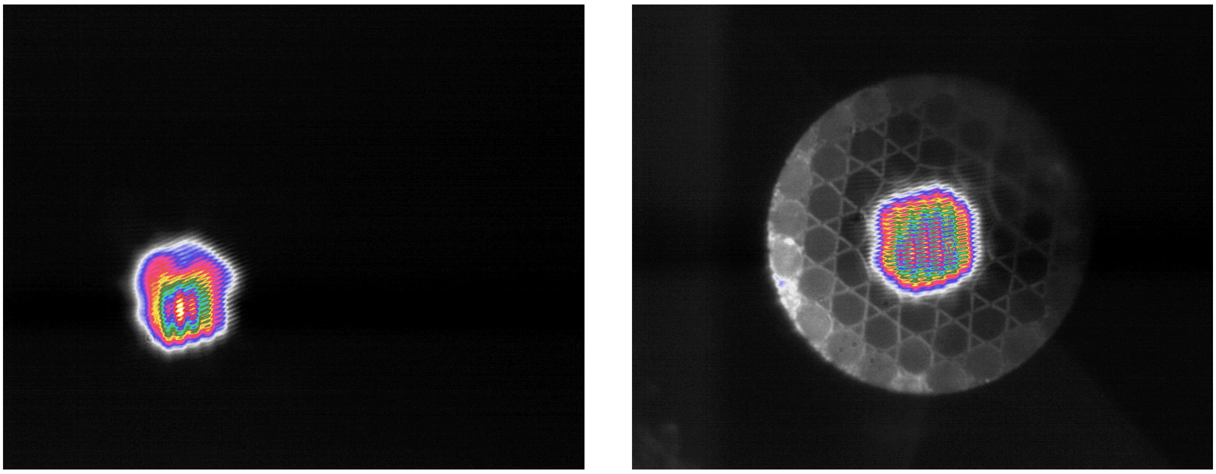


Figure 2.14: *Images captured with the final horizontal microscope setup. A mode with* $\sim$ 80% *of transmission efficiency (left) and a guided mode with the kagome structure visible (right).*

A final optimization was performed without the ocular lens by translating the objective slightly past the working distance and placing the camera approximately 2 m away. With this configuration, the magnification achieved is about $\times 70$ (more than doubling the magnification of the horizontal microscope). At this distance, the light scattered from the kagome structure becomes very faint, complicating the focusing procedure as its signal approaches the noise level. However, employing the FLIR camera, which has a lower read noise and a higher sensitivity, the kagome lattice remained resolvable, as shown in Figure 2.15. This figure shows clearly that the near-field mode assumes a hexagonal shape, perfectly fitting the core: as expected the field inherit the symmetry of the medium in which it is confined.

The advantage of this extended projection distance is the significant expansion of the mode, the pixel grid interference patterns disturb the shape far less, allowing finer spatial details to be resolved, further improving the alignment procedure. With this method an efficiency of 94% was achieved. The resulting mode (Figure 2.16 left) is still slightly asymmetric, indicating a residual superposition of higher order modes with the fundamental one.

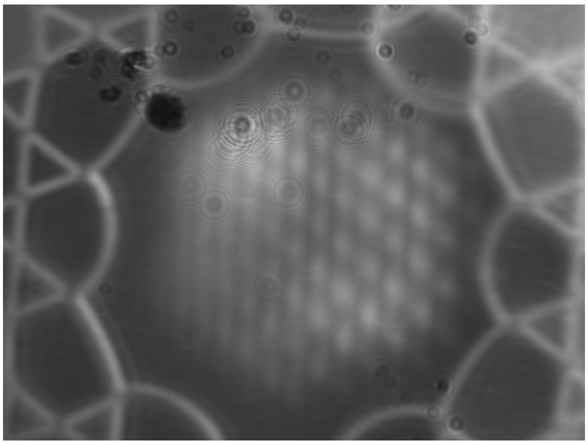


Figure 2.15: *Images obtained with the FLIR camera using the microscope objective without the ocular lens. The mode clearly exhibits a hexagonal shape. Thanks to the FLIR high sensitivity and low read noise, the kagome structure remains visible even in this configuration. Artifacts in the top-left corner are dust grain deposited on the camera sensor.*

Finally, a sweep of laser nominal power was conducted, which changes also the beam waist at the fiber injection. At every tested power the injection alignment was re-optimized, and the best mode profile was obtained at 6.5 W, with an efficiency of 96%. The mode is shown in Figure 2.16 right, in this case the shape is far more symmetric than the previous iterations, which, together with the high transmission efficiency, suggests a significant high share of the fundamental LP_{01} mode.

2.2.3.2 Chamber fiber

After validating the effectiveness of the alignment procedure using the test fiber, the same method was applied to the fiber installed inside the vacuum chamber. Unfortunately, it was not possible to properly illuminate the fiber, and the kagome structure could not be visualized. Using the same 635 nm LED employed for the test fiber, several configurations of focusing lenses and LED positions were tested, however, no light was successfully coupled into the kagome structures. In addition, due to physical constraints imposed by the vacuum chamber geometry, it was not possible to directly illuminate the fiber facet from the outside. One possible limitation might be the presence of a translucent plastic casing surrounding the fiber. The casing may diffuse the light of the LED preventing it to

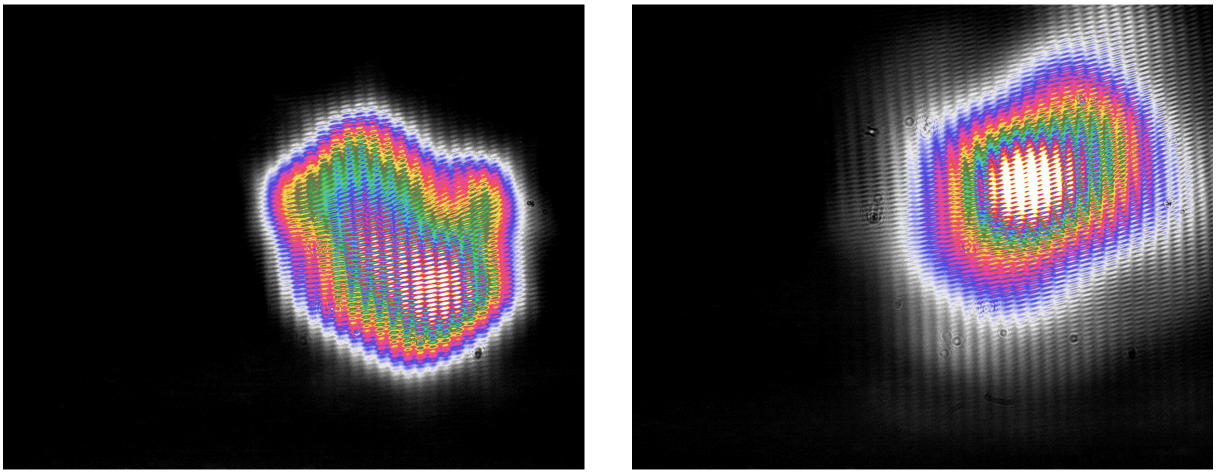


Figure 2.16: *Images obtained with the objective without the ocular lens. Mode with* 94% *of injection efficiency (left). Optimized mode with nominal laser power set to 6.5 W which achieved* 96% *of transmission efficiency (right).*

reach the kagome structure. Unlike the images obtained with the test fiber, in this case the jacket appears illuminated, suggesting that the light propagated in the jacket rather than reaching the fiber kagome structure.

Furthermore, the need to create a virtual image outside the chamber required the use of multiple lenses (see Section 2.2.2.3). While this approach enabled imaging of the fiber facet, it also introduced non-negligible chromatic aberrations. Figures 2.17 left and 2.17 right show respectively a preliminary image of the focused fiber and the same configuration with light guided through the fiber. In the latter case, the optical mode is in focus while the fiber facet appears out of focus.

The mode obtained after the injection alignment procedure is shown in Figure 2.18, which exhibited $\sim 50\%$ of transmission efficiency. It should be noted that this efficiency values was measured after the light propagated through a chamber window, two lenses and two mirrors, all of which introduce transmission and reflection losses. Moreover, the fiber installed in the chamber is longer than the test fiber. Although propagation losses associated with the additional length should be negligible, the increased length requires several bends in the fiber routing, which are known to induce leakage losses in optical fibers.

The best mode obtained after the optimization procedure is displayed in Figure 2.18. It shows a clear asymmetry, indicating the presence of a superposition of various LP modes. Although, it should also be considered that, in order to enter the vacuum chamber, the fiber must be tightened in a ferrule by a swage-lock fitting. This mechanical stress can deform the fiber microstructure, altering its guiding properties and degrading the mode shape quality.

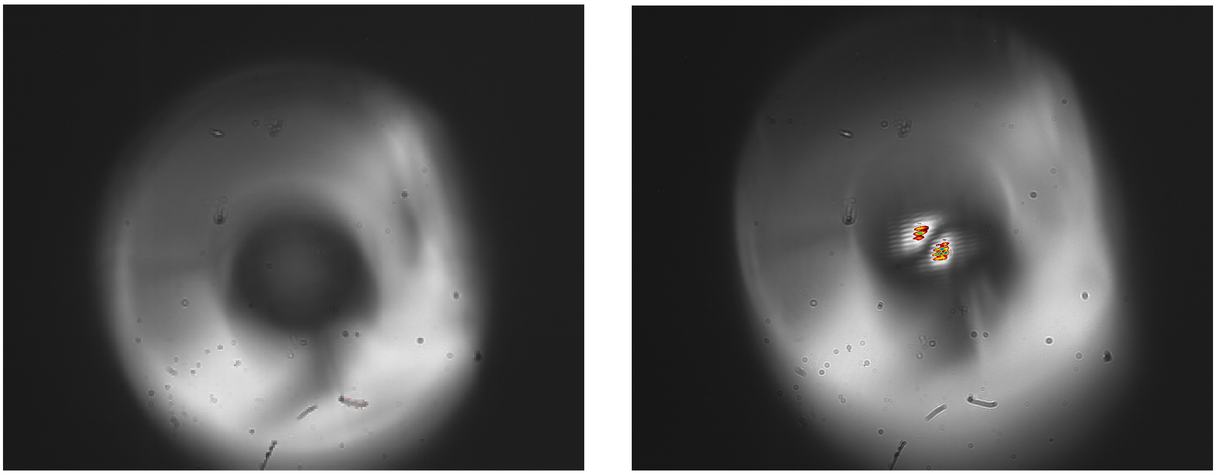


Figure 2.17: *Preliminary images taken with a zoom about* $\times 10$ *of the fiber facet in the vacuum chamber (left) and the fiber with a non-optimized mode (right). In contrast to the test fiber images, here, the kagome structure is not illuminated, whereas the outer jacket is. The mode exhibits a clear* LP_{11} *profile. Note that focusing the mode bring the fiber out of focus, this makes the mode seem non-centered with respect to the fiber core.*

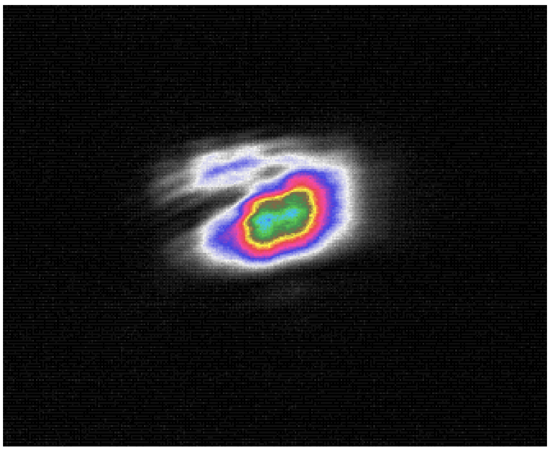


Figure 2.18: *Final mode profile obtained after the injection optimization (magnification about* $\times 10$*). The mode shape is a bit deformed, meaning that it is a superposition of fundamental LP modes. The image is cropped.*

2.2.4 ODT loading efficiency calculation

To have a theoretically calculated reference of the capture efficiency of the ODT, a small model has been developed in two regimes: shallow trap, in which the trap depth is assumed much smaller than the thermal energy, and the non-shallow trap in which the trap depth is assumed comparable to the thermal energy.

2.2.4.1 Model description

The ODT beam is modelled as a Gaussian beam propagating along the z-axis, its local intensity distribution as a function of the beam total power P is given by:

$$I(x, y, z) = \frac{2P}{\pi w(z)^2} \exp\left[-2\frac{x^2 + y^2}{w(z)^2}\right], \tag{2.9}$$

where the beam radius $w(z)$ evolves along the propagation axis according to:

$$w(z) = w_0\sqrt{1 + \left(\frac{z}{z_R}\right)^2}. \tag{2.10}$$

Using (2.9) in (1.55), the ODT potential energy can be expressed as

$$U_{\text{dip}}(\mathbf{r}) = \alpha\frac{2P}{\pi w(z)^2} \exp\left[-2\frac{x^2 + y^2}{w(z)^2}\right], \tag{2.11}$$

where α represents the real part of the polarizability expressed in units of $1/(2\epsilon_0 c)$:

$$\alpha = -\frac{1}{2\epsilon_0 c}\text{Re}[\alpha'], \tag{2.12}$$

with α' being the polarizability defined as in Section 1.3.
The maximum trap depth U_0 is located at the beam waist w_0 position, which is assumed to be located at the fiber tip ($z = 0$), and it is given by:

$$U_0 = \alpha I_{\text{max}}\,, \tag{2.13}$$

where $I_{\text{max}} = 2P/(\pi w_0^2)$ is the maximum intensity.
Following the molasses cooling stage, the atomic cloud is assumed in thermodynamic equilibrium, therefore, the phase space distribution is described by the spatial profile ($n_{\text{MOL}}(\mathbf{r})$) of the atomic cloud coupled with the Maxwell-Boltzmann velocity distribution:

$$f(\mathbf{r}, \mathbf{v}) = n_{\text{MOL}}(\mathbf{r})\left(\frac{M}{2\pi k_B T}\right)^{3/2} \exp\left[-\frac{Mv^2}{2k_B T}\right], \tag{2.14}$$

where M is the mass of a ^{87}Rb atom, T is the sample temperature, and the total atom number is normalized via

$$N_{\text{MOL}} = \int d^3r d^3v f(\mathbf{r}, \mathbf{v}). \tag{2.15}$$

An atom positioned at the spatial coordinate $\mathbf{r}$ is successfully trapped by the ODT if its transverse kinetic energy (in the x-y plane) is lower than the local depth of the trap:

$$v_x^2 + v_y^2 < \frac{2|U_{\text{dip}}(\mathbf{r})|}{M}. \tag{2.16}$$

For the v_z velocity component, it is assumed that all the atoms having $v_z < 0$ are captured, eventually reaching the fiber tip.
Therefore, the total number of captured atoms N_{ODT} is evaluated by integrating over the phase-space volume of trapping success:

$$N_{\text{ODT}} = \int d^3r\, n_{\text{MOL}}(\mathbf{r}) \times \frac{1}{2} \iint_{v_x^2+v_y^2<\frac{2|U_{\text{dip}}|}{M}} dv_x dv_y \left(\frac{M}{2\pi k_B T}\right) \exp\left[-\frac{M(v_x^2+v_y^2)}{2k_B T}\right], \tag{2.17}$$

where the factor 1/2 is introduced by the trapping criterion on the longitudinal component v_z of the velocity.
Passing to polar coordinates in the transverse velocity plane $v_r^2 = v_x^2 + v_y^2$, $dv_x dv_y = \pi d(v_r^2)$ and defining the dimensionless variable $x = (Mv_r^2)/(2k_BT)$, the velocity integral evaluates directly:

$$\frac{1}{2}\int_0^{\frac{|U_{\text{dip}}(\mathbf{r})|}{k_BT}} dx\, e^{-x} = \frac{1}{2}\left(1 - \exp\left[-\frac{|U_{\text{dip}}(\mathbf{r})|}{k_BT}\right]\right). \tag{2.18}$$

Consequently, the general expression for the ODT capture efficiency can be expressed as:

$$\frac{N_{\text{ODT}}}{N_{\text{MOL}}} = \frac{1}{2N_{\text{MOL}}} \int d^3r\, n_{\text{MOL}}(\mathbf{r}) \left(1 - \exp\left[-\frac{|U_{\text{dip}}(\mathbf{r})|}{k_BT}\right]\right). \tag{2.19}$$

2.2.4.2 Shallow trap approximation

In the limit where the thermal energy of the atomic cloud is significantly greater than the local optical potential in the overlapping region ($|U_{\text{dip}}(\mathbf{r})| \ll k_BT$), the exponential term can be linearized to first order:

$$1 - \exp\left(-\frac{|U_{\text{dip}}(\mathbf{r})|}{k_BT}\right) \simeq \frac{|U_{\text{dip}}(\mathbf{r})|}{k_BT}, \tag{2.20}$$

Equation (2.19) simplifies to:

$$N_{\text{ODT}} \simeq \int d^3r\, n_{\text{MOL}}(\mathbf{r}) \frac{|U_{\text{dip}}(\mathbf{r})|}{2k_BT} = \frac{U_0}{2k_BT} \int d^3r\, n_{\text{MOL}}(\mathbf{r}) \frac{w_0^2}{w(z)^2} \exp\left[-2\frac{x^2+y^2}{w(z)^2}\right]. \tag{2.21}$$

Since the transverse waist of the dipole beam ($w(z) \lesssim 100\ \mu$m) is significantly smaller than the transverse radii of the atomic cloud ($R_x, R_y \sim 1\,$mm), the atomic density can be considered constant over the beam profile, and the integration along z can be decoupled from the x and y integration, obtaining:

$$\iint dxdy \, \exp\left[-2\frac{x^2+y^2}{w(z)^2}\right] = \frac{\pi w(z)^2}{2}. \tag{2.22}$$

Substituting this result in equation (2.21), the terms $w(z)^2$ cancel out, eliminating the explicit dependence on the beam divergence:

$$N_{\mathrm{ODT}} \simeq \frac{U_0}{2k_BT}\frac{\pi w_0^2}{2}\int dz \, n_{\mathrm{MOL}}(0,0,z). \tag{2.23}$$

Adopting a uniform-density ellipsoid model to represent the cloud, characterized by radii R_x, R_y, R_z and a constant core density

$$n_{\mathrm{MOL}} = \frac{3N_{\mathrm{MOL}}}{4\pi R_x R_y R_z}, \tag{2.24}$$

integration along the longitudinal axis over the ellipsoid boundaries ($\pm R_z$) yields the ODT capture efficiency:

$$\frac{N_{\mathrm{ODT}}}{N_{\mathrm{MOL}}} \simeq \frac{3}{8}\frac{U_0}{k_BT}\frac{w_0^2}{R_xR_y}. \tag{2.25}$$

In this regime, the loading efficiency is independent of the absolute distance between the cloud center and the fiber tip (d), and independent of the minimum waist w_0, since the product $U_0w_0^2 = 2\alpha P/\pi$ depends solely on the total laser power. The primary experimental figure of merit to optimize is therefore the ratio $N_{\mathrm{MOL}}/(T \cdot R_xR_y)$.

2.2.4.3 Non-shallow trap

When the local trap depth becomes comparable to or greater than the kinetic energy of the atoms, the linear truncation introduces an overestimation. In this case the full Taylor series is needed:

$$1-\exp\left[-\frac{|U_{\mathrm{dip}}(\mathbf{r})|}{k_BT}\right] = \sum_{q=1}^{\infty}\frac{(-1)^{q+1}}{q!}\left[\frac{U_0}{k_BT}\frac{w_0^2}{w(z)^2}\exp\left(-2\frac{x^2+y^2}{w(z)^2}\right)\right]^q. \tag{2.26}$$

Under the same assumption of a highly localized beam relative to the cloud size ($w(z)/\sqrt{q} \ll R_x, R_y$), spatial integration over the transverse plane for the q-th order term yields:

$$\iint dxdy \, \exp\left[-2q\frac{x^2+y^2}{w(z)^2}\right] = \frac{\pi w(z)^2}{2q}. \tag{2.27}$$

Inserting this back into the volume integral yields the following series representation:

$$N_{\mathrm{ODT}} = \frac{1}{2}\sum_{q=1}^{\infty}\frac{(-1)^{q+1}\pi}{2q(q!)}\left(\frac{U_0}{k_BT}\right)^q w_0^2 \int dz\, n_{\mathrm{MOL}}(0,0,z)\left(\frac{w_0^2}{w(z)^2}\right)^{q-1}. \tag{2.28}$$

Using $w(z)^2/w_0^2 = 1 + z^2/z_R^2$, assuming constant density, and assuming the center of the atomic cloud is positioned at an axial displacement $z = d$ from the fiber tip, the longitudinal integral becomes:

$$\frac{3N_{\mathrm{MOL}}}{4\pi R_xR_yR_z}\int_{d-R_z}^{d+R_z} dz\left(1+\frac{z^2}{z_R^2}\right)^{1-q}. \tag{2.29}$$

Under the physical condition where the longitudinal radius of the cloud is significantly narrower than the Rayleigh range ($R_z \ll z_R$), the integrand remains approximately constant across the integration volume and can be evaluated at the cloud center $z = d$. This approximation leads to the result

$$\frac{3N_{\mathrm{MOL}}}{4\pi R_xR_yR_z}\int_{d-R_z}^{d+R_z} dz\left(1+\frac{z^2}{z_R^2}\right)^{1-q} = \frac{3N_{\mathrm{MOL}}}{2\pi R_xR_y}(1+d^2/z_R^2)^{1-q}. \tag{2.30}$$

Using (2.30) in (2.28), by defining the dimensionless parameter ξ, which evaluates the ratio of the trap depth at the cloud center to the thermal energy:

$$\xi = \frac{U_0}{k_BT\left(1+\frac{d^2}{z_R^2}\right)}, \tag{2.31}$$

the total captured atom fraction becomes:

$$\frac{N_{\mathrm{ODT}}}{N_{\mathrm{MOL}}} = \frac{3}{8}\frac{w_0^2}{R_xR_y}\left(1+\frac{d^2}{z_R^2}\right)\left[-\sum_{q=1}^{\infty}\frac{(-\xi)^q}{q\cdot q!}\right]. \tag{2.32}$$

The infinite series inside the brackets evaluates to (see Power Series section in [33])

$$-\sum_{q=1}^{\infty}\frac{(-\xi)^q}{q\cdot q!} = \gamma_E + \log(\xi) + \Gamma(0,\xi), \tag{2.33}$$

where γ_E is the Euler–Mascheroni constant. The final analytical solution for the non-shallow ODT capture efficiency is

$$\frac{N_{\mathrm{ODT}}}{N_{\mathrm{MOL}}} = \frac{3}{8}\frac{w_0^2}{R_xR_y}\left(1+\frac{d^2}{z_R^2}\right)\Big[\gamma_E + \log(\xi) + \Gamma(0,\xi)\Big]. \tag{2.34}$$

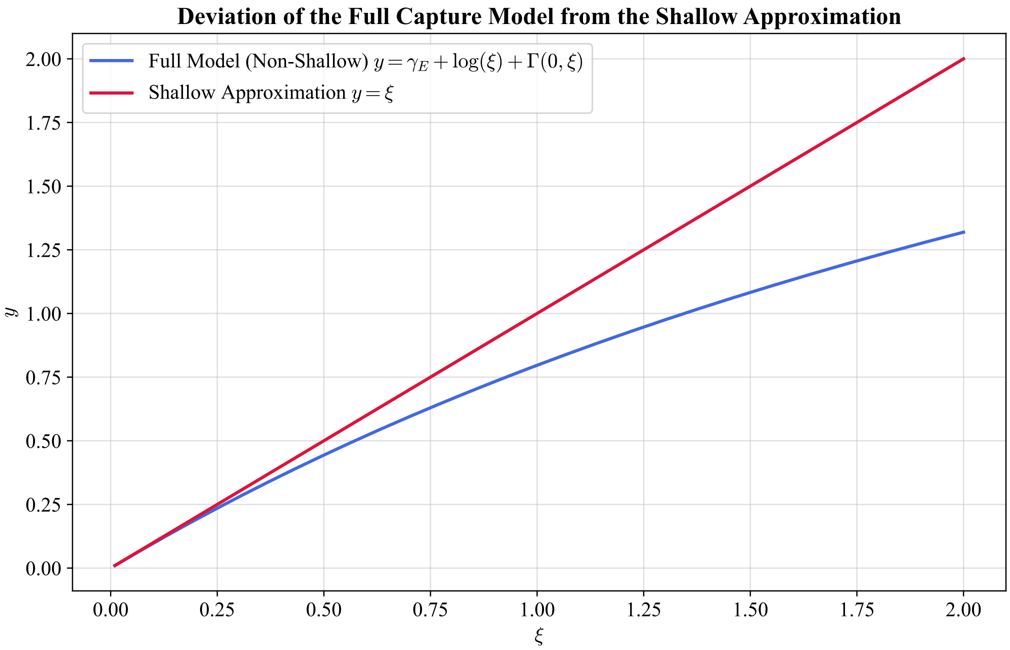


Figure 2.19: *Comparison between the relative ODT capture efficiency calculated using the shallow $y = \xi$ (red line) and non-shallow $y = \gamma_E + \log(\xi) + \Gamma(0, \xi)$ (blue line) models.*

The shallow trap result (2.25) is recovered by the substitution:

$$\gamma_E + \log(\xi) + \Gamma(0, \xi) \to \xi, \tag{2.35}$$

corresponding also to the limit $\xi \to 0$. The comparison between the models is shown in Figure 2.19.

2.3 Experimental Procedures and analysis methodology

The first step of the experimental sequence is the capture of atoms in the MOT, using the optimal parameters previously determined in for this apparatus [34]. The MOT is realized with an axial magnetic field gradient of $B' = 9.32$ G/cm (corresponding to a current of 8 A), a cooler laser detuning of $\Delta = -13$ MHz, and all compensation coil currents set to 0 mA. The MOT is loaded for a total duration of 5 s.
After the loading phase, the magnetic field is switched off, initiating the optical molasses phase. In the absence of the magnetic field, the atomic temperature decreases, however the confinement force is lost, and the atomic cloud begins to expand. At the same time, under the influence of gravity, the cloud starts to fall while experiencing a viscous-like damping force due to its interaction with the laser light. The temperature achieved during the molasses phase scales as $I/|\Delta|$, where I is the laser intensity and Δ is the detuning [25]. However, this relation remains valid as long as $|\Delta|$ is sufficiently small to not reduce significantly the optical cycle rate $\Gamma_{\mathrm{p}} \propto 1/\Delta^2$. Additionally, the detuning must not be so large to induce a significant excitation probability of the neighboring $F = 2 \rightarrow F' = 2$ transition. Furthermore, optimal molasses performance is obtained when the magnetic field is effectively zero. For this reason 5 ms after the MOT magnetic field is switched off, the shim coil currents are adjusted to compensate for the Earth's magnetic field and any residual magnetic fields present in the experimental environment. The molasses is performed with a detuning different from that used for the MOT phase, it is changed with a ramp of 20 steps with a total duration of 5 ms. The ramp starts 50 μs after the shim coils are changed to their molasses value.

2.3.1 Atom Imaging

Two different CMOS cameras were used to image the atoms inside the vacuum chamber:

- **Teledyne FLIR BFS-U3-04S2M-CS**, monochromatic camera mounting the Sony IMX287 sensor, with a resolution of 720×540, and a pixel size of 6.9 μm $\times$ 6.9 μm.

- **Allied Vision Manta G-419 NIR**, monochromatic camera mounting the CMOSIS/ams CMV4000 CMOS sensor, with a resolution of 2048×2048, and a pixel size of 5.5 μm $\times$ 5.5 μm.

The FLIR camera was primarily used during the optimization of the molasses phase, whereas, the Manta camera was employed for measurements of the atom capture efficiency of the ODT. The Manta was preferred with respect to the FLIR because of its higher

sensitivity at 780 nm and its higher pixel density, which helped the imaging of the small region where the ODT is present.
Both cameras were equipped with a Thorlabs MVL50M23 adjustable objective lens with a focal length of 50 mm, and a minimum working distance of 200 mm, corresponding to a minimum magnification of ~ 0.33. The objective was adjusted to focus on the HCPCF inside the vacuum chamber, ensuring that the atomic cloud, located at a similar distance, was also in focus. Both cameras were externally triggered by signals sent by the FPGA control system.
Since the temperature determination of the atomic cloud requires knowledge of its physical dimensions (see Section 2.3.2), the FLIR camera had to be calibrated.
The first calibration method was based on the measure of the diameter in pixel of the fiber inside the vacuum chamber. By turning on the MOT beam, the fiber gets illuminated, and a picture is taken. Using this method the fiber diameter of ~ 316 μm was found to correspond to 8 pixels. This yielded a calibration factor of 39.5 μm/px, corresponding to a magnification of approximately 0.17. However, the measurement uncertainty was significant: assuming an uncertainty of one pixel on each edge of the fiber image, the measured diameter becomes (8 ± 2) px, resulting in a calibration factor of (40 ± 10) μm/px. Consequently, this method provides only a rough estimate of the imaging magnification.
A second and more accurate calibration method was therefore employed. A mirror was positioned in front of the camera without modifying the objective setting, and a ruler with a visible scale was positioned so that it was in focus. This ensured that the ruler was located at the same distance from the objective as the fiber inside the vacuum chamber. With this method 2 cm corresponded to (471 ± 2) px. Since the instrumental uncertainty associated with a single measurement was negligible, the calibration was performed by taking five independent measurement, with the ruler repositioned at focus each time. This procedure yielded a final calibration factor of (43.4 ± 0.3) μm/px, corresponding to a magnification of 0.16.

2.3.2 Temperature measurements

Temperature measurements were performed to optimize the molasses parameters, with the goal of achieving the lowest possible temperature.
The temperature of the atomic sample was determined by measuring its ballistic expansion during free fall. Assuming a non-interacting gas and considering a single spatial coordinate, if an atom is found at x_0 at $t = 0$, after a free expansion time t its position is given by

$$x(t) = x_0 + v_x t. \tag{2.36}$$

Taking into account the distributions of both positions and velocities, and assuming that these quantities are statistically independent, the variance of their sum is equal to the sum of their variances. Therefore,

$$\sigma_x^2(t) = \sigma_{x,0}^2 + \sigma_{v_x}^2 t^2, \tag{2.37}$$

where $\sigma_{x,0}$ is the standard deviation of the position distribution at $t = 0$.
According to classical energy equipartition theorem, the temperature associated with a given spatial direction is related to the squared mean velocity along that direction:

$$\frac{1}{2} m \langle v_x^2 \rangle = \frac{1}{2} k_B T_x, \tag{2.38}$$

where m is the atomic mass and k_B is the Boltzmann constant.
In the frame of reference of the cloud center of mass, the atomic cloud is globally at rest, thus $\langle v_x \rangle = 0$ and

$$\langle v_x^2 \rangle = \sigma_{v_x}^2 = \frac{k_B T_x}{m}. \tag{2.39}$$

Substituting this result into 2.37 one gets:

$$\sigma_x^2(t) = \sigma_{x,0}^2 + \frac{k_B T_x}{m} t^2. \tag{2.40}$$

Using the atomic mass of the ^{87}Rb

$$m = m_A(^{87}R) = 86.9 \text{ u} \approx 1.443 \times 10^{-25} \text{ kg}, \tag{2.41}$$

and acquiring images at different times of flight (TOF), equation (2.40) can be fitted obtaining T_x as a fitting parameter.
The complete acquisition and analysis procedure is summarized below:

1. First a MOT is loaded, followed by the molasses phase. For the molasses different parameters and durations were tested in order to find the best configuration that minimized the temperature (see Section 3.1)

2. After the molasses stage, all cooling beams are switched off, allowing the atomic cloud to expand and fall freely. Using the FLIR camera images are taken at different TOFs, obtaining the intensity distribution $I(TOF)$. For each TOF a corresponding background image $I_{\text{bkg}}(TOF)$ is also acquired. The background image is taken by waiting a sufficiently long time (1 s) ensuring that the cloud has completely disappeared and only the background signal remains.

   It should be noted that the measurements are destructive since the flash heats the sample. Therefore, it is not possible to take multiple TOF images during the same experimental run, therefore each image requires a new MOT loading and molasses sequence.

3. The background is subtracted

$$I_{\text{clean}}(TOF) = I(TOF) - I_{\text{bkg}}(TOF), \tag{2.42}$$

and the associated Poisson uncertainty is calculated as

$$\sigma_{\text{N}} = \sqrt{I(TOF) + I_{\text{bkg}}(TOF)}. \tag{2.43}$$

4. A two-dimensional Gaussian is fitted to each image (TRF minimization algorithm) to extract the cloud widths σ_x and σ_y along the two principal axis.

5. Equation (2.40) is fitted independently for both directions, obtaining the temperatures T_x and T_y. Then, the final temperature T is calculated as their average.

2.3.3 ODT loading measurements

In the analysis of the ODT, two main figures of merits are taking into account, namely the loading efficiency η and the intensity integral of the ODT peak, that is directly proportional to the number of captured atoms. They are measured by comparing fluorescence images acquired with the ODT either present or absent during the MOT and molasses stages. Since only a small fraction (order of 1/1000) of the atoms is expected to be captured by the ODT, the analysis relies on differential imaging and careful background normalization to isolate the signal of the trapped atoms from fluctuations in the total atom number. The complete analysis procedure is summarized below.

1. Two identical experimental sequences of MOT and molasses are consecutively performed with the same parameters. In one sequence, the ODT is kept on throughout the entire cycle starting from the MOT loading phase, whereas in the other the ODT remains off. After a TOF of 30 ms, fluorescence images of the atomic distribution denoted by I_{on} and I_{off} are taken with the Manta camera. During the TOF the atom not trapped in the ODT expands enhancing the visibility of those confined in the ODT. Each pair of measurements is repeated five times

2. These images are cropped to a region of interest of 530×380 pixels centered on the atomic cloud. This removes the region where the HCPCF is visible as well as peripheral areas containing only background signal.

3. To improve the signal-to-noise ratio, the images corresponding to each experimental condition are averaged:

$$\bar{I}_{\text{on}} = \langle I_{\text{on}} \rangle \qquad \bar{I}_{\text{off}} = \langle I_{\text{off}} \rangle \,.$$

This is important because the number of atom loaded into the ODT is expected to be small.

4. The averaged images are converted into one dimensional density profiles by integrating along the columns. This further increases the signal-to-noise ratio.

5. The total number of atoms in the molasses exhibits slow fluctuations even when the experimental parameters are unchanged. These variations are attributed to slow drifts in the chamber pressure and to thermal effects in optical components, such as the AOMs. Since the ODT signal is small, due to the correlated nature of the drifts, despite the average of different images the difference would result in a background different from zero that is comparable to the ODT signal amplitude. To compensate this effect, a rescaling procedure is applied to the background images. This method computes a scale factor, k, which enforces a zero integrated difference outside the ODT region.

   The procedure is implemented as follows:

   i. An exclusion region centered on the ODT position, $x_{\mathrm{ODT}} \pm R_{\mathrm{window}}$ is defined. All remaining positions constitute the outside region.

   ii. A linear background $L_i(x)$ is estimated by interpolating the edges of each intensity profile. After subtracting the background, the integrated signal outside the ODT region is computed for both the ODT-on and ODT-off profiles ($A_{\mathrm{on}}^{\mathrm{out}}$ and $A_{\mathrm{off}}^{\mathrm{out}}$) as

$$A_i^{\mathrm{out}} = \int_{\mathrm{outside}} \left(\bar{I}_i(x) - L_i(x)\right) dx \qquad i \in \{\mathrm{on}, \mathrm{off}\} \tag{2.44}$$

   iii. The normalization factor is given by the ratio of these integrated areas:

$$k = \frac{A_{\mathrm{on}}^{\mathrm{out}}}{A_{\mathrm{off}}^{\mathrm{out}}} \tag{2.45}$$

6. The final rescaled profile, which isolates the signal of the atoms trapped in the ODT, is obtained as:

$$\Delta\bar{I}(x) = \bar{I}_{\mathrm{on}}(x) - k \cdot \bar{I}_{\mathrm{off}}(x) \tag{2.46}$$

7. Two independent fits consisting of a Gaussian function and a linear background are performed. The first fit is applied to the ODT-off profile $\bar{I}_{\mathrm{off}}(x)$, while the second is applied to the differential profile $\Delta\bar{I}(x)$.
   The areas $A_{\mathrm{off}}^{\mathrm{Gaus}}$ and $A_{\mathrm{ODT}}^{\mathrm{Gaus}}$ are calculated from the fit results, and the capture efficiency is calculated as

$$\eta = \frac{A_{\mathrm{ODT}}^{\mathrm{Gaus}}}{k \times A_{\mathrm{off}}^{\mathrm{Gaus}}} \tag{2.47}$$

Chapter 3

Result And Discussion

Following the experimental and analysis procedures explained in Chapter 2, the ODT capture efficiency was characterized and optimized. This chapter presents the corresponding experimental results.
The first stage of the optimization process consists of minimizing the molasses temperature through the adjustment of the cooling laser detuning, the cooling laser intensity, and the compensation magnetic field.
Subsequently, the ODT loading is optimized by exploiting the compensation coils to displace the MOT after loading. This optimization aims to maximize the spatial overlap between the atomic cloud and the ODT beam, thereby enhancing the capture efficiency.

3.1 Molasses temperature optimization

Before performing any optimization, the six cooling beams were realigned to minimize their overlap mismatch at the MOT position. This procedure is crucial since thermal drifts can induce misalignments in the optical components over time.
Preliminary values for the compensation coil currents were obtained prior to the realignment:

$$\begin{cases} I_{\mathrm{x}} &= 80 \text{ mA}, \\ I_{\mathrm{y}} &= -200 \text{ mA}, \\ I_{\mathrm{z}} &= -400 \text{ mA}. \end{cases} \tag{3.1}$$

These values were used as the starting point for the subsequent optimization procedure.

3.1.1 Detuning frequency optimization

The first optimization step consisted of measuring the molasses temperature for different values of the cooling laser detuning. The detuning was varied from -24.4 MHz ($4.0\ \Gamma/2\pi$) to -184 MHz ($30.4\ \Gamma/2\pi$), with a step size of -11.4 MHz.

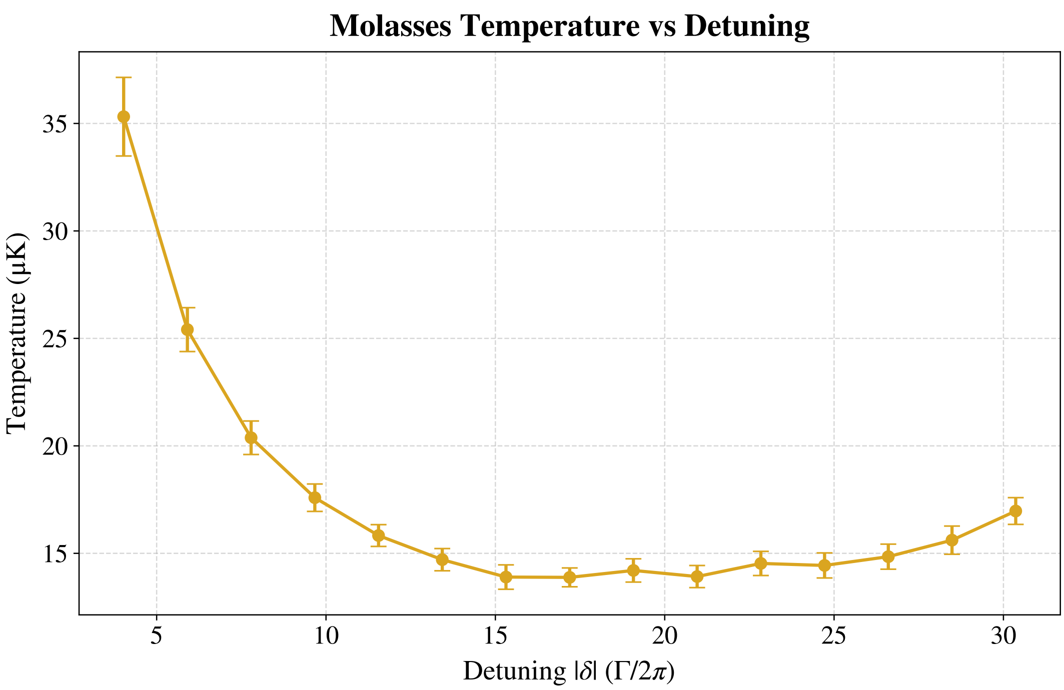


Figure 3.1: *Optical molasses temperature as a function of laser detuning. The horizontal axis represents the absolute value of the applied red detuning* $|\delta|$*, normalized to the natural linewidth* $\Gamma/2\pi$*. The data exhibit optimal sub-Doppler cooling in the region* $|\delta| \approx 15 - 20$*, reaching a minimum temperature of* 13.9 ± 0.4 μK*. For detunings up to* $|\delta| \approx 15$*, the temperature follows the* $1/|\delta|$ *dependence expected for an ideal Sisyphus cooling. At larger detunings, the curve flattens and eventually begins to rise.*

The resulting average temperature is shown in Figure 3.1 as a function of the detuning expressed in units of $\Gamma/2\pi$. At small detunings, the data follow the expected $1/|\delta|$ dependence typical of the sub-Doppler cooling. As the detuning increases, two phenomena become relevant: a significant drop in the optical scattering rate $\Gamma_p \propto 1/\delta^2$ and the approach of the transition $F = 2 \rightarrow F' = 2$ located 267 MHz below the cooling transition $F = 2 \rightarrow F' = 3$ (see Figure 2.2). These effects cause the temperature curve to flatten in the region $|\delta| \approx 15 - 20$ and eventually to rise again for larger detunings.

The lowest average temperature was obtained for a normalized detuning of $|\delta| = 17.2$ corresponding to an absolute detuning of $|\delta| = 104.2$ MHz. Figure 3.2 displays the images obtained at various TOF of the falling atomic cloud for this optimal configuration, while Figure 3.3 shows the corresponding fit of equation (2.37). The resulting temperatures along the Cartesian axes are $T_x = (14.1 \pm 0.5)$ μK and $T_y = (13.5 \pm 0.7)$ μK. Their weighted average yields a final temperature of $T = (13.9 \pm 0.4)$ μK.

As observed in Figure 3.3, the data points at long TOF diverge appreciably from the linear trend, consistently falling below the expected values. This behavior is systematically observed across all measurements. The hypothesis is that at sufficiently long TOF the cloud radius becomes non-negligible compared to the probe beam radius, violating the assumption of uniform illumination. Consequently, the outer region of the cloud interacts

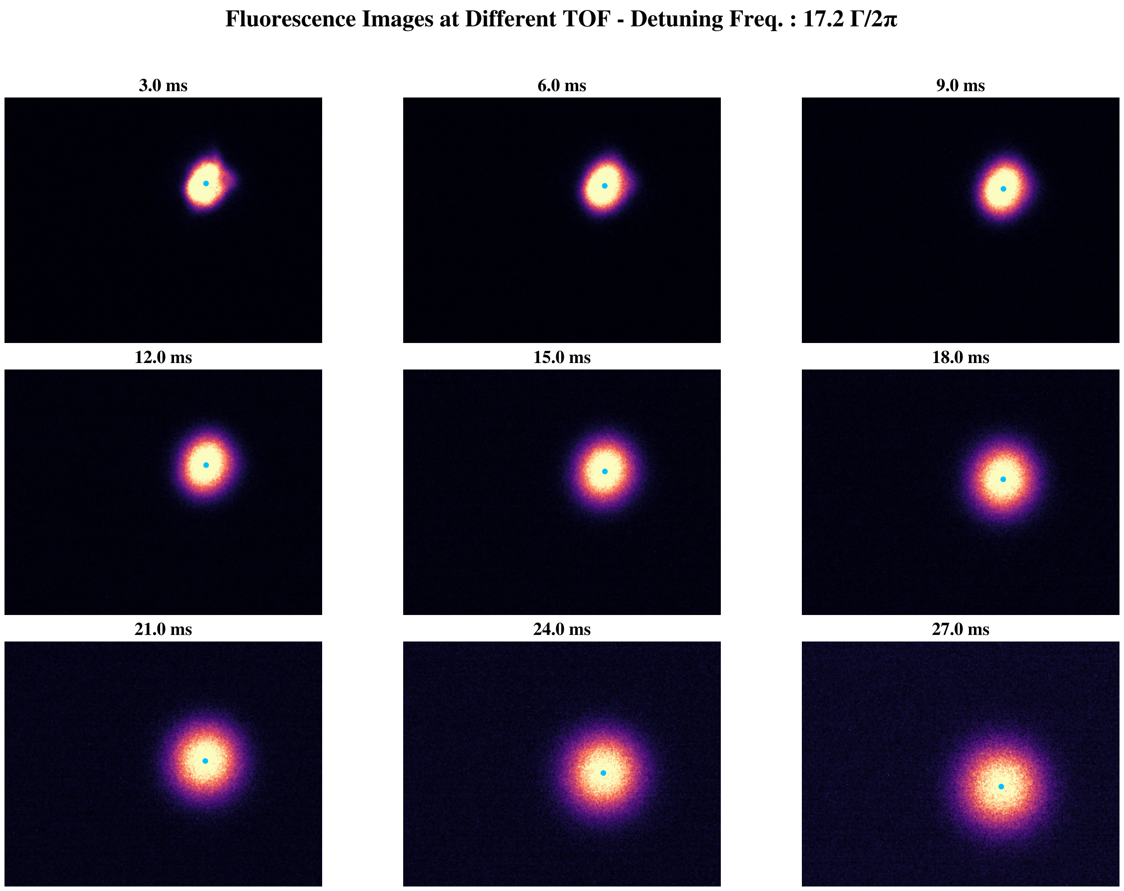


Figure 3.2: *Fluorescence images of the atomic cloud acquired during Time-of-Flight expansion for times ranging from 3.0 ms to 27.0 ms for* $|\delta| = 17.2\ \Gamma/2\pi$. *The blue markers indicate the center of the cloud obtained through the Gaussian fit.*

with a lower intensity field, causing the Gaussian fit to systematically underestimate the true variance. Fortunately, this effect is expected to become less prominent as the temperature is lowered.

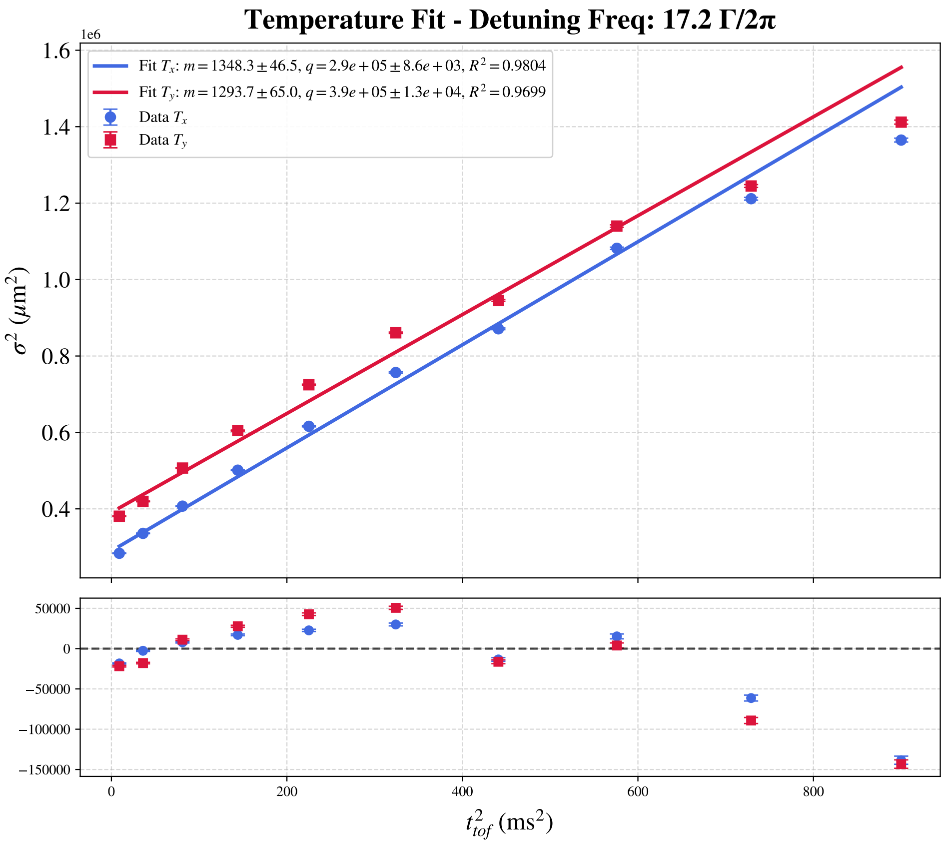


Figure 3.3: *Optical molasses temperature fit for* $|\delta| = 17.2\ \Gamma/2\pi$ *with residuals. The large residuals of the last three data points from the linear fit are attributed to the cloud expansion, which makes its radial size non-negligible compared to the probe beam profile.*

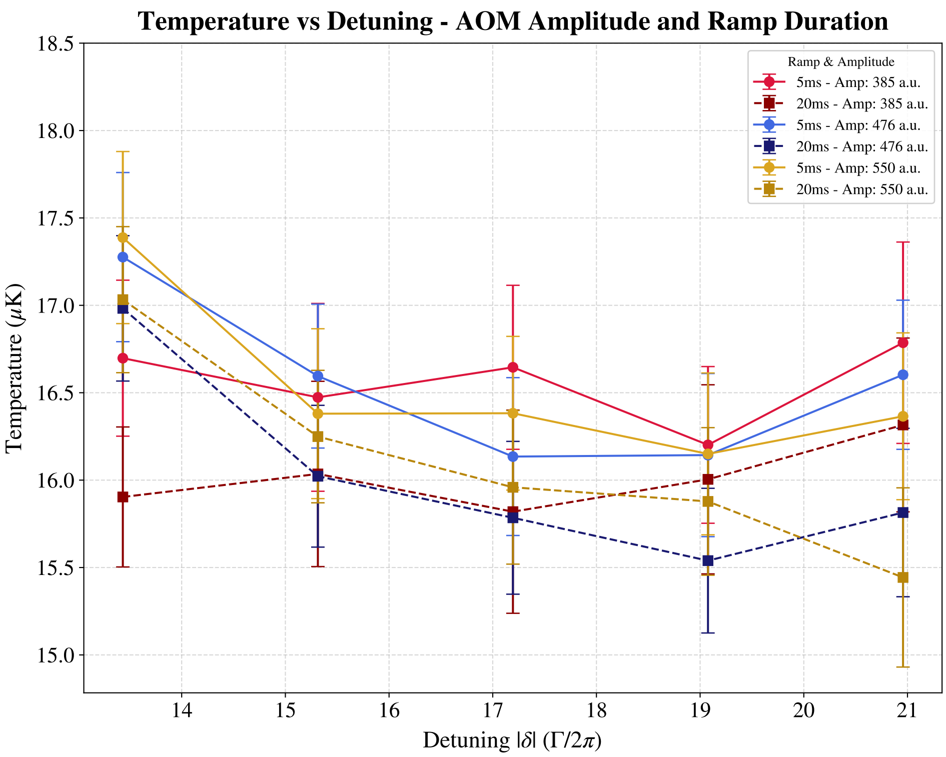


Figure 3.4: *Optical molasses temperature versus cooling detuning for different AOM amplitude and ramp durations. No clear optimal AOM amplitude is observed as the variations remain within the uncertainty. Temperatures corresponding to a ramp duration of 20 ms (dashed lines) are systematically lower than those obtained with a 5 ms ramp (solid lines).*

3.1.2 AOM amplitude optimization

As a second optimization step, the influence of the cooling beam power was investigated by varying the AOM drive amplitude and the duration of its transition ramp, which is applied at the beginning of the molasses phase. The AOM amplitude, expressed in arbitrary units (a.u.), controls the RF amplitude driving the AOM. Three values were tested: 385, 475 and 550 a.u., equivalent to 5.8 mW, 6.6 mW, 6.3 mW of cooling laser power measured after the collimator of the MOT z-beam. The MOT is realized at 550 a.u. Additionally, ramp durations of 5 ms and 20 ms were tested for the transition from the MOT value to the molasses value. These measurements were repeated for different values of the cooling detuning. The optimization results are shown in Figure 3.4. Although small variations in the expected values of the temperature can be observed for different values of the AOM amplitude, these differences fall within the experimental error bars. Therefore, no statistically significant dependence of the molasses temperature on the AOM amplitude was found, and no clear optimum could be identified within the tested range.

Conversely, as shown in Figure 3.4, the temperatures obtained with a 20 ms ramp are systematically lower than those obtained with a 5 ms ramp, yielding an average temperature reduction of approximately 0.5 μK. This improvement was relatively small, and was

considered sufficient for the scope of the experiment. Since further optimization on the order of a few tenths of a microkelvin was not required, no other ramp times were tested. Consequently, a ramp duration of 20 ms was selected for the subsequent measurements.
It is worth noting that during the ODT loading characterization, this optimization was found to introduce additional fluctuations into the system. For this reason, the subsequent optimization of the compensation coil currents was carried out using an AOM amplitude of 385 a.u. (corresponding to the lowest temperature measured at $|\delta| = 17.2\ \Gamma/2\pi$) with a ramp duration of 20 ms. However, these parameters were later readjusted during the ODT optimization. Nevertheless, the measurements presented in this section revealed that their influence on the molasses temperature is limited to a few tenths of a microkelvin, and therefore does not significantly affect the final achieved temperature.

3.1.3 Compensation field optimization

The final stage of the molasses optimization procedure consisted of adjusting the compensation magnetic field. The compensation coil currents are set to the molasses values 5 ms after the MOT magnetic field had been switched off.
The optimization was performed sequentially. First, the z-axis current was optimized while keeping the x- and y-axes currents fixed to their preliminary values (3.1). Then the x-axis current was optimized using the newly found optimal z-value. Finally, the y-axis current was optimized using the optimal currents for both the z- and x-axes. Quadratic fits were applied to the experimental data to extract the optimal current values and their corresponding minimum temperatures. The results are shown in Figures 3.5, 3.6 and 3.7 for the z-, x-, and y-axis coils respectively.
The resulting optimized currents, used in the subsequent ODT loading stage, are:

$$\begin{cases} I_x &= -15\ \text{mA}, \\ I_y &= -185\ \text{mA}, \\ I_z &= -230\ \text{mA}. \end{cases} \tag{3.2}$$

The corresponding generated magnetic field can be calculated using the calibration (2.1):

$$\begin{cases} B_x &= -21\ \text{mG}, \\ B_y &= -241\ \text{mG}, \\ B_z &= -345\ \text{mG}. \end{cases} \tag{3.3}$$

These parameters yielded a minimized molasses temperature of $T = (9.6 \pm 0.1)\ \mu$K. The z-axis compensation coil current deviated the most from its preliminary value, while the x-axis current underwent a smaller but still appreciable correction. In contrast, the y-axis coil current remained close to its initial value.

It should be noted that the operating current was set to $I_z = -230$ mA, rather than the true optimum of -223.2 mA identified in Figure 3.5. Given the proximity of both values to the vertex of the optimization parabola, this discrepancy introduces no significant systematic error into the measurements.

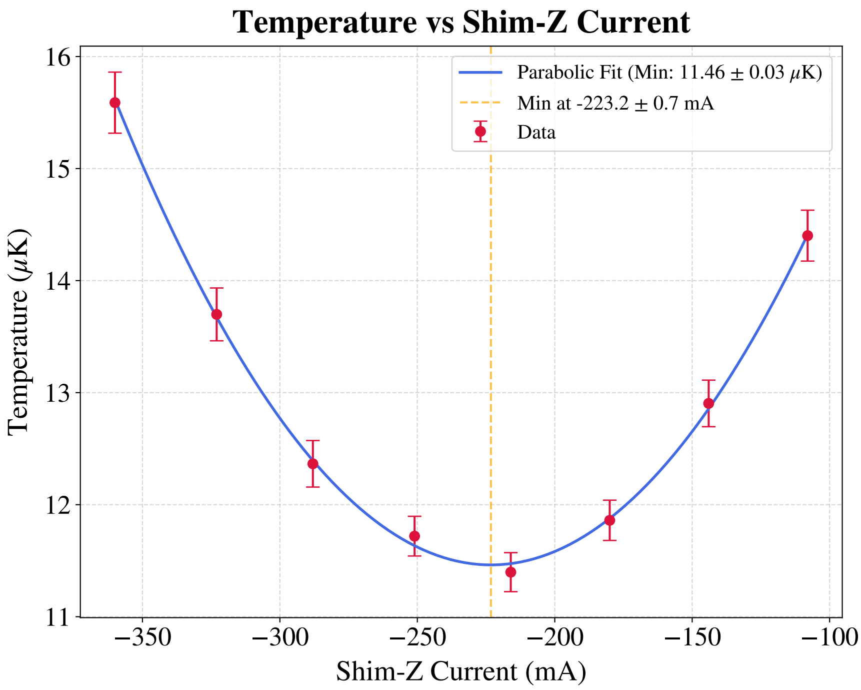


Figure 3.5: *Optical molasses temperature versus Shim-Z current. The parabolic fit yields a minimum temperature* $T = (11.46 \pm 0.03)\ \mu\mathrm{K}$ *at a current* $I_z = (-223.2 \pm 0.7)$ *mA.*

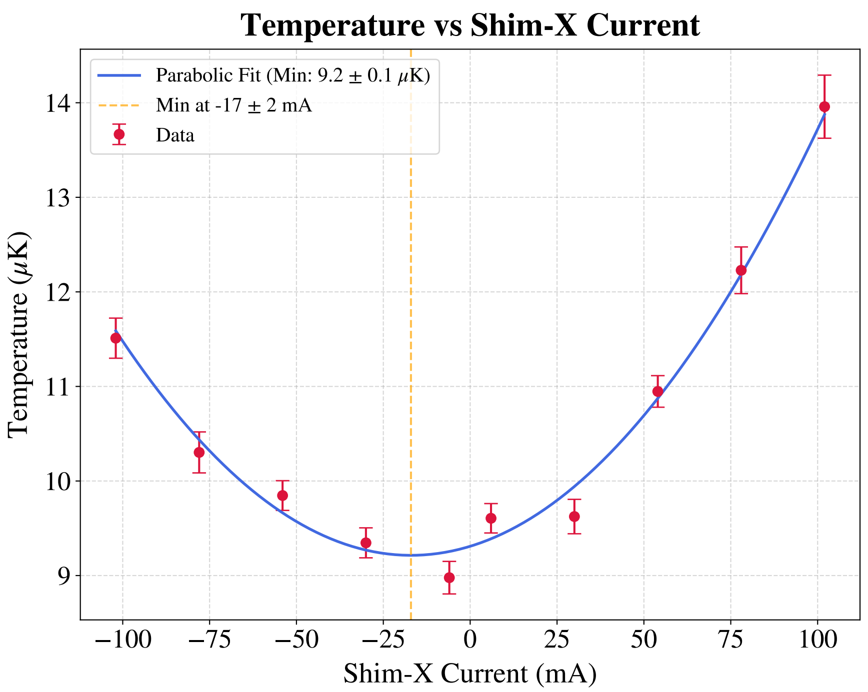


Figure 3.6: *Optical molasses temperature versus Shim-X current. The parabolic fit yields a minimum temperature* $T = (9.2 \pm 0.1)$ μK *at a current* $I_x = (-17 \pm 2)$ *mA.*

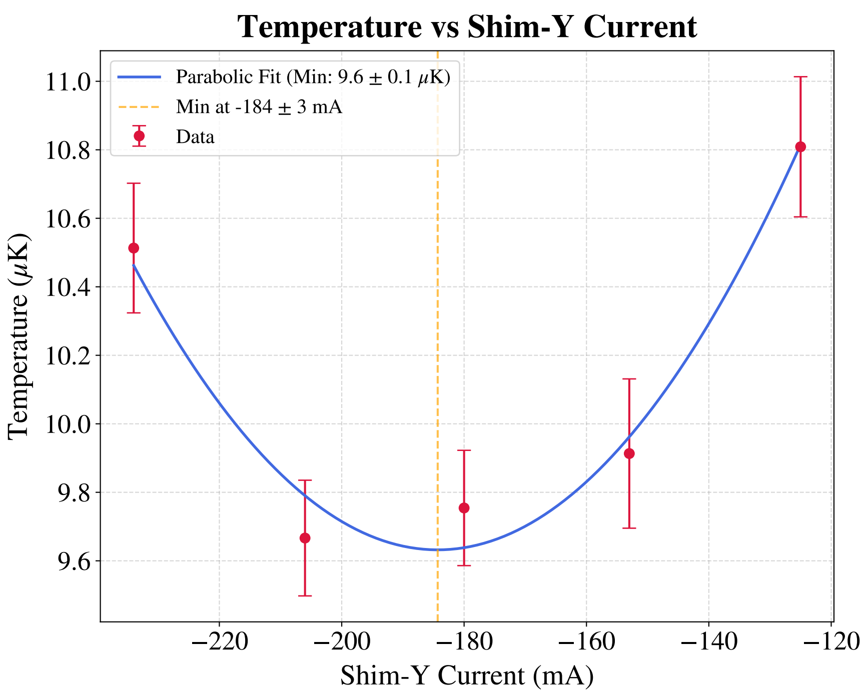


Figure 3.7: *Optical molasses temperature versus Shim-Y current. The parabolic fit yields a minimum temperature of* $T = (9.6 \pm 0.1)$ μK *at a current* $I_y = (-184 \pm 3)$ *mA.*

3.2 ODT loading optimization

The optimal parameters obtained for the molasses were used for the optimization of the ODT loading. This procedure primarily focused on tuning the compensation magnetic field. The underlying concept is to first load the MOT and then, prior to the molasses phase, exploit the compensation coils to spatially shift the MOT position.
The MOT forms at the minimum of the magneto-optical potential, which can be shifted by adjusting the compensation coil currents, thereby displacing the trapped atomic cloud. Moving the MOT away from the optimal intersection volume of the beams inevitably causes some atom loss and a temperature increase. However, this is a necessary trade-off to achieve a better spatial overlap with the ODT beam, which ultimately maximizes the number of atoms transferred into the dipole trap. As long as this spatial shift is small, the optimal molasses parameters derived in the previous sections remain effective at nullifying the residual magnetic field, requiring no further adjustments.
Specifically, after the initial MOT loading phase, the compensation coil currents are ramped to their ODT loading values over a duration of 200 ms, discretized in 20 steps. This smooth translation is crucial to minimize atom loss and heating during the transport process.
Similarly to the molasses phase, preliminary current values for the ODT loading, obtained prior to the optical realignment and the molasses optimization, were available as a starting point:

$$\begin{cases} I_x &= -200 \text{ mA} \\ I_y &= -350 \text{ mA} \\ I_z &= 50 \text{ mA}. \end{cases} \tag{3.4}$$

During the measurements, the ODT laser power prior to injection was measured to be 1.4 W. The transmission efficiency, measured at low injected power just before lens L3 (see the schematic in Figure 2.11), was approximately 42%, corresponding to an optical power of about 0.59 W. Because several optical components precede L3, the actual power at the fiber facet is likely higher. To account for this uncertainty, a conservative relative error of 10% was assigned, yielding an estimated power at the fiber facet of $P = 0.59 \pm 0.06$ W.

3.2.1 Imaging system optimization

Initial measurements were performed using the FLIR camera, employing the optimal parameters derived from the molasses temperature optimization and the preliminary compensation coil currents for the ODT loading. The data were analyzed to characterize the ODT capture efficiency using a procedure analogous to that described in Section 2.3.3, though without background rescaling. The resulting intensity difference images are shown

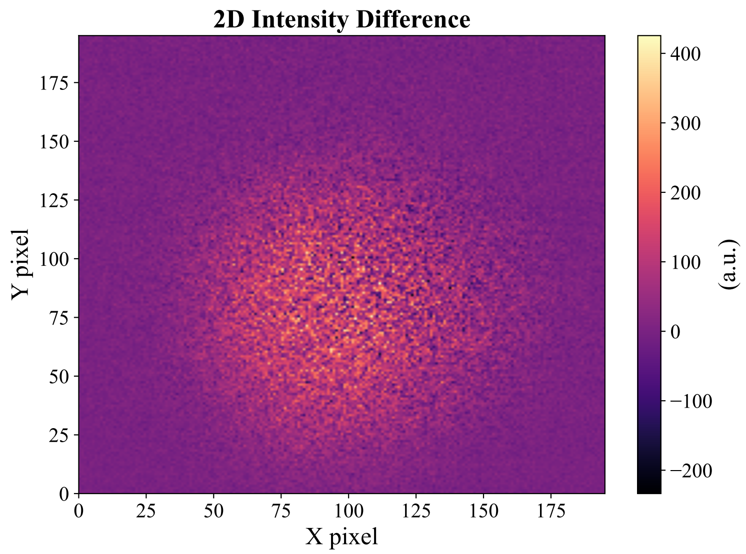


Figure 3.8: *Intensity difference 2D image, obtained with the FLIR camera and a 20 ms AOM amplitude ramp duration during the molasses phase. The intensity distribution does not show any brighter area that can be attributed to the presence of the ODT.*

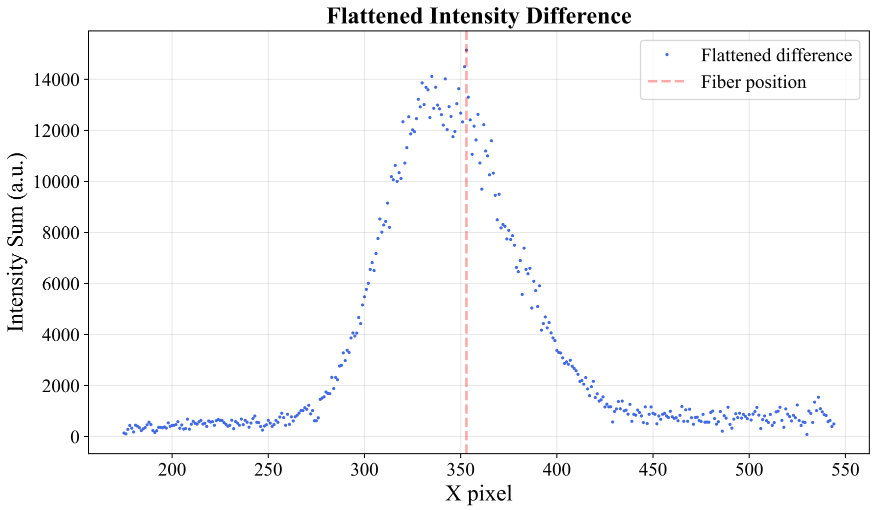


Figure 3.9: *Intensity difference 1D integrated profile, obtained with the FLIR camera and a 20 ms AOM amplitude ramp during the molasses phase. The red vertical line indicates the approximate position of the fiber center. A few higher-intensity pixels are recognizable corresponding to the ODT position, but the averaging and subtracting procedures are not effective in removing the background.*

in Figure 3.8 (2D image) and Figure 3.9 (1D integrated profile). Visual inspection of the 2D image does not reveal any distinct brighter region attributable to the atoms captured by the ODT. Conversely, the 1D integrated profile shows a localized intensity increase spanning ~ 4 pixels around the fiber position. Subsequent measurements revealed a similar behavior, confirming that this localized peak corresponds to the atoms trapped in the ODT. However, the signal amplitude was often too small to be reliably distinguished from the noise. Since the 1D integration is explicitly intended to enhance the signal-to-noise ratio (SNR), the persistence of a poor SNR highlighted a critical issue. Furthermore, the presence of a residual background pedestal significantly larger than the ODT signal indicates underlying oscillatory noise in the system that is not averaged by the acquisition of five images, suggesting either a large standard deviation or a correlated shot-to-shot drift. Beyond the noise, the magnitude of the ODT signal itself was a primary concern. Even far from an optimal atomic-cloud-ODT overlap, a higher initial capture was expected. Additionally, the limited spatial extent of the ODT signal (occupying only a few pixels) posed a severe analytical challenge. Such poor spatial resolution precludes reliable 2D Gaussian fitting, and extracting the signal area via direct pixel summation introduces systematic errors. In summary, the diagnostic capabilities were limited by two main factors: first, the residual background in the subtracted images overwhelmed the weak ODT signal; second, the small waist of the ODT beam resulted in an inadequate number of illuminated pixels for robust analysis.

To address the poor SNR, the molasses parameters were systematically investigated. It was found that the noise was strongly dependent on the AOM amplitude ramp during the molasses phase. In particular, reducing the ramp duration from 20 ms to 5 ms significantly improved the background subtraction and increased the SNR. Furthermore, to minimize intensity fluctuations induced by the AOM driver, the target amplitude was set to 480 a.u. (6.6 mW of cooling laser power measured on the z-beam). This value corresponds to the saturation regime of the AOM diffraction efficiency, which acts as a working point with a minimum derivative of the power over the amplitude of the RF driving signal ($dP/dA \approx 0$). Operating in this saturation region intrinsically suppresses amplitude noise, thereby stabilizing the actual optical power delivered during the molasses phase.

The results obtained with this optimized configuration are shown in Figures 3.10 (2D difference) and 3.11 (1D integrated profile). In the 2D plot, a distinct brighter strip representing the atoms captured in the ODT is now recognizable. By comparing the 1D integrated profiles in Figures 3.9 and 3.11, it is evident that the pedestal in the 5 ms ramp case is suppressed, showing a maximum amplitude and area approximately half those of the 20 ms ramp case. This reduction is sufficient to make the ODT signal emerge clearly from the residual pedestal and noise.

Despite the improved SNR, a critical issue remained: the insufficient spatial resolution provided by the FLIR camera setup. To overcome this limitation, subsequent measurements were performed using the Manta camera, which features a smaller pixel pitch. Additionally, the Manta camera was mounted closer to the experimental chamber to exploit the minimum working distance of the MVL50M23 objective, maximizing the optical magnification. Comparing the apparent fiber width measured by both cameras, it was deduced that the object size in pixels increased by a factor of ~ 2.25 with the Manta setup. This scaling factor arises from the combined contribution of the higher optical magnifi-

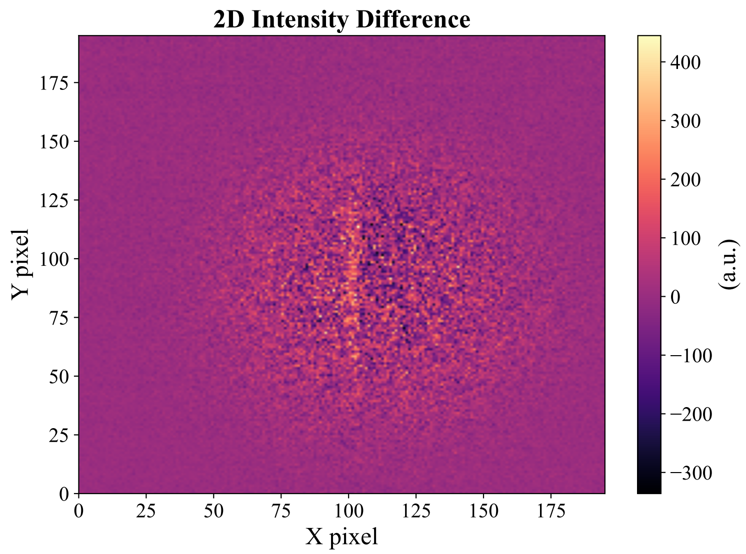


Figure 3.10: *Intensity difference 2D image, obtained with the FLIR camera and a 5 ms AOM amplitude ramp duration during the molasses phase. A brighter area corresponding to the ODT is clearly visible.*

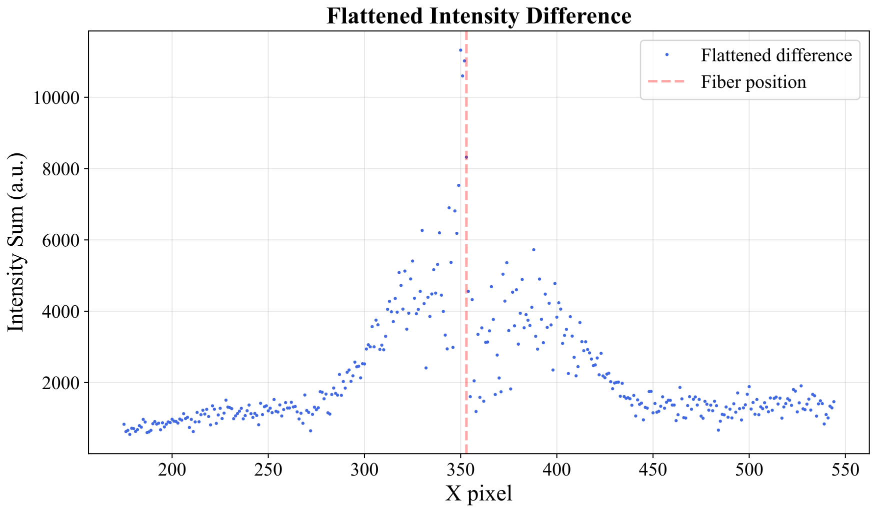


Figure 3.11: *Intensity difference 1D integrated profile, obtained with the FLIR camera and a 5 ms AOM amplitude ramp during the molasses phase. The red vertical line indicates the approximate position of the fiber center. With the 5 ms ramp, the averaging and the subtraction become more effective in separating the ODT signal from the background. However, the problem of the small number of pixels representing the ODT signal is still present.*

cation and the smaller pixel size, finally yielding a sufficiently resolved ODT signal for robust analysis, as shown in Figure 3.12.

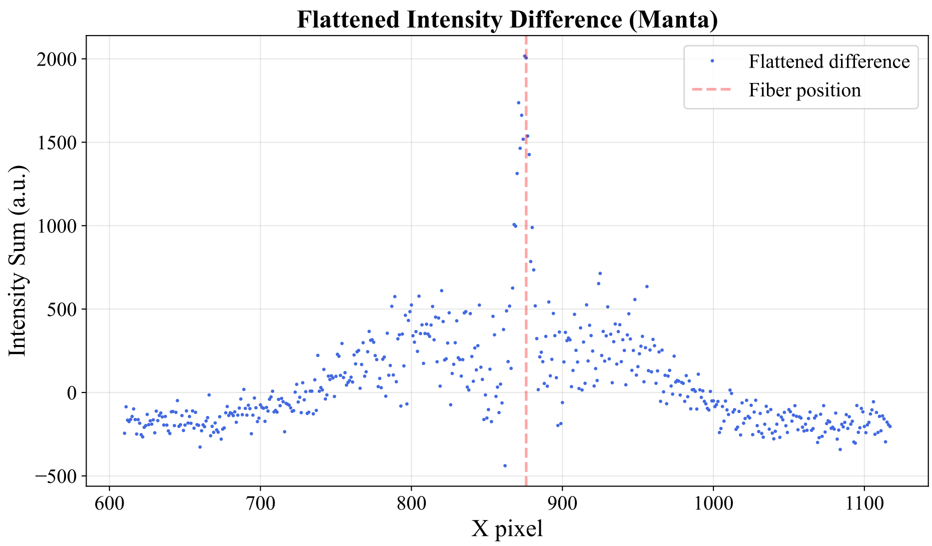


Figure 3.12: *Intensity difference 1D integrated profile, obtained with the Manta camera and a 5 ms AOM amplitude ramp during the molasses phase. The combination of the Manta's smaller pixel size and the closer mounting position effectively increased the spatial resolution of the ODT signal.*

3.2.2 Capture efficiency optimization

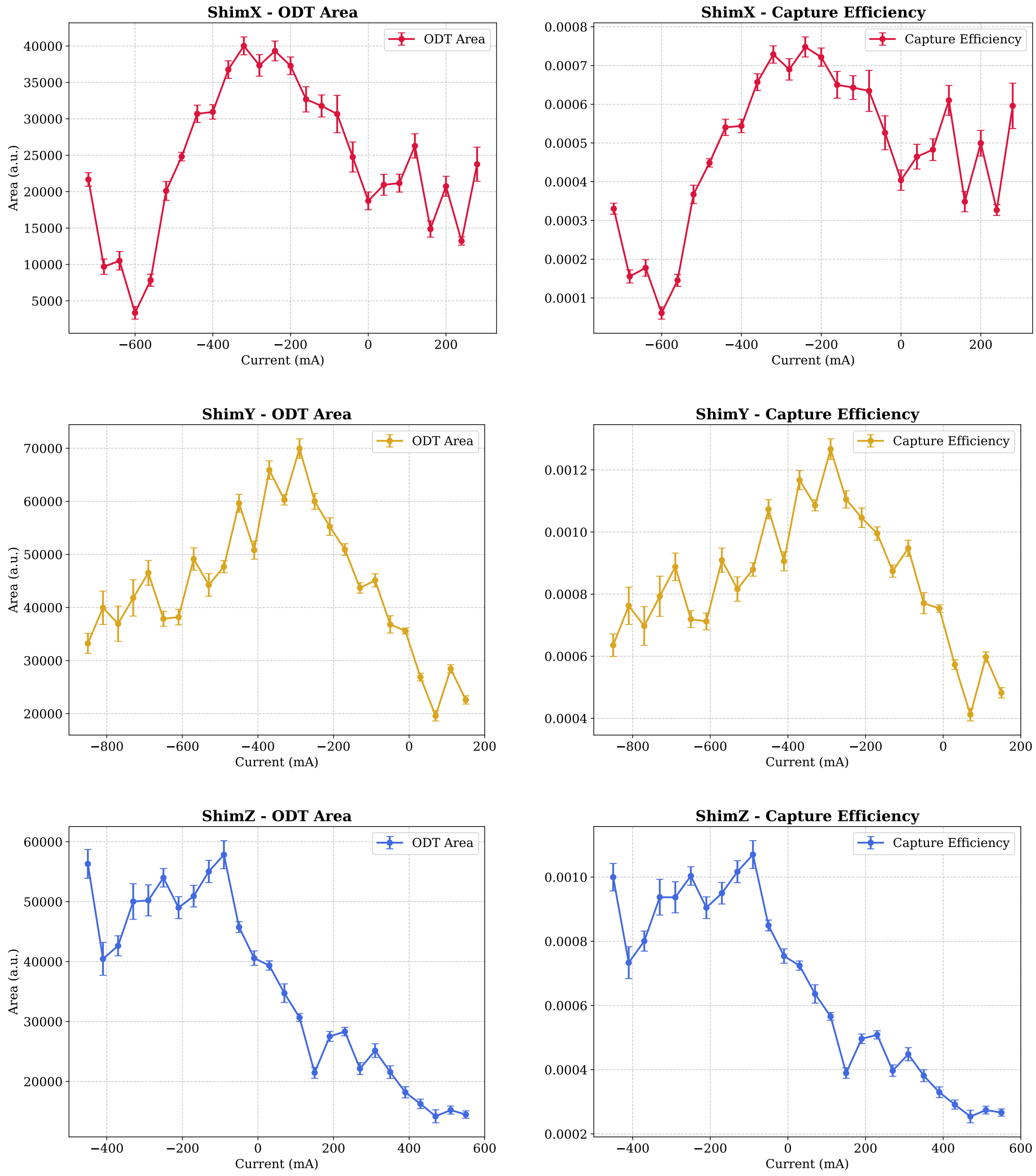
The systematic optimization of the compensation coil currents was carried out following the complete analysis procedure detailed in Section 2.3.3, utilizing the Manta camera. The procedure was performed sequentially along the spatial axes in the order $x \to z \to y$. During each scan, the currents of the remaining non-optimized coils were maintained at the preliminary values established in Equation (3.4).

The capture efficiencies extracted from these scans are summarized in Figure 3.13b for the x-, y-, and z-coils. Similarly, Figure 3.13a displays the evaluated ODT signal area for each scan. This area is directly proportional to the total number of captured atoms, making it the physical quantity of fundamental interest. However, since the camera was not calibrated for absolute atom number measurements, this quantity can only be expressed in arbitrary units. Nonetheless, these values remain perfectly comparable provided the camera gain is kept constant. As evident from the plots, the behavior of the capture efficiency mirrors the trend of the ODT area.

This initial optimization yielded the following optimal currents:

$$\begin{cases} I_x &= -220 \text{ mA}, \\ I_y &= -290 \text{ mA}, \\ I_z &= -90 \text{ mA}. \end{cases} \tag{3.5}$$

To confirm the optimization achieved in the initial measurements, a second run was per-



(a) *ODT area.* (b) *Capture efficiency.*

Figure 3.13: *ODT area (left) and capture efficiency (right) for the compensation coil current scans along x-, y-, and z-axes (performed in the order $x \rightarrow z \rightarrow y$). The qualitative agreement between the two observables is clearly visible for all the investigated axes.*

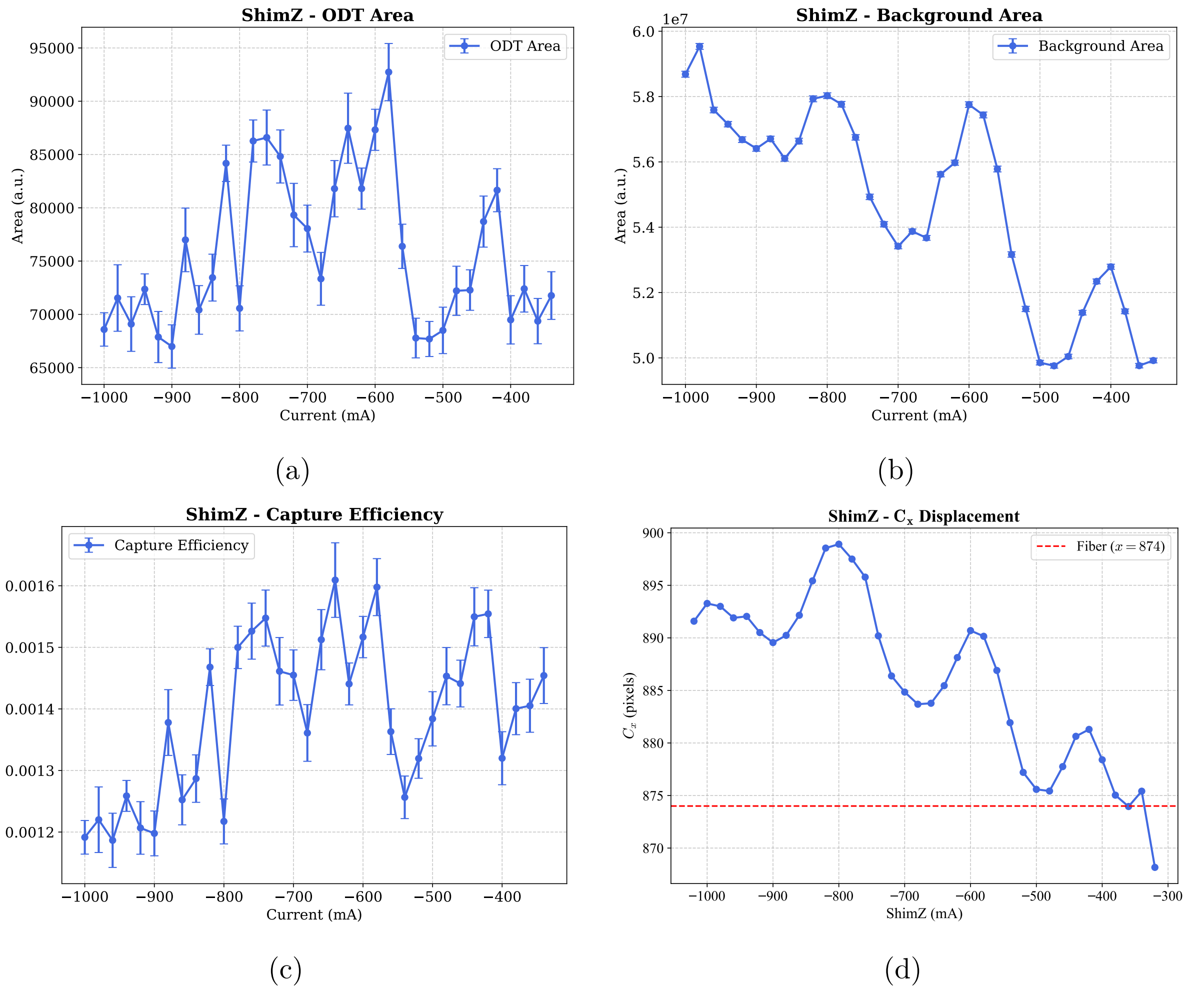


Figure 3.14: *Optimization results for the second set of measurements of the z-compensation coil current: (a) ODT signal area, (b) background area, (c) capture efficiency, (d) atomic cloud center position projected on the camera's x-axis.*

formed, yielding unexpected results regarding the z-axis compensation coil (Figure 3.14). As clearly shown in Figure 3.14a, the signal does not follow an expected parabolic or bell-shaped profile with a clear maximum. Instead, the ODT area and the capture efficiency exhibit an oscillatory behavior. Starting from a current of -1000 mA, the signal increases to a first local maximum at -800 mA, drops until ~ -700 mA, and then rises again to reach subsequent local maxima at ~ -600 mA and ~ -400 mA.

A similar periodic trend is observed in the MOT background area which represents the atomic cloud area measured at the end of the TOF with the ODT turned off (Figure 3.14b). This area exhibits peaks at ~ -800 mA, ~ -600 mA, and ~ -400 mA, matching the ODT area oscillation with good precision. While a proportional increase in the captured ODT fraction is naturally expected when the initial MOT atom number is higher, the oscillatory nature of this trend is peculiar. Tracking the x-centroid of the atomic cloud on the camera's image plane (Figure 3.14d) reveals an analogous spatial oscillation, a phenomenon not observed during the scans of the other two compensation axes. The

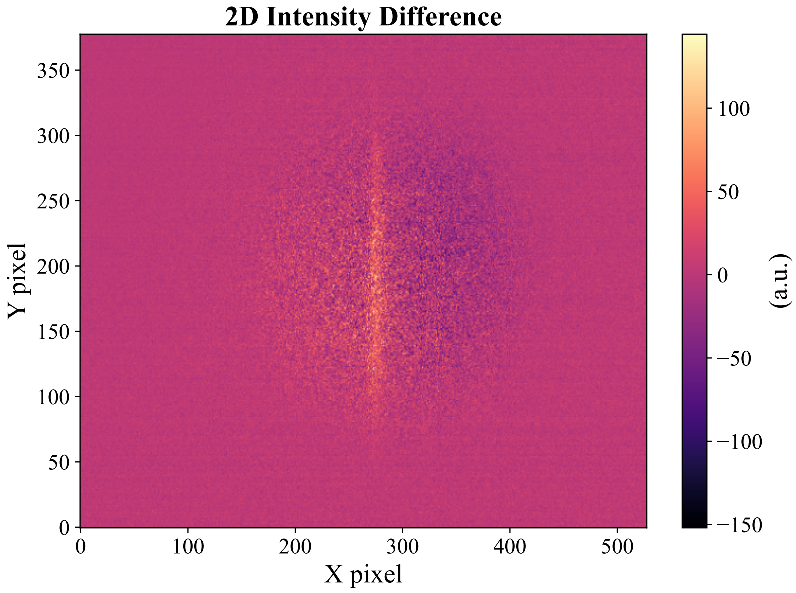


Figure 3.15: *Intensity difference 2D image obtained with the optimal currents:* $I_{\mathrm{x}} = -220$ mA*,* $I_{\mathrm{y}} = -280$ mA*, and* $I_{\mathrm{z}} = -600$ mA*, corresponding to a capture efficiency of* $\sim 0.16\%$*.*

physical origin of this oscillatory behavior remains unclear. This trend might be caused by the complex spatial intensity distribution of the cooling beams near the trap center, combined with residual misalignments. However, dedicated experimental investigations are required to determine the exact cause of these fluctuations and the consequent variations in the capture efficiency.

Finally, the optimal compensation currents and corresponding generated magnetic field component were determined to be:

$$\begin{cases} I_x &= -220 \text{ mA}, \\ I_y &= -290 \text{ mA}, \\ I_z &= -600 \text{ mA}. \end{cases} \qquad \begin{cases} B_x &= -308 \text{ mG}, \\ B_y &= -377 \text{ mG}, \\ B_z &= -900 \text{ mG}. \end{cases} \tag{3.6}$$

The corresponding MOT position translation is calculated as $t_\alpha = B_\alpha / B'$ for $\alpha = x, y$, and $t_z = -B_z/2B'$, where $2B' = 9.32$ G/cm is the axial gradient of the MOT quadrupole field. The calculation yields:

$$\begin{cases} t_x &= -661 \ \mu\text{m}, \\ t_y &= -809 \ \mu\text{m}, \\ t_z &= 966 \ \mu\text{m}. \end{cases} \tag{3.7}$$

This configuration maximized the ODT trap area, corresponding to the highest number of captured atoms, and resulted in a capture efficiency of $\eta = (0.160 \pm 0.005)\%$. The associated 2D difference image is shown in Figure 3.15.

This result must be compared to the theoretical expectations derived from the shallow and non-shallow trap models introduced in Section 2.2.4. Evaluating these models with the data obtained at the best configuration of the molasses: $w_0 = 19.0\ \mu$m, $P = (0.59 \pm 0.06)$ W, $R_x R_y \approx (1.043 \pm 0.004)$ mm^2, $T = (9.60 \pm 0.10)\ \mu$K yields:

- Shallow trap model: $\eta = (0.45 \pm 0.05)\%$;
- Non-shallow trap model: $\eta = (0.41 \pm 0.04)\%$.

Given $\xi = 0.39 \pm 0.04$, the predictions of the two models are in close agreement as expected from Figure 2.19. Note that in Section 2.2.4, the model includes a factor of 1/2 to account for the fraction of atoms with $v_z < 0$, since the model originally aimed to estimate the number of atoms guided by the ODT into the fiber core. For these measurements, this factor was omitted because our interest lies in the total number of captured atoms, rather than exclusively those reaching the fiber tip with a negative v_z velocity component.
The experimental efficiency is of the same order of magnitude as the theoretical calculation, confirming that the experimental apparatus operates correctly and that the chosen parameters are near optimal. However, the theoretical efficiency remains ~ 2.5 times larger than the measured one. This discrepancy may arise from model limitations. For instance, the model assumes a Gaussian ODT beam, which is an excellent approximation for the fundamental LP_{01} mode, however, the beam transmitted by the fiber also contains higher-order modes, as detailed in Section 2.2.3.2. Nevertheless, before attributing the deviation entirely to theoretical limitations, potential procedural artifacts must be investigated. In particular, in the current procedure, the imaging procedure at the end of the TOF requires 600 μs of time with the ODT switched off. This dead time could lead to atom loss, resulting in an artificially reduced measured trapping efficiency.

Conclusions

This thesis work implemented and characterized an experimental apparatus for an Optical Dipole Trap (ODT) designed to trap a sample of ^{87}Rb atoms, which was integrated into a pre-existing setup based on a Magneto-Optical Trap (MOT) and optical molasses cooling. The ODT was realized using 1064 nm laser light coupled into a kagome Hollow-Core Photonic Crystal Fiber (HCPCF). A crucial objective of this work was the optimization and reliable light injection into the HCPCF. Standard injection procedures relying solely on power monitoring frequently resulted in light coupling into the silica jacket or into higher-order modes. To overcome this, a custom-built optical microscope was employed for near-field monitoring. On a test fiber, this approach enabled coupling into a fundamental LP_{01}-like mode with minor deformations, while also achieving $\sim 96\%$ transmission efficiency. The application of the same procedure to the HCPCF installed in the vacuum chamber yielded a final operating efficiency of $\sim 40\%$ with a partially asymmetric mode profile. This reduced performance on the in-chamber fiber is likely due to mechanical stresses applied to the fiber itself by the vacuum sealing system.

Once the ODT was operative, the experimental focus shifted to measurements inside the vacuum chamber. First, the molasses stage temperature was optimized through systematic investigation of the cooling beam detuning, the Acousto-Optic Modulator (AOM) ramp durations and amplitudes, and the cancellation of residual magnetic fields via compensation coils. These adjustments led to a minimum x-y-averaged atomic cloud temperature of $T = (9.6 \pm 0.1)$ μK.

The final characterization focused on the atom loading efficiency from the molasses into the ODT. The spatial overlap between the atomic cloud and the ODT beam was optimized by utilizing the compensation coils during the MOT phase to displace the atomic cloud from its optimal position, thereby increasing the overlap with the ODT. The successful isolation of the trapped atom signal relied on the implementation of a higher-resolution imaging system and, most importantly, on the mitigation of the intensity noise introduced by the AOM. This enhanced the signal-to-noise ratio, enabling the correct optimization of the ODT loading process, leading to a maximum observed capture efficiency of $(0.160 \pm 0.005)\%$.

Although numerically small, the observed capture efficiency is consistent with the order of magnitude predicted by the non-shallow trap theoretical model presented in this thesis. Applying the experimental parameters of the molasses and the trap geometry to the model yields an expected capture efficiency of $(0.41 \pm 0.04)\%$. The agreement on the order of magnitude suggests that the primary limitations to the capture efficiency are given by the intrinsic properties of the apparatus such as the distance between the fiber facet and the MOT center, the beam geometrical properties and the ODT optical power, together with the minimum temperature achieved. However, the experimental value remains approximately one-third of the theoretical expectation. It should be investigated whether this discrepancy comes from procedural artifacts such as the 600 μs delay between the end of the Time-of-Flight (TOF) sequence and the conclusion of the imaging flash.

Beyond addressing this discrepancy, the results obtained in this thesis establish a robust foundation for the future development of the experiment. The identification of the optimal parameters for both the optical molasses and the ODT loading sequence provides a reliable experimental baseline. This ensures that these operational stages are fully characterized and will not require fundamental recalibration in subsequent stages. Furthermore, the developed injection setup and near-field inspection protocol will be instrumental for introducing a second, counter-propagating 1064 nm beam. This configuration is necessary to realize an optical conveyor belt, enabling the efficient loading and deep confinement of atoms directly within the hollow core of the HCPCF.

Bibliography

[1] A. Yariv and P. Yeh. *Photonics: Optical Electronics in Modern Communications*. Oxford series in electrical and computer engineering. Oxford University Press, 2007. ISBN: 9780195179460. URL: `https://books.google.it/books?id=B2xwQgAACAAJ`.

[2] Milton Abramowitz and Irene A. Stegun. *Handbook of Mathematical Functions with Formulas, Graphs, and Mathematical Tables*. New York: Dover Publications, 1965.

[3] D. Gloge. "Weakly Guiding Fibers". In: *Appl. Opt.* 10.10 (Oct. 1971), pp. 2252–2258. DOI: `10.1364/AO.10.002252`. URL: `https://opg.optica.org/ao/abstract.cfm?URI=ao-10-10-2252`.

[4] Bahaa E. A. Saleh and Malvin Carl Teich. *Fundamentals of Photonics*. 3rd. Hoboken, NJ: John Wiley & Sons, 2019. ISBN: 978-1-119-50687-4.

[5] L. Allen *et al.* "Orbital angular momentum of light and the transformation of Laguerre-Gaussian laser modes". In: *Phys. Rev. A* 45 (11 June 1992), pp. 8185–8189. DOI: `10.1103/PhysRevA.45.8185`. URL: `https://link.aps.org/doi/10.1103/PhysRevA.45.8185`.

[6] J. C. Knight *et al.* "All-silica single-mode optical fiber with photonic crystal cladding". In: *Opt. Lett.* 21.19 (Oct. 1996), pp. 1547–1549. DOI: `10.1364/OL.21.001547`. URL: `https://opg.optica.org/ol/abstract.cfm?URI=ol-21-19-1547`.

[7] R. F. Cregan *et al.* "Single-Mode Photonic Band Gap Guidance of Light in Air". In: *Science* 285.5433 (1999), pp. 1537–1539. DOI: `10.1126/science.285.5433.1537`. URL: `https://www.science.org/doi/abs/10.1126/science.285.5433.1537`.

[8] Benoît Debord *et al.* "Hollow-Core Fiber Technology: The Rising of "Gas Photonics"". In: *Fibers* 7.2 (2019). ISSN: 2079-6439. DOI: `10.3390/fib7020016`. URL: `https://www.mdpi.com/2079-6439/7/2/16`.

[9] Francesco Poletti. "Nested antiresonant nodeless hollow core fiber". In: *Opt. Express* 22.20 (Oct. 2014), pp. 23807–23828. DOI: `10.1364/OE.22.023807`. URL: `https://opg.optica.org/oe/abstract.cfm?URI=oe-22-20-23807`.

[10] J. von Neumann and E. Wigner. "Über merkwürdige diskrete Eigenwerte". In: *Phys. Z.* 30 (1929), pp. 465–467.

[11] F. Couny *et al.* "Generation and Photonic Guidance of Multi-Octave Optical-Frequency Combs". In: *Science* 318.5853 (2007), pp. 1118–1121. DOI: `10.1126/science.1149091`. eprint: `https://www.science.org/doi/pdf/10.1126/science.1149091`. URL: `https://www.science.org/doi/abs/10.1126/science.1149091`.

[12] Y. Y. Wang *et al.* "Low loss broadband transmission in hypocycloid-core Kagome hollow-core photonic crystal fiber". In: *Opt. Lett.* 36.5 (Mar. 2011), pp. 669–671. DOI: `10.1364/OL.36.000669`. URL: `https://opg.optica.org/ol/abstract.cfm?URI=ol-36-5-669`.

[13] Francois Couny, Fetah Benabid, and Philip S. Light. “Large Pitch Kagome-Structured Hollow-Core PCF”. In: *2007 Conference on Lasers and Electro-Optics (CLEO)*. 2007, pp. 1–2. DOI: 10.1109/CLEO.2007.4453155.

[14] Thomas D. Bradley *et al.* “Modal content in hypocycloid Kagome; hollow core photonic crystal fibers”. In: *Opt. Express* 24.14 (July 2016), pp. 15798–15812. DOI: 10.1364/OE.24.015798. URL: https://opg.optica.org/oe/abstract.cfm?URI=oe-24-14-15798.

[15] E. A. J. Marcatili and R. A. Schmeltzer. “Hollow Metallic and Dielectric Waveguides for Long Distance Optical Transmission and Lasers”. In: *Bell System Technical Journal* 43.4 (1964), pp. 1783–1809. DOI: https://doi.org/10.1002/j.1538-7305.1964.tb04108.x. eprint: https://onlinelibrary.wiley.com/doi/pdf/10.1002/j.1538-7305.1964.tb04108.x. URL: https://onlinelibrary.wiley.com/doi/abs/10.1002/j.1538-7305.1964.tb04108.x.

[16] M.J.F. Digonnet *et al.* “Understanding air-core photonic-bandgap fibers: analogy to conventional fibers”. In: *Journal of Lightwave Technology* 23.12 (2005), pp. 4169–4177. DOI: 10.1109/JLT.2005.859406.

[17] Kiarash Aghaie *et al.* “Classification of the Core Modes of Hollow-Core Photonic-Bandgap Fibers”. In: *Quantum Electronics, IEEE Journal of* 45 (Oct. 2009), pp. 1192–1200. DOI: 10.1109/JQE.2009.2019767.

[18] J. W. Nicholson *et al.* “Spatially and spectrally resolved imaging of modal content in large-mode-area fibers”. In: *Opt. Express* 16.10 (May 2008), pp. 7233–7243. DOI: 10.1364/OE.16.007233. URL: https://opg.optica.org/oe/abstract.cfm?URI=oe-16-10-7233.

[19] Rudolf Grimm, Matthias Weidemüller, and Yurii B. Ovchinnikov. *Optical dipole traps for neutral atoms*. 1999. arXiv: physics/9902072 [physics.atom-ph]. URL: https://arxiv.org/abs/physics/9902072.

[20] Daniel A. Steck. *Quantum and Atom Optics*. Revision 0.16.8, accessed: 2026-06-01. 2026. URL: http://steck.us/teaching.

[21] J.J. Sakurai and J. Napolitano. *Modern Quantum Mechanics*. Cambridge University Press, 2017. ISBN: 9781108422413. URL: https://books.google.it/books?id=010yDwAAQBAJ.

[22] Claude Cohen-Tannoudji. “Atomic Motion in Laser Light”. In: *Fundamental Systems in Quantum Optics*. Ed. by J. Dalibard, J. M. Raimond, and J. Zinn-Justin. Vol. LIII. Les Houches Summer School. Elsevier Science Publishers, 1992.

[23] Harold J. Metcalf and Peter Straten. *Laser Cooling and Trapping*. Graduate Texts in Contemporary Physics. Springer New York, NY, 1999. DOI: https://doi.org/10.1007/978-1-4612-1470-0.

[24] T. A. Savard, K. M. O’Hara, and J. E. Thomas. “Laser-noise-induced heating in far-off resonance optical traps”. In: *Phys. Rev. A* 56 (2 Aug. 1997), R1095(R)–R1098(R). DOI: 10.1103/PhysRevA.56.R1095. URL: https://link.aps.org/doi/10.1103/PhysRevA.56.R1095.

[25] J. Dalibard and C. Cohen-Tannoudji. “Laser cooling below the Doppler limit by polarization gradients: simple theoretical models”. In: *J. Opt. Soc. Am. B* 6.11 (Nov. 1989), pp. 2023–2045. DOI: 10.1364/JOSAB.6.002023. URL: https://opg.optica.org/josab/abstract.cfm?URI=josab-6-11-2023.

[26] M. Marchesini. "A new magneto-optical trap for cold rubidium atoms in hollow-core fibres". Ph.D. thesis. University of Bologna, 2025.

[27] M-Marqe. *MOT_setup_cads-at-DIFA_unibo*. Available online. 2024. URL: https://github.com/M-Marqe/MOT_setup_CADs-at-DIFA_UniBo.

[28] Y. Sasaki *et al.* "Polarization-maintaining and absorption-reducing fibers". In: *Optical Fiber Communication*. Optica Publishing Group, 1982, ThCC6. DOI: 10.1364/OFC.1982.ThCC6. URL: https://opg.optica.org/abstract.cfm?URI=OFC-1982-ThCC6.

[29] Daniel A. Steck. *Rubidium 85 D Line Data*. http://steck.us/alkalidata. Version revision 2.3.4. Revision 2.3.4, 8 August 2025. Aug. 2025.

[30] Daniel A. Steck. *Rubidium 87 D Line Data*. http://steck.us/alkalidata. Revision 2.3.4, 8 August 2025. Aug. 2025.

[31] Matteo Marchesini *et al.* "All-fiber, near-infrared, laser system at 780 nm for atom cooling". In: *Opt. Continuum* 3.10 (Oct. 2024), pp. 1868–1879. DOI: 10.1364/OPTCON.525738. URL: https://opg.optica.org/optcon/abstract.cfm?URI=optcon-3-10-1868.

[32] A. E. Siegman. "How to (Maybe) Measure Laser Beam Quality". In: *DPSS (Diode Pumped Solid State) Lasers: Applications and Issues*. Optica Publishing Group, 1998, MQ1. DOI: 10.1364/DLAI.1998.MQ1. URL: https://opg.optica.org/abstract.cfm?URI=DLAI-1998-MQ1.

[33] *NIST Digital Library of Mathematical Functions*. http://dlmf.nist.gov/. Release 1.2.x. F. W. J. Olver, A. B. Olde Daalhuis, D. W. Lozier, B. I. Schneider, R. F. Boisvert, C. W. Clark, B. R. Miller, B. V. Saunders, H. S. Cohl, and M. A. McClain, eds. 2026.

[34] Letizia Loparco. "Characterization of a magneto-optical trap of 87 Rb atoms". Supervisor: Prof. Francesco Minardi. MA thesis. University of Bologna, Italy, 2025. URL: https://amslaurea.unibo.it/id/eprint/35347.